Stochastic Degradation-Aware Optimal Fleet Management with Application to Electric Vehicles

A Comprehensive Reference Document

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Abstract

This document scopes a research project and collects the information needed to write an article. The goal is a general-purpose model of the management of a fleet over a planning horizon, and an algorithm that solves it.

Each member of the fleet is a collection of components. Each day, a member serves a mission (a route), goes to the depot for maintenance (a partial or full repair of each component), or stays idle. Serving a mission damages the components of the member. Each component has its own stochastic degradation model, and the damage that a day adds depends on the mission served. We show a concrete application to fleets of electric vehicles, such as buses and cars.

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1 Overview of the problem

We consider a fleet of F members. Each member has L components, and each component degrades stochastically.

- **Horizon.** The planning horizon has $T = H_1 + H_2$ days, the time steps of the model. The first H_1 days form the *transitory phase*, which starts from the current state of the fleet. The last H_2 days form the *operating phase*, which is the schedule to repeat.
- **Activities.** On each day, every member performs at most one activity: it serves one of M available missions or spends the day in the depot. A member with neither activity is idle.
- **Damage.** Serving a mission damages the components of the member. The damage follows a stochastic degradation process that depends on the mission and on the component.
- **Interventions.** On a depot day, each component receives, independently of the other components, an imperfect repair, a replacement or no intervention. An imperfect repair removes part of the accumulated damage, and a replacement resets the component to the state of a new part. A repair or a replacement is an *intervention*.

The damage, or degradation level, of component ℓ of member i on day k is a random variable $D_{i\ell k}$. Its distribution belongs to a parametric family chosen for each component. A component fails when $D_{i\ell k}$ exceeds a threshold τ. The schedule must keep the probability of this event below ε on every day.

The schedule must also be repeatable without end. For this purpose, a *loop constraint* compares each component at the end of the horizon with the same component at the end of the transitory phase. Every statistic that certifies the reliability of the component must be no larger at the end of the horizon. So, at the end of the operating phase, the fleet is in a condition at least as good as at its start.

Table 1 collects the notation used throughout this document. All variables are general and independent of any specific degradation model.

1.1 Scheduling and maintenance variables

Three families of binary decision variables encode the schedule. The *assignment variable* x_{ijk} determines which activity fleet member i performs on day k: serving mission $j \geq 1$, or spending the day in the depot ($j = 0$). Each member performs at most one activity per day,

$$\sum_{j=0}^{M} x_{ijk} \;\leq\; 1 \qquad \forall\, i,\, k, \tag{1}$$

and each mission must be covered by exactly one member,

$$\sum_{i=1}^{F} x_{ijk} \;=\; 1 \qquad \forall\, j \geq 1,\, k. \tag{2}$$

A member whose assignment row is all zeros on day k is *idle*: it serves no mission and is not in the depot.

- **Members on missions.** By (1) and (2), exactly M members serve missions on every day. Since $F > M$, the other $F - M \geq 1$ members are in the depot or idle.
- **Depot day or idle day.** Because (1) is an inequality, the choice between a depot day and an idle day is free. An idle day costs nothing, and the depot-day cost C_M prices this choice (Section 1.4).
- **No damage.** Idle days and depot days add no damage, because a component accumulates damage only on the days on which its member serves a mission (Section 1.2).
- **Explicit depot day.** A depot day must be chosen explicitly, and repairs and replacements need one.

Table 1: Notation and variable definitions.

Symbol	Description
Index sets	
$i \in \{1, \dots, F\}$	Fleet member
$j \in \{0, 1, \dots, M\}$	Activity ($j = 0$: depot day; $j \geq 1$: mission; a member with no activity is idle)
$k \in \{1, \dots, T\}$	Time step (day)
$\ell \in \{1, \dots, L\}$	Component
Scalar and fleet-level parameters	
F	Number of fleet members
H_1, H_2	Lengths of the transitory phase (days $1, \dots, H_1$) and of the operating phase (days $H_1 + 1, \dots, T$)
T	Horizon length, $T = H_1 + H_2$
M	Number of missions (must satisfy $F > M$)
L	Number of components per fleet member
Cost functions	
C_M	Depot-day cost: $\{x_{i0k}\}_{i,k} \to \mathbb{R}_{\geq 0}$
C_R	Repair cost: $\{z_{i\ell k}\}_{i,\ell,k} \to \mathbb{R}_{\geq 0}$
C_{rep}	Replacement cost: $\{x^r_{i\ell k}\}_{i,\ell,k} \to \mathbb{R}_{\geq 0}$
C_D	Damage-penalty cost: $u \to \mathbb{R}_{\geq 0}$
Component-level parameters	
$\tau_{i\ell}$	Failure threshold of the damage; the subscript $i\ell$ is often dropped ($\tau = 1$ in the Palmgren–Miner normalisation)
$\varepsilon_{i\ell}$	Reliability level: $\Pr(D_{i\ell k} > \tau_{i\ell}) \leq \varepsilon_{i\ell}$ for all k; the subscript $i\ell$ is often dropped
$\rho_{i\ell}$	Repair efficiency $\in (0, 1]$
Initial and reference values	
$D^{\text{start}}_{i\ell}$	Initial damage of component ℓ of member i (known distribution with mean in $[0, \tau)$; the reliability level is required from day 1 on)
$D^{\text{new}}_{i\ell}$	Damage just after a replacement (known distribution; typically near zero)
Stochastic state	
$D_{i\ell k}$	Random variable: damage of component ℓ of member i on day k
$\Delta D^{(j)}_{i\ell k}$	Random damage increment to component ℓ of member i when mission j is served on day k
$\xi_{i\ell k}$	Tracked state of component ℓ of member i on day k: the statistics of $D_{i\ell k}$ that certify its reliability and, under ARD_1, their anchors (Section 1.3)
Decision variables	
$x_{ijk} \in \{0, 1\}$	1 if member i is assigned to activity j on day k
$x^m_{i\ell k} \in \{0, 1\}$	1 if component ℓ of member i undergoes imperfect repair on day k
$x^r_{i\ell k} \in \{0, 1\}$	1 if component ℓ of member i is replaced on day k
$z_{i\ell k} \geq 0$	Damage removed by imperfect repair (in expected-damage units)
$u \geq 0$	Damage-penalty variable

On a depot day ($x_{i0k} = 1$), two more binary variables control the intervention on each component, independently of the other components. The *repair variable* $x^m_{i\ell k}$ indicates whether component ℓ undergoes imperfect repair, and the *replacement variable* $x^r_{i\ell k}$ indicates full replacement. Both require a depot day and are mutually exclusive:

$$x^m_{i\ell k} + x^r_{i\ell k} \;\le\; x_{i0k} \qquad \forall\, i,\, \ell,\, k. \tag{3}$$

Several components of one member may receive interventions on the same depot day. A component on a depot day with

$$x^m_{i\ell k} = x^r_{i\ell k} = 0 \tag{4}$$

receives no intervention, and its damage does not change. So, on a depot day, each component is repaired, replaced, or rests without an intervention, and exactly one of the three cases holds.

Remark 1.1 (One row instead of three). Row (3) has the same binary solutions as the three rows $x^m_{i\ell k} \le x_{i0k}$, $x^r_{i\ell k} \le x_{i0k}$ and $x^m_{i\ell k} + x^r_{i\ell k} \le 1$. Its linear relaxation is tighter: the point

$$x_{i0k} = x^m_{i\ell k} = x^r_{i\ell k} = 0.5 \tag{5}$$

satisfies the three rows but not (3). With $x^m_{i\ell k}, x^r_{i\ell k} \ge 0$ and $x_{i0k} \le 1$, row (3) describes the convex hull of the four binary solutions

$$(x_{i0k}, x^m_{i\ell k}, x^r_{i\ell k}) \in \{(0,0,0), (1,0,0), (1,1,0), (1,0,1)\}. \tag{6}$$

1.2 Damage dynamics

The damage of each component evolves day by day. Its change on a day depends on the activity of the member that carries the component. We write $\Delta D^{(j)}_{i\ell k}$ for the *random damage increment* of component ℓ of member i when the member serves mission j on day k.

We assume that the distribution of each increment is known, from the degradation model of the component. It may depend on i, ℓ, j and k. On an idle day or a depot day, the component accumulates no damage.

On each day, exactly one of the following mutually exclusive cases applies to every component (i, ℓ):

(i) **Mission.** Member i serves mission $j \ge 1$ ($x_{ijk} = 1$). The damage grows by a random increment that depends on the mission and on the component:

$$D_{i\ell k} \;=\; D_{i\ell,k-1} \;+\; \Delta D^{(j)}_{i\ell k}. \tag{7}$$

(ii) **Imperfect repair.** The member is on a depot day ($x_{i0k} = 1$) and component ℓ undergoes imperfect repair ($x^m_{i\ell k} = 1$). The repair removes a fraction of the accumulated damage. The repair efficiency $\rho_{i\ell}$ and the repair law in use determine this fraction. Abstractly:

$$D_{i\ell k} \;=\; \mathcal{R}\big(D_{i\ell,k-1},\; \rho_{i\ell}\big), \tag{8}$$

where $\mathcal{R}$ denotes the repair operator. Under ARD_1 the operator also uses the damage just after the last intervention; the notation leaves this argument implicit. The formulation offers two repair operators, or repair laws, chosen per component.

- ARD_∞ (arithmetic reduction of deterioration with infinite memory) removes the fraction $\rho_{i\ell}$ of the whole damage.
- ARD_1 (arithmetic reduction of deterioration of order 1) removes the fraction $\rho_{i\ell}$ of the damage accumulated since the last intervention, repair or replacement (Section 2.3.4).

See Section 2.3 for more details. The repair-cost variable $z_{i\ell k}$ is the expected damage that the repair removes:

$$z_{i\ell k} \;=\; \mathbb{E}\big[D_{i\ell,k-1}\big] - \mathbb{E}\big[D_{i\ell k}\big], \qquad \text{when } x^m_{i\ell k} = 1. \tag{9}$$

On a day without a repair, the equality is not imposed. Then $z_{i\ell k}$ is a free non-negative variable, and the cost minimisation drives it to zero. Equivalently, $z_{i\ell k} = 0$ when $x^m_{i\ell k} = 0$.

(iii) **Full replacement.** The member is on a depot day and component ℓ is replaced ($x^r_{i\ell k} = 1$). The damage is reset:

$$D_{i\ell k} \;=\; D^{\text{new}}_{i\ell}. \tag{10}$$

(iv) **Idle day, or depot day with no intervention on this component.** The damage is unchanged:

$$D_{i\ell k} \;=\; D_{i\ell,k-1}. \tag{11}$$

Cases (ii) and (iii) are mutually exclusive: a component cannot be both repaired and replaced on the same day. Both require a depot day ($x_{i0k} = 1$). Since exactly one case applies on each day, one equation describes the dynamics:

$$D_{i\ell k} \;=\; D_{i\ell,k-1}\big(1 - x^m_{i\ell k}\big)\big(1 - x^r_{i\ell k}\big) + \sum_{j=1}^{M} x_{ijk}\,\Delta D^{(j)}_{i\ell k} \;+\; \mathcal{R}\big(D_{i\ell,k-1},\, \rho_{i\ell}\big)\, x^m_{i\ell k} \;+\; D^{\text{new}}_{i\ell}\, x^r_{i\ell k}, \tag{12}$$

with initial condition $D_{i\ell,0} = D^{\text{start}}_{i\ell}$.

Horizon and phases. We write the horizon as $H = [H_1, H_2]$, with $T = H_1 + H_2$ days in two phases. A single value H means

$$H_1 = H_2 = H. \tag{13}$$

The transitory phase, days $1, \ldots, H_1$, starts from the current state of the fleet. The operating phase, days $H_1 + 1, \ldots, T$, is the schedule to repeat without end. The two phases may have different lengths and different increment laws.

The increment laws of the operating phase repeat with period H_2. In the *endless schedule*, the transitory phase runs once, and then the operating phase repeats. Day $T + nH_2 + h$ carries the decisions and the increment laws of day $H_1 + h$, for

$$n \geq 0 \quad \text{and} \quad 1 \leq h \leq H_2. \tag{14}$$

The days of the transitory phase read their increment laws in one of two ways.

- **No transitory laws.** When the transitory phase has no increment laws of its own, its days read the laws of the operating phase cyclically. For example, with $H = [2, 3]$, days $1, \ldots, 5$ read the operating laws $1, 2, 1, 2, 3$. With a single value H, both phases read the same H laws, one phase after the other.
- **Separate transitory laws.** In the same example, days $1, 2$ read the transitory laws, and days $3, 4, 5$ read the operating laws $1, 2, 3$.

Degradation models of the formulation. Each component has one of five degradation models: Gaussian, inverse Gaussian, Wiener, Gamma or rainflow. Section 2.2.2 lists the statistics that each model tracks and its reliability row.

- **Gaussian and Wiener.** Both the Gaussian and the Wiener model use Gaussian increments. The Gaussian model tracks a variance that a repair reduces. The Wiener model fits components whose uncertainty does not depend on the route and is not reduced by a repair. Without a reason to choose the Wiener model, choose the Gaussian model.
- **Rainflow.** The rainflow model is the distribution-free pipeline of Section 2.1.4. It makes no assumption on the distribution of the increments. Its reliability guarantee holds for every increment law that has the measured summary statistics. The cost of this generality is a looser bound than the bound that an exact distribution function gives.

State after a replacement. A replacement installs a new part. Its damage $D_{i\ell}^{\text{new}}$ is described by the same statistics as the initial damage: a known damage when its variance is 0, a random one otherwise. The default is a new part with zero damage. On a replacement day every tracked statistic takes its replacement value, and so does every anchor (Section 2.3.4).

The replacement state must satisfy every row that holds on a replacement day, in the form built at the component. A linear form of a reliability row is stricter than the exact row (Section 2.1.7). So a replacement state that passes only the exact row would make every replacement of the component infeasible. An input check prevents this (Section 5.2).

1.3 Stochastic loop constraint

For the schedule to be repeatable without end, the reliability constraint $\Pr(D_{i\ell k} > \tau) \leq \varepsilon$ must hold on every day of the endless schedule, not only on the T days of the horizon. So the condition of the fleet at the end of the operating phase must be no worse than at its start. A first formalisation compares the probabilities of failure at the start and at the end of the operating phase, component by component:

$$\Pr\big(D_{i\ell T} > \tau\big) \;\leq\; \Pr\big(D_{i\ell H_1} > \tau\big) \qquad \forall\, i,\, \ell. \tag{15}$$

Together with the reliability constraint on every day, condition (15) requires that the operating phase ends with a failure probability no larger than the one it starts with.

However, a comparison of two survival functions at a single threshold is a summary of the state that loses information. By itself, it does not guarantee that the inequality still holds when the schedule is repeated. Section 2.1.5 shows this failure for bound-based certificates.

The failure occurs already for the exact probabilities whenever the law of the state has more than one free parameter. For a Gaussian state, a change of the mean balanced by an opposite change of the variance can satisfy (15) over one repetition, while the failure probability $\Phi\big((\mu - \tau)/\sigma\big)$ rises above ε in a later repetition.

Repeating the schedule is guaranteed to keep the failure probability below ε whenever the comparison uses a quantity with two properties:

- the quantity bounds the failure probability monotonically;
- the dynamics (12) propagate it monotonically.

One such comparison is the dominance requirement

$$D_{i\ell T} \leq_{\text{st}} D_{i\ell H_1}, \tag{16}$$

in the usual stochastic order of Definition 2.7. Under ARD_∞ this requirement *chains*: it carries over from one repetition of the operating phase to the next. The reason is that the dynamics is then a coordinatewise non-decreasing map of the state and of independent random terms (Proposition 2.1). The formulation compares a few statistics of the state instead, one by one.

Tracked state. Each degradation model certifies the reliability of a component through a few statistics of its damage $D_{i\ell k}$.

- The inverse Gaussian and Gamma models track the mean.
- The Gaussian and Wiener models track the mean and the variance.
- The rainflow model tracks the mean and, except under the Markov bound, one statistic of the chosen tail bound (Sections 2.1.4 and 2.2.2).

Under the repair law ARD_1, each statistic except the Wiener variance also has an *anchor*: its value just after the last intervention (Section 2.3.4). The *tracked state* $\xi_{i\ell k}$ collects the statistics and their anchors on day k. Each component also has a *start of the repetition* $y_{i\ell}$. It is a vector over the same coordinates as $\xi_{i\ell k}$, anchors included: the tracked state at the start of every repetition of the operating phase. Three groups of rows tie $y_{i\ell}$ to the tracked state, one row per coordinate:

- **Start rows.** The start is at least the state of day H_1:

$$y_{i\ell} \;\geq\; \xi_{i\ell H_1} \qquad \forall\, i,\, \ell \quad \text{(componentwise)}. \tag{17}$$

- **Rows of day H_1+1.** Every row of day H_1+1 that reads the state of the day before reads $y_{i\ell}$ in place of $\xi_{i\ell H_1}$. So each rule of Section 1.3.1 applies to $y_{i\ell}$, and

$$\xi_{i\ell,H_1+1} \;\geq\; \pi_{H_1+1}\big(y_{i\ell}\big), \tag{18}$$

  with the notation of Section 1.3.1.
- **Loop rows.** The operating phase ends at most at its start:

$$\xi_{i\ell T} \;\leq\; y_{i\ell} \qquad \forall\, i,\, \ell \quad \text{(componentwise)}. \tag{19}$$

For a Gaussian component with mean μ and variance v, for example, the loop rows are $\mu_T \leq y^{\mu}$ and $v_T \leq y^{v}$, plus the same rows for the two anchors under ARD_1. The variable $y_{i\ell}$ has no day index. It may lie above $\xi_{i\ell H_1}$, so the repetitions may start higher than the state that the transitory phase reaches. This matters when the component is replaced on day H_1: the replacement fixes $\xi_{i\ell H_1}$, but not $y_{i\ell}$ (Section 1.3.3).

1.3.1 One-sided encoding of the dynamics

Each day, every tracked statistic of a component follows one rule:

- **Mission.** A mission adds an increment.
- **Repair.** A repair applies the repair map of Section 2.3.4.
- **Replacement.** A replacement resets the statistic (Section 1.2).

The formulation writes each rule as a *one-sided* row: an inequality ($\geq$), not an equality. The solver may choose a value above the result of the rule, never a value below it. A value above the result of the rule carries *slack*. So a tracked statistic is an upper bound on the value that the rule gives.

Example: with $\mu_{i\ell,k-1} = 0.3$ and a mission increment of 0.2, the rule gives 0.5; the rows accept $\mu_{i\ell k} = 0.5$ and also $\mu_{i\ell k} = 0.6$.

Replacement rows and anchor updates are equalities. The rows have two forms with the same feasible set at binary points (Section 5.1).

- **Big-M row.** A binary switches the row off through a large constant M.
- **Indicator constraint.** A row that the solver enforces only when a binary is 1.

Fix a schedule, that is, values of every binary, and one component. The following objects are used here and in the rest of this section.

- The *full tracked state* ξ_k collects the values on day k of every tracked statistic of the component and of every anchor. ξ_0 holds the initial values and the starting anchors.
- The *day map* π_k gives, for each coordinate, the value that the rule of day k assigns: $s+\delta s_k$ and g on a day without intervention; $a\,s+(1-a)\,g$ for both s and g on a repair day; the replacement value s^{new} for both on a replacement day.
- The *reference trajectory* is $\zeta_0 = \xi_0$, $\zeta_k = \pi_k(\zeta_{k-1})$: the rows with every inequality replaced by an equality.
- The *true statistics* θ_k are the true values on day k of the tracked statistics (the mean, the variance, the Hoeffding budget or the cumulant generating function of the damage $D_{i\ell k}$) and of their anchors (their values just after the last intervention).
- The *certificate set* $\mathcal{C}_{i\ell}$ is the set of values of the tracked statistics at which the reliability row of the component holds in its exact form, side rows included. Anchors do not enter it.

For every model of the formulation, three facts hold (Section 2.2.2 states them model by model):

(M1) Each repair factor a lies in $[0,1]$, and each increment δs_k is non-negative.

(M2) On day 0 the true statistics are at most the initial values and starting anchors, $\theta_0 \le \xi_0$. On every later day the true statistic is at most the day map applied to the true statistics of the day before: $\theta_k \le \pi_k(\theta_{k-1})$. This inequality is an equality for every statistic, except the cumulant generating function of the Chernoff bound, whose map is an upper bound (Proposition 2.2).

(M3) If values $s \in \mathcal{C}_{i\ell}$ are at least the true statistics of day k, then $\Pr(D_{i\ell k} > \tau) \le \varepsilon$.

The set $\mathcal{C}_{i\ell}$ is closed, and it is a down-set: if $s \in \mathcal{C}_{i\ell}$ and

$$0 \le s' \le s \quad \text{componentwise}, \tag{20}$$

then $s' \in \mathcal{C}_{i\ell}$. The reference values are non-negative, $\zeta_k \ge 0$: the initial and replacement values are non-negative (input checks), and by (M1) every day map keeps non-negative values non-negative. Table 2 lists the reliability rows in chance-bound form. The derivation is in Appendix A.1, page 74.

Table 2: Reliability rows of the formulation in chance-bound form. $\kappa_G = \Phi^{-1}(1-\varepsilon)$, $\kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}$, $\kappa = \ln(1/\varepsilon)$, $c = \kappa b_\ell/3$.

Model or bound	**Row in chance-bound form**	**Source**
Gaussian, Wiener	$\mu + \kappa_G\sqrt{v} \le \tau$	Section 2.2.2
Inverse Gaussian	$\mu \le \tau/\bar{q}$	Theorem 2.13
Gamma	$\mu \le A_1(\beta^*\tau, \varepsilon)/\beta^*$	Proposition 2.19
Rainflow, Markov	$\mu \le \varepsilon\,\tau$	Table 8
Rainflow, Cantelli	$\mu + \kappa_C\sqrt{v} \le \tau$	Table 8
Rainflow, Hoeffding	$\mu + \sqrt{\kappa R/2} \le \tau$	Table 8
Rainflow, Bernstein	$\mu + c + \sqrt{c^2 + 2\kappa v} \le \tau$	Table 8
Rainflow, Chernoff	$K \le s\,\tau + \ln\varepsilon$	Table 8

Lemma 1.2 (Monotone day maps)**.** *Each day map π_k is non-decreasing in each coordinate of the full tracked state, anchors included.*

The proof is in Appendix A.1, page 74.

Proposition 1.3 (Domination)**.** *Every feasible solution satisfies*

$$\xi_k \ge \pi_k(\xi_{k-1}) \quad \textit{and} \quad \theta_k \le \zeta_k \le \xi_k \qquad \textit{for } k = 1, \dots, T. \tag{21}$$

The proof is in Appendix A.1, page 74.

On every day $k = 1, \dots, T$ the tracked values satisfy the reliability row in the form chosen for the component. So $\xi_k \in \mathcal{C}_{i\ell}$, and $\Pr(D_{i\ell k} > \tau) \le \varepsilon$ on the days of the horizon. The derivation is in Appendix A.1, page 74. Theorem 1.4 below extends this to every day of every later repetition of the operating phase.

Meaning of the reported values. The output of a solve reports the tracked values ξ_k, not the reference values. By Proposition 1.3, each reported statistic and anchor is an upper bound on its reference value, and so on its true statistic.

The costs computed from them are upper bounds on the costs computed from the reference values: the repair cost (Proposition 1.9) and the damage penalty (Section 1.4). Against the true costs there is one exception, for a starting anchor above the true one (Section 1.4).

The minimisation removes slack only where slack costs something.

- **Mean.** A larger mean raises the damage penalty $C_D\,u$ where u is binding (where a row that defines u holds with equality), and raises the repair cost $C_R\,z$ of the next repair, unless a replacement comes first. Under ARD_∞ it also raises the cost of the later repairs up to the next replacement; under ARD_1 the anchor takes the slack at the next intervention, and the later repairs do not change.

- **Other statistics.** A larger variance, Hoeffding budget or cumulant generating function tightens the reliability row and so restricts the schedule.

Elsewhere the solver may return any value above the reference value. With $C_D = 0$ the damage penalty is absent, and the reported states may carry more slack. Slack never affects safety (Theorem 1.4).

1.3.2 Perpetual safety

In the endless schedule (Section 1.2), day $T + nH_2 + h$ carries the binaries and the increment laws of day $H_1 + h$. So every repetition applies the same day maps $\pi_{H_1+1}, \dots, \pi_T$. We write

$$\Pi_h = \pi_{H_1+h} \circ \cdots \circ \pi_{H_1+1} \tag{22}$$

for the first h days of the operating phase and $\Pi = \Pi_{H_2}$ for the whole phase.

Theorem 1.4 (Perpetual safety)**.** *Let a solution satisfy every row of the problem. Then, for every component and every day $k \geq 1$ of the endless schedule, the true statistics θ_k and the reference values ζ_k are at most a tracked value of the horizon that lies in $\mathcal{C}_{i\ell}$. Therefore $\zeta_k \in \mathcal{C}_{i\ell}$ and $\Pr(D_{i\ell k} > \tau) \leq \varepsilon$ on every day $k \geq 1$.*

The proof is in Appendix A.1, page 74.

- **Models and repair laws.** The theorem holds for every model and both repair laws, because it uses only the facts (M1)–(M3) and Lemma 1.2.
- **Encodings and forms.** The theorem holds in both encodings and in every form of the reliability rows.
- **Phase lengths.** The theorem needs no relation between H_1 and H_2: the transitory phase may be shorter or longer than the operating phase, because H_1 only marks the day at which the operating phase starts.
- **Slack.** Slack in the tracked values does not weaken the theorem: the proof compares the reference trajectory with the tracked values, whatever slack they carry.

Why the anchors enter the loop. Under ARD_1 a larger anchor gives a weaker repair, so an anchor that grows over one repetition weakens the repairs of the next one. A loop row on the mean alone would accept an unsafe schedule. The derivation is in Appendix A.1, page 75.

For the same reason, the loop constraint compares the tracked statistics one by one and never a scalar function of them, such as a bound on the failure probability (Section 2.1.5 gives an example). The loop constraint states an order between tracked statistics; it is not a stochastic order between damage laws.

1.3.3 What the one-sided rows accept

Call a schedule *safe* for a component when the reference trajectory of the endless schedule lies in $\mathcal{C}_{i\ell}$ and in the natural ranges of the statistics on every day. The natural range of a statistic is an upper bound of its value at every point that satisfies the reliability row (Section 5.1). The one-sided rows relate to safety as follows.

- *They never accept an unsafe schedule* (Theorem 1.4).
- *They accept every schedule that equality rows accept*, and more. Equality rows force $\xi_k = \zeta_k$; Proposition 1.5 (a) shows that such a schedule satisfies the one-sided rows too. The first example below shows a safe schedule that only the one-sided rows accept.
- *They may reject a safe schedule*, as equality rows may. Proposition 1.5 (both parts) excludes this for a component in the exact form with one tracked statistic and the law ARD_∞. Otherwise a rejection can occur in two cases:
  - tracked statistics, anchors included, move in opposite directions over the first repetition (the second example below);

– a linear form is stricter than the exact row.

A replacement on day H_1 causes no rejection: the start of the repetition y may lie above the replacement state (Section 1.3). A check of all schedules of 42 small test fleets (176 912 schedules; 8 Nov 2026) compared the rows with the repeated exact trajectory. The rows refuse none of the safe schedules and accept no unsafe one. Loop rows that compare day T with day H_1 instead (without the start of the repetition) refuse 1512 of these safe schedules, all with a replacement on day H_1.

Proposition 1.5 (Acceptance)**.** *Fix a schedule and a component, use the exact reliability form, and let the schedule be safe for the component. Let* $\bar{\zeta} = \zeta_{H_1}$.

(a) If $\Pi(\bar{\zeta}) \le \bar{\zeta}$*, the rows of the component, the loop rows included, accept the schedule with* $\xi_k = \zeta_k$ *for every* k *and* $y = \bar{\zeta}$.

(b) If $\Pi(\bar{\zeta}) \ge \bar{\zeta}$*, the rows of the component accept the schedule.*

The proof is in Appendix A.1, page 75.

A component in the exact form with one tracked statistic under ARD_∞ accepts every safe schedule, whatever happens on day H_1. The derivation is in Appendix A.1, page 76. Under ARD_1 the anchor is a second coordinate of the state. The coordinates can move in opposite directions over a repetition, so $\Pi(\bar{\zeta})$ and $\bar{\zeta}$ may be incomparable. Then neither part applies, and it is not proven that the rows accept every safe schedule.

Example 1.6 (One statistic, accepted only by the one-sided rows)**.** One tracked mean, law ARD_∞, $\rho = 0.5$. The operating phase is a mission ($\delta\mu = 0.2$) followed by a repair, so

$$\Pi(y) = 0.5\,(y + 0.2) = 0.5\,y + 0.1, \tag{23}$$

and $\Pi(y) \le y$ if and only if $y \ge 0.2$. Let $\bar{\zeta} = 0.1$.

- Equality rows:
$$\Pi(0.1) = 0.15 > 0.1, \tag{24}$$
so equality rows reject the schedule.
- One-sided rows: they accept it with $y = 0.2$. The tracked values over one repetition, from y, are $0.4, 0.2$, and the reliability row is checked at 0.4.
- Reference: the reference trajectory is $0.1, 0.3, 0.15, 0.35, 0.175, \ldots$. Its values at the start of each repetition rise to 0.2, and its peaks rise to 0.4. So the schedule is safe whenever $0.4 \in \mathcal{C}_{i\ell}$.

The drawback is in the reported cost: each repair is charged $\rho \cdot 0.4 = 0.2$, while the reference removal rises from 0.15 toward 0.2.

Example 1.7 (Two statistics that move in opposite directions, rejected)**.** Cantelli bound, $\varepsilon = 0.1$ ($\kappa_C^2 = 9$), $\tau = 1$, $\rho = 0.5$, law ARD_∞. The operating phase is a mission, which adds 0.2 to the mean and 0.009 to the variance, followed by a repair. So

$$\Pi(\mu, v) = (0.5\,\mu + 0.1,\ 0.25\,v + 0.00225), \tag{25}$$

with fixed point $(0.2, 0.003)$. Let $\bar{\zeta} = (0.5, 0)$.

- Reference: over the repetitions the mean falls to 0.2 and the variance rises to 0.003. The first peak is $(0.7, 0.009)$, with
$$9 \cdot 0.009 = 0.081 \le 0.3^2 = 0.09. \tag{26}$$
Every later peak has a mean of at most 0.55 and a variance below 0.012, with
$$9 \cdot 0.012 = 0.108 \le 0.45^2 = 0.2025. \tag{27}$$
The schedule is safe.

- One-sided rows: the start rows and the loop rows need

$$y \geq \xi_{H_1} \geq (0.5, 0) \tag{28}$$

with $\Pi(y) \leq y$, so $y_v \geq 0.003$. The tracked peak is then at least $(0.7, 0.012)$, where

$$9 \cdot 0.012 = 0.108 > 0.09. \tag{29}$$

Rejected.
- Equality rows: $\Pi(\bar{\zeta}) = (0.35, 0.00225)$ has a larger variance than $\bar{\zeta}$. Rejected as well.

1.3.4 Replacements required by the loop constraint

A component may be replaced on any depot day of its member, any number of times, and a replacement costs $C_{\text{rep},i\ell}$ and nothing else (Section 1.4). The loop constraint can require at least one replacement in the operating phase, for two reasons. The first is the Wiener variance, which falls only by replacement (Section 2.2.2). The second is the repair law ARD_1.

Proposition 1.8 (ARD_1). *Let a component follow ARD_1 with $\rho < 1$, and let its member serve, in the operating phase, at least one mission with a positive mean increment for the component. Then every schedule that satisfies the loop constraint* (19) *replaces the component at least once in the operating phase.*

The proof is in Appendix A.1, page 76.

The requirement is a property of the repair law, not a side effect of the formulation. Without a replacement, each repair leaves in place the share $1 - \rho$ of the damage added since the last intervention. So the anchor rises with every repair, and the mean grows without bound over the repetitions of the operating phase.

Example: $\rho = 0.5$, and a mission adds 0.2 to the mean between two repairs; each repair leaves the mean 0.1 higher than the previous one. With $\rho = 1$ a repair returns the state to the anchor, and no replacement is needed.

When replacements are not allowed, the member of such a component cannot serve, in the operating phase, a mission with a positive mean increment for the component. If the fleet cannot cover the missions without such a mission, the problem is infeasible.

1.4 Cost function

The objective is to minimise the total cost of operating the fleet over the planning horizon. Four terms contribute to the cost:

$$\min \quad C_M\left(\{x_{i0k}\}_{i,k}\right) \; + \; C_R\left(\{z_{i\ell k}\}_{i,\ell,k}\right) \; + \; C_{\text{rep}}\left(\{x^r_{i\ell k}\}_{i,\ell,k}\right) + \; C_D\left(u\right). \tag{30}$$

We assume that all four cost functions are non-negative and non-decreasing in each argument. So the cost never decreases when the schedule has more depot days, removes more damage, replaces more components or accepts a higher aggregate damage.

The first three terms are the following.

- **Depot-day cost C_M.** The first term charges for every depot day of the fleet. It is paid once per member and depot day ($x_{i0k} = 1$), whatever the number of components repaired or replaced on that day. It prices the visit to the depot, for example the member out of service and the set-up of the work. An idle day costs nothing.
- **Repair cost C_R.** The second term is the cost of imperfect repairs: the cost of a repair grows with the damage removed, $z_{i\ell k}$.
- **Replacement cost C_{rep}.** The third term charges for every replacement.

The last term is the *damage penalty*, a regularisation term. The variable u is the largest aggregate expected damage over all members and days:

$$u \;=\; \max_{i,k}\; \phi\Big(\mathbb{E}[D_{i1k}],\; \ldots,\; \mathbb{E}[D_{iLk}]\Big), \tag{31}$$

where

$$\phi\colon \mathbb{R}^{L} \to \mathbb{R}_{\geq 0} \tag{32}$$

is a non-decreasing aggregation function. The formulation offers two penalty types.

- **Default type.** It takes ϕ as the sum of the expected component damages, which gives one linear row per member and day.
- **Quadratic type.** It takes ϕ as the sum of their squares, which gives one convex quadratic row per member and day. The quadratic type is used only with the exact form of the reliability rows, because the linear forms of those rows (Section 2.1.7) serve to keep the whole model linear.

The damage penalty sets the balance between low operating costs and an even spread of damage across the fleet. It prices the largest damage of a member, so the minimisation spreads damage across the fleet. It also removes slack from the tracked means where u is binding (Section 1.3.1).

By Proposition 1.3 and $\zeta_k \geq 0$, the tracked means satisfy

$$\mu_{i\ell k} \geq \zeta^{\mu}_{i\ell k} \geq 0. \tag{33}$$

So both penalty types computed from them are upper bounds on the same quantity computed from the reference values. The coefficient satisfies $C_D \geq 0$, because a negative C_D would reward damage.

A variable that feeds only a cost term with a zero coefficient is not built, and neither are the rows that define it. For example, with $C_D = 0$ the variable u and its rows are dropped. At a component whose repair-cost coefficient is zero, the variables $z_{i\ell k}$ and their rows are dropped.

Objective modes. Let c_k be the cost of day k (the depot-day, repair and replacement terms of that day), and let

$$J_{\text{init}} = \sum_{k \leq H_1} c_k \qquad \text{and} \qquad J_{\text{op}} = \sum_{k > H_1} c_k \tag{34}$$

be the cost of the transitory phase and the cost of the operating phase. The formulation offers three objectives, and Table 3 compares them.

Table 3: Objective modes of the formulation.

Mode	Objective	Meaning	Transitory phase
Total	$J_{\text{init}} + J_{\text{op}} + C_D\, u$	the cost of the T modelled days	priced by J_{init}
Operating average	$J_{\text{op}}/H_2 + C_D\, u$	the cost per day of the operating phase	no day cost, but still constrained and part of u
Evaluation total	$J_{\text{init}} + \frac{E-H_1}{H_2}\, J_{\text{op}} + C_D\, u$	the projected cost over an evaluation horizon of E days	priced by J_{init}

- **Operating average.** It is the long-run cost per day of the endless schedule, in which the cost of the transitory phase spreads over infinitely many days. It is comparable across values of H_2, whereas a raw objective over T days is not. The solver chooses the schedule of the transitory phase for feasibility and, when $C_D > 0$, for the damage penalty, because u also covers the transitory days.

- **Evaluation total.** The E days hold the transitory phase once and then $(E - H_1)/H_2$ repetitions of the operating phase; a fraction of a repetition counts in proportion. Because it prices the transitory phase, it compares solves with different H_1. It needs $E > H_1$: for a smaller E the operating cost would have a negative weight.

The damage penalty is added once in every mode, because u is the maximum over the whole horizon. Example: $H_1 = 5$, $H_2 = 10$, $E = 365$. Then the evaluation total weights the operating cost by 36, and the operating average by 1/10.

In the operating average, the repair-cost variable is not built on the transitory days, because their cost does not enter the objective. The repair cost of those days is computed after the solve, from the tracked mean and its anchor. The objective value is computed from the tracked values, so it is an upper bound on the cost of the schedule computed from the reference values (Section 1.3.1).

Repair cost. On a repair day, $z_{i\ell k}$ is the expected damage that the repair removes (Eq. (9)). The formulation writes it with the tracked mean μ and its anchor g^μ, with $g^\mu \equiv 0$ under ARD_∞:

$$z_{i\ell k} \;\geq\; \rho\,(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1}) - \rho\,\tau\,(1 - x^{m}_{i\ell k}), \qquad z_{i\ell k} \geq 0. \tag{35}$$

- **Repair day.** The first row reads

$$z_{i\ell k} \geq \rho\,(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1}), \tag{36}$$

  the amount that the repair map removes from the mean.
- **Any other day.** The right-hand side of the first row is at most 0, because

$$0 \leq \mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1} \leq \mu_{i\ell,k-1} \leq \tau. \tag{37}$$

  A replacement day has $x^m_{i\ell k} = 0$, so $z_{i\ell k} = 0$ at the optimum, and a replacement costs $C_{\mathrm{rep},i\ell}$ only. The replaced part is discarded, so its damage does not change the price of the new part.

The variable $z_{i\ell k}$ appears in no row other than (35), and in the objective with a positive coefficient. The minimisation therefore drives it to the largest of its lower bounds.

A McCormick envelope is the standard set of linear rows for the product of a bounded variable and a binary. Its upper rows, $z \leq \rho\,(\mu - g^\mu)$ and $z \leq \rho\,\tau\,x^m$, would only remove points that are never optimal, so they are not built.

A solution that is not optimal (for example, the best one found at the time limit) may carry a larger z. Its reported cost is then larger, never smaller.

Proposition 1.9 (What z measures)**.** *In every feasible solution, on every repair day k of a component, under both laws,*

$$z_{i\ell k} \;\geq\; \rho\,(\mu_{k-1} - g^{\mu}_{k-1}) \;\geq\; \rho\,(\zeta^{\mu}_{k-1} - \zeta^{g}_{k-1}), \tag{38}$$

where ζ^μ and ζ^g are the mean and its anchor on the reference trajectory (Section 1.3.1). When the starting anchor equals the true mean level after the last intervention before the horizon, the right-hand side is the expected damage that the repair removes.

The proof is in Appendix A.1, page 76.

The default starting anchor may lie above the true one (Section 2.3.4), and such an anchor models a weaker repair.

- **First repair.** Then, for the first repair of the horizon, when no intervention precedes it, the charged removal follows the starting anchor and may be below the true removal.
- **Later repairs.** From the first intervention on, the reference gap $\zeta^\mu - \zeta^g$ equals the true gap, the sum of the mean increments since that intervention, so the bound holds again.

Under both laws, then, the reported repair cost

$$\sum C_R\, z \tag{39}$$

is an upper bound on the expected repair cost of the schedule, with this exception for a starting anchor above the true one.

Under ARD_1 the tracked mean and the tracked anchor are both upper bounds, and their difference is still an upper bound on the reference difference. The reason is that the anchor carries the slack that the mean had at the last intervention, and the slack of the mean does not decrease until the next intervention. The bound is exact when the tracked mean has no slack.

1.5 Abstract problem formulation

The fleet management problem collects the objective (30), the scheduling constraints (1)–(3), the damage dynamics (12), the definition (9) of the repair-cost variable, the reliability constraint and the loop constraint (19):

$$\min_{x,\,x^m,\,x^r,\,z,\,u} \quad C_M\big(\{x_{i0k}\}\big) + C_R\big(\{z_{i\ell k}\}\big) + C_{\mathrm{rep}}\big(\{x^r_{i\ell k}\}\big) + C_D(u) \tag{40a}$$

$$\text{s.t.} \quad \sum_{j=0}^{M} x_{ijk} \le 1 \qquad \forall\, i,\, k, \tag{40b}$$

$$\sum_{i=1}^{F} x_{ijk} = 1 \qquad \forall\, j \ge 1,\, k, \tag{40c}$$

$$x^m_{i\ell k} + x^r_{i\ell k} \le x_{i0k} \qquad \forall\, i,\, \ell,\, k, \tag{40d}$$

$$D_{i\ell k} = D_{i\ell,k-1}(1 - x^m_{i\ell k})(1 - x^r_{i\ell k}) + \sum_{j=1}^{M} x_{ijk}\, \Delta D^{(j)}_{i\ell k} + \\ + \mathcal{R}(D_{i\ell,k-1}, \rho_{i\ell})\, x^m_{i\ell k} + D^{\mathrm{new}}_{i\ell}\, x^r_{i\ell k} \qquad \forall\, i,\, \ell,\, k, \tag{40e}$$

$$D_{i\ell,0} = D^{\mathrm{start}}_{i\ell} \qquad \forall\, i,\, \ell, \tag{40f}$$

$$\Pr(D_{i\ell k} > \tau) \le \varepsilon \qquad \forall\, i,\, \ell,\, k, \tag{40g}$$

$$y_{i\ell} \ge \xi_{i\ell H_1}, \quad \xi_{i\ell T} \le y_{i\ell} \quad \text{(componentwise)} \qquad \forall\, i,\, \ell, \tag{40h}$$

$$z_{i\ell k} = x^m_{i\ell k}\,\big(\mathbb{E}[D_{i\ell,k-1}] - \mathbb{E}[D_{i\ell k}]\big) \qquad \forall\, i,\, \ell,\, k, \tag{40i}$$

$$u = \max_{i,k}\ \phi\big(\mathbb{E}[D_{i1k}], \ldots, \mathbb{E}[D_{iLk}]\big), \tag{40j}$$

$$x_{ijk},\, x^m_{i\ell k},\, x^r_{i\ell k} \in \{0,1\}, \quad z_{i\ell k} \ge 0, \quad u \ge 0. \tag{40k}$$

Here $\xi_{i\ell k}$ is the tracked state and $y_{i\ell}$ the start of the repetition of Section 1.3. The rows of day H_1+1 that read the state of the day before read $y_{i\ell}$ in place of $\xi_{i\ell H_1}$. When $C_D = 0$, the variable u and row (40j) are not built (Section 1.4). The constraints have three groups.

- The fleet rows (assignment and depot days) do not depend on the degradation models.
- The common rules of the dynamics (repair laws, one-sided encoding, replacement, linear forms of the reliability rows) hold for every model.
- Each model adds its tracked statistics, day maps and reliability row (Section 2.2.2).

The rows of the repair cost, of the damage penalty and of the loop constraint complete the problem. All rows hold for every day $k = 1, \ldots, T$ unless a section says otherwise.

Problem (40) is a stochastic combinatorial optimisation problem. It does not fix the degradation model (the distribution of $\Delta D^{(j)}_{i\ell k}$), the repair law (the operator $\mathcal{R}$), the aggregation

function ϕ or the cost structure (the functions $C_M, C_R, C_{\text{rep}}, C_D$). A concrete fleet fixes all of them. For a fleet of electric vehicles, Section 3.1 gives the degradation family of each component, and Section 4 gives linear cost functions.

2 Literature review

2.1 Stochastic degradation and fatigue models

Stochastic models of component degradation form two families. *Damage-accumulation models* track a cumulative damage variable $X(t)$, whose distribution evolves over time. Failure occurs when $X(t)$ exceeds a threshold. *Remaining-life models* describe the number of load cycles that a component can sustain before failure. Empirical laws relate the applied stress to the fatigue life.

2.1.1 Damage-accumulation processes

Table 4 summarises four widely used stochastic processes for cumulative damage. In each case $X(t)$ denotes the cumulative damage at time t, with $X(0) = 0$. All four processes have independent, stationary increments, so they are Lévy processes.

Table 4: Stochastic damage-accumulation models. Parameters: $\alpha, \beta > 0$ (Gamma shape rate / inverse scale), μ (drift), σ^2 (diffusion coefficient), λ (shape rate of the inverse Gaussian (IG) process), ν (Poisson intensity), F_Y (jump-size distribution). Mon.: monotone sample paths; CP: compound Poisson.

Model	**Increment** $X(t) - X(s)$	**Marginal** $X(t)$	**Mon.**	**Closure under +**
Gamma process [1]	$\mathrm{Ga}\big(\alpha(t-s),\, \beta\big)$	$\mathrm{Ga}(\alpha t,\, \beta)$	Yes	Same rate β
Wiener process [39]	$\mathcal{N}\big(\mu(t-s),\, \sigma^2(t-s)\big)$	$\mathcal{N}(\mu t,\, \sigma^2 t)$	No	Always
Inverse Gaussian process [42]	$\mathrm{IG}\big(\mu(t-s),\, \lambda(t-s)^2\big)$	$\mathrm{IG}(\mu t,\, \lambda t^2)$	Yes	Same μ^2/λ
Compound Poisson [13, 36]	$\sum_{n=N(s)+1}^{N(t)} Y_n$	$\mathrm{CP}(\nu t,\, F_Y)$	If $Y \geq 0$	Always

Gamma process. Abdel-Hameed [1] introduced the Gamma process, and van Noortwijk [35] surveyed it in depth. It is the most widely used model for monotone degradation (wear, corrosion, crack growth). The process $\{X(t)\}$ has non-negative independent increments

$$X(t) - X(s) \sim \mathrm{Ga}(\alpha(t-s),\, \beta), \tag{41}$$

where $\alpha > 0$ is the shape rate and $\beta > 0$ is the inverse scale (rate). Its mean and variance at time t are

$$\mathbb{E}[X(t)] = \alpha t/\beta \quad \text{and} \quad \mathrm{Var}[X(t)] = \alpha t/\beta^2. \tag{42}$$

The Gamma family is closed under convolution when the rate parameter β is shared: if $X_1 \sim \mathrm{Ga}(\alpha_1, \beta)$ and $X_2 \sim \mathrm{Ga}(\alpha_2, \beta)$ are independent, then

$$X_1 + X_2 \sim \mathrm{Ga}(\alpha_1 + \alpha_2, \beta). \tag{43}$$

Wiener process (Brownian motion with drift). The Wiener degradation model [39] sets $X(t) = \mu t + \sigma W(t)$, where $W(t)$ is a standard Brownian motion, $\mu > 0$ the drift rate, and σ^2 the diffusion coefficient, with $\sigma > 0$. Increments are normally distributed:

$$X(t) - X(s) \sim \mathcal{N}(\mu(t-s),\, \sigma^2(t-s)). \tag{44}$$

The normal family is closed under addition without conditions. Unlike the other three models, the Wiener process is *not* monotone: its sample paths can decrease. So it suits degradation with short-term fluctuations. Its first-passage time to a fixed threshold τ follows an inverse Gaussian distribution:

$$T_\tau \sim \mathrm{IG}(\tau/\mu,\ \tau^2/\sigma^2). \tag{45}$$

Inverse Gaussian process. Ye and Chen [42] proposed the IG process as a degradation model. It has non-negative increments

$$X(t) - X(s) \sim \mathrm{IG}(\mu(t-s),\ \lambda(t-s)^2), \tag{46}$$

with mean μt and variance $\mu^3 t/\lambda$. Its increments have heavier tails than those of the Gamma process, so it suits data with occasional large damage jumps. The IG distribution is closed under convolution when the summands share the dispersion parameter μ^2/λ. However, the sum of two independent IG *processes* with different parameters is not an IG process [42].

Compound Poisson process. The compound Poisson process

$$X(t) = \sum_{n=1}^{N(t)} Y_n \tag{47}$$

models degradation by shocks [13]. The shocks arrive according to a Poisson process $N(t)$ with rate ν. Each shock causes a random damage Y_n, and the Y_n are independent draws from one distribution F_Y.

When $Y_n \geq 0$, the process is monotone. The sum of independent compound Poisson processes is always a compound Poisson process. This model suits discrete damage events, for example impacts and overloads, rather than continuous wear. Wang [36] gives a recent application to structural reliability.

2.1.2 Discrete-time formulation

The fleet management setting of Section 1 samples the continuous-time processes of Table 4 once a day. On each day k, the damage of component (i, ℓ) grows by a random increment $\Delta D_{i\ell k}^{(j)}$, whose distribution depends on the mission j served that day. The cumulative damage after k days is therefore a random walk

$$D_{i\ell k} \;=\; D_{i\ell,0} \;+\; \sum_{t=1}^{k} \Delta D_{i\ell t}, \tag{48}$$

where the increments $\Delta D_{i\ell t}$ are independent but *not* identically distributed (different missions may be served on different days). Table 5 restates the four models in this discrete-time setting.

The closure conditions decide whether the marginal distribution of $D_{i\ell k}$ stays in the same parametric family when increments of different missions are summed. When the condition holds, the distribution of $D_{i\ell k}$ has a closed form in the cumulative parameters. This closed form is essential for an efficient evaluation of the reliability constraint $\Pr(D_{i\ell k} > \tau) \leq \varepsilon$. When the condition does not hold, for example for a sum of Gamma increments with different rates, the marginal must be computed numerically or approximated.

In the fleet management problem of Section 1, the increment of each day has parameters that depend on the mission. Closure therefore requires one value of the shared parameter for all missions of a component: the rate β for Gamma, or the dispersion μ^2/λ for inverse Gaussian.

- **Gamma.** The shape parameters may vary freely across missions, and they add up.

Table 5: Discrete-time specialisation of the damage-accumulation models. Each row gives the distribution of a one-day increment ΔD and the resulting marginal of $D_k = \sum_{t=1}^{k} \Delta D_t$ when all increments share the parameters required for closure.

Model	**Increment** ΔD_t	**Marginal** D_k (homogeneous)	**Closure condition**
Gamma	$\mathrm{Ga}(\alpha_t,\, \beta)$	$\mathrm{Ga}\big(\sum_t \alpha_t,\, \beta\big)$	Same rate β
Gaussian	$\mathcal{N}(\mu_t,\, \sigma_t^2)$	$\mathcal{N}\big(\sum_t \mu_t,\, \sum_t \sigma_t^2\big)$	Always
Inv. Gaussian	$\mathrm{IG}(\mu_t,\, \lambda_t)$	$\mathrm{IG}\big(\sum_t \mu_t,\, \frac{(\sum_t \mu_t)^3}{\sum_t (\mu_t^3/\lambda_t)}\big)$	Same μ^2/λ
Compound Poisson	$\sum_{n=1}^{N_t} Y_n$	$\mathrm{CP}\big(\sum_t \nu_t,\, F_Y\big)$	Always; same F_Y keeps the jump law

- **Gaussian.** The mean and variance parameters may vary freely across missions, and they add up. No restriction is needed: normal random variables are closed under addition without conditions.
- **Inverse Gaussian.** The means add up, and the shared dispersion d determines the cumulative shape $\lambda_{1:k}$,

$$\lambda_{1:k} = \Big(\sum_t \mu_t\Big)^2 \Big/ d, \qquad d = \mu_t^2/\lambda_t \text{ for every } t. \tag{49}$$

2.1.3 Remaining-life models: the S–N curve and Palmgren–Miner rule

The stress–life (S–N) approach relates the stress amplitude σ_a to the number of cycles to failure N_f. It started with Wöhler's experiments in the 1850s. The most common analytical form is Basquin's law [3]:

$$\sigma_a^{\,m}\, N_f \;=\; C, \tag{50}$$

where $m > 0$ is the fatigue exponent and $C > 0$ is a material constant, both typically fitted from experimental data. Equivalently, $N_f = C\,\sigma_a^{-m}$, which appears as a straight line on a log–log plot.

Under variable-amplitude loading, the Palmgren–Miner linear damage rule [23, 27] defines the cumulative damage fraction as

$$D_{\mathrm{PM}} \;=\; \sum_i \frac{n_i}{N_{f,i}}, \tag{51}$$

where n_i is the number of applied cycles at stress level $\sigma_{a,i}$ and $N_{f,i}$ is the corresponding life from the S–N curve. Failure is predicted when $D_{\mathrm{PM}} \geq 1$. The rule assumes that damage accumulates linearly and independently of the order of the loads. Experiments have questioned this approximation: the critical damage sum D^*_{PM} at failure is scattered around 1 rather than exactly 1. Stochastic extensions treat D_{PM}, D^*_{PM} or both as random variables [6, 9, 16, 25, 33].

Two sources of randomness enter the Palmgren–Miner accumulation:

(i) **S–N scatter.** At a given stress level σ_a, the fatigue life N_f is a random variable, not a deterministic constant. Its distribution reflects the variability of the material, the surface finish and the test conditions. The increment $\Delta D_{\mathrm{PM}} = n/N_f$ therefore inherits this randomness.

(ii) **Random critical damage sum.** Experimental evidence shows that the critical value D^*_{PM} at failure is itself scattered (typically between 0.5 and 2.0) rather than exactly 1 [33]. A model can replace the deterministic threshold with a random one or, equivalently, divide the damage by a random capacity.

In the fleet management framework of Section 1, source (i) is the relevant one: the threshold is fixed at $\tau = 1$ and the randomness enters through the increments. We therefore focus on the distributional assumptions for N_f and the resulting distribution of ΔD_{PM}. Table 6 summarises the three most common choices.

Table 6: Distributional assumptions for S–N scatter and their effect on the Palmgren–Miner increment $\Delta D_{\mathrm{PM}} = n/N_f$. In all cases, the location $\mu(\sigma_a)$ or the Weibull scale $\lambda(\sigma_a)$ depends on the stress level through Basquin's law (50); the scatter parameters σ, k and β do not.

Model	**Life N_f**	**Increment ΔD_{PM}**	**Mon.**	**Closure under $+$**
Log-normal	$\ln N_f \sim \mathcal{N}(\mu,\, \sigma^2)$	$\mathrm{LogN}(\ln n - \mu,\, \sigma^2)$	Yes	No
Weibull	$N_f \sim \mathrm{Wei}(k,\, \lambda)$	Inv. Weibull (Fréchet)	Yes	No
Gumbel	$\ln N_f \sim \mathrm{Gumbel}(\mu,\, \beta)$	Log-Gumbel	Yes	No

Log-normal. The log-normal law is the most widely used assumption [25]. At a fixed stress level σ_a, the fatigue life satisfies

$$\ln N_f \sim \mathcal{N}(\mu(\sigma_a),\, \sigma^2), \tag{52}$$

where $\mu(\sigma_a) = \ln C - m \ln \sigma_a$ comes from Basquin's law and σ measures the scatter. Since $\ln(n/N_f) = \ln n - \ln N_f$, each increment is log-normally distributed:

$$\Delta D_{\mathrm{PM}} \sim \mathrm{LogN}(\ln n - \mu,\, \sigma^2), \tag{53}$$

with mean and variance

$$n\, e^{-\mu+\sigma^2/2} \quad \text{and} \quad n^2 e^{-2\mu+\sigma^2}(e^{\sigma^2} - 1). \tag{54}$$

The log-normal family is *not* closed under summation: the sum of independent log-normals has no standard parametric form. In practice, the accumulated damage

$$D_{\mathrm{PM},k} = \sum_{t=1}^{k} \Delta D_{\mathrm{PM},t} \tag{55}$$

is often approximated by the log-normal law with the same first two moments (the Fenton–Wilkinson method). The quality of this approximation falls as the number of summands or the coefficient of variation grows.

Weibull. The Weibull law is widely adopted in fatigue standards, for example Eurocode and ASTM. It sets

$$N_f \sim \mathrm{Wei}(k,\, \lambda(\sigma_a)), \tag{56}$$

with shape k and a scale λ that depends on the stress level. The reciprocal $1/N_f$ follows an inverse Weibull (Fréchet) distribution, so $\Delta D_{\mathrm{PM}} = n/N_f$ is a scaled Fréchet variable. The sum of independent Fréchet variables has no closed form, so the marginal of $D_{\mathrm{PM},k}$ cannot be computed analytically.

Gumbel (extreme-value type I). Castillo and Fernández-Canteli [9] proposed a compatible statistical model for the whole S–N field. It uses a Gumbel (minimum) distribution for $\ln N_f$:

$$\ln N_f \sim \mathrm{Gumbel}(\mu(\sigma_a),\, \beta). \tag{57}$$

The increment $\Delta D_{\mathrm{PM}} = n/N_f$ then follows a log-Gumbel distribution. As with the other two families, the log-Gumbel is not closed under summation.

Contrast with damage-accumulation models. Unlike the processes of Table 4, *none* of the standard distributional assumptions for S–N scatter gives increments that are closed under summation. The marginal distribution of $D_{\mathrm{PM},k}$ after k loading blocks must therefore always be computed by convolution (in general not analytically), by Monte Carlo simulation or by approximations based on moments.

In the fleet management formulation, the reliability constraint $\Pr(D_{\mathrm{PM},k} \geq 1) \leq \varepsilon$ therefore has no closed form in the cumulative parameters alone. In the Gamma and Gaussian models, by contrast, the cumulative distribution function (CDF) of the marginal is directly available.

The upper bounds of Section 2.2 (Markov, Cantelli, Chernoff) still apply. The mean and the variance of $D_{\mathrm{PM},k}$ add up over the increments, whatever the distributional family. So the Cantelli bound gives a conservative and tractable constraint.

If the scatter is small, a deterministic S–N curve, for example the mean curve or a design percentile, may be an acceptable approximation. The problem then reduces to the deterministic Miner's rule, and all constraints become linear.

2.1.4 Conservative surrogates for the empirical damage-increment distribution

In the remaining-life setting, the increment of mission j is the Palmgren–Miner contribution

$$\Delta D^{(j)} = \sum_i n_i / N_{f,i} \tag{58}$$

of Equation (51). In our intended use, its distribution is *measured*, not postulated. Rainflow counting is the standard method that extracts the load cycles from a load history. A rainflow count [25] of the load histories recorded for mission j gives an empirical probability density of the non-negative increment $\Delta D^{(j)}$.

We choose *not* to fit a parametric law to this density. A fit ties the model to one data set and one functional form, and so it weakens the generality that the abstract formulation seeks.

Instead, we summarise the density by a few *functionals*, that is, numbers computed from the distribution (moments, support, selected quantiles), and we replace it by a *conservative* surrogate. Here, conservative means that the surrogate never *under*states the damage. Then the reliability constraint $\Pr(D_{i\ell k} > \tau) \leq \varepsilon$ and the loop constraint (19) stay valid by construction.

Two features of the problem limit the admissible options.

- **Accumulated damage.** Both constraints act on the *accumulated* damage

$$D = \sum_t \Delta D_t \tag{59}$$

  of Equation (48), not on a single increment. So the surrogate must be evaluated after summation.
- **No closure under summation.** None of the standard S–N scatter families is closed under summation (see the contrast above). So the law of D in general has no closed form, and it must be bounded or replaced by a surrogate from a family closed under convolution.

Route I, defined below, never needs the moments or the support of the accumulated damage D in advance. It uses only information *per mission*, and the schedule combines this information through the assignment counts.

Fix a component (i, ℓ) and an accumulation window, and write $D = D_{i\ell}$ for its damage over that window. Let

$$n_{ij} = \sum_t x_{ijt} \tag{60}$$

be the number of days on which member i serves mission j. Model the repeated executions of mission j on component ℓ as independent draws from an increment law of the mission. This law has mean $\mu_{\ell j}$, variance $v_{\ell j}$, support $[0, b_{\ell j}]$ and cumulant generating function

$$\psi_{\ell j}(s) = \ln \mathbb{E}\big[e^{s\Delta D^{(j)}}\big], \tag{61}$$

where $\mathbb{E}[e^{sX}]$ is the moment generating function (MGF) of X. Independence across days then gives

$$\mathbb{E}[D] = \sum_j n_{ij}\,\mu_{\ell j}, \qquad \mathrm{Var}[D] = \sum_j n_{ij}\,v_{\ell j}, \qquad \ln M_D(s) = \sum_j n_{ij}\,\psi_{\ell j}(s). \tag{62}$$

- **Linearity.** Each functional of D is therefore a sum of quantities of the missions, weighted by the counts. It is linear in the schedule.
- **Support.** The support enters through the support $b_{\ell j}$ of each mission; only the closed form of Bernstein's inequality merges the supports into one shared value (see below).
- **Repetitions.** Repetitions enter the description of the tail additively. Serving mission j a total of n_{ij} times multiplies its cumulant generating function by n_{ij}, and different missions add up. So a mission that causes little damage keeps its small $\psi_{\ell j}$ beside a mission that causes much damage; the two are not averaged.

These per-mission functionals support two routes.

- **Route I** bounds the upper tail of D directly from the functionals in (62).
- **Route II** replaces each increment by a stochastically larger surrogate from a family closed under convolution. It then carries the order to the sum by the following standard closure property of the usual stochastic order $\leq_{\mathrm{st}}$ (Definition 2.7).

Proposition 2.1 (Closure of $\leq_{\mathrm{st}}$ under independent convolution)**.** *Let $\Delta D_t \leq_{\mathrm{st}} \widetilde{\Delta D}_t$ for $t = 1, \ldots, k$, with $\{\Delta D_t\}_t$ mutually independent and $\{\widetilde{\Delta D}_t\}_t$ mutually independent. Then*

$$\sum_t \Delta D_t \leq_{\mathrm{st}} \sum_t \widetilde{\Delta D}_t. \tag{63}$$

More generally, the conclusion holds with the sum replaced by any coordinatewise non-decreasing map g (see Shaked and Shanthikumar [32, §1.A]).

The proof is in Appendix A.2, page 77.

Proposition 2.1 makes Route II usable beyond a single sum. The dynamics (12) is a coordinatewise non-decreasing map of the increments and of the independent reset variables. So a stochastic upper bound on each increment gives a bound on the whole state $D_{i\ell k}$, repairs and replacements included, and hence on $\Pr(D_{i\ell k} > \tau)$ on every day. Table 7 collects the resulting options together with their guarantees and trade-offs.

The Route I bounds are elementary, and this is why they need no assumption on the distribution. Let $D \geq 0$ and $\tau > 0$.

- **Markov.** Markov's inequality reads

$$\Pr(D > \tau) \leq \mathbb{E}[D]/\tau. \tag{64}$$

- **Cantelli.** The one-sided Chebyshev (Cantelli) bound reads, for $\tau > \mathbb{E}[D]$,

$$\Pr(D > \tau) \;\leq\; \frac{\mathrm{Var}[D]}{\mathrm{Var}[D] + (\tau - \mathbb{E}[D])^2}. \tag{65}$$

Both are the *tight* bounds that the first moment and the first two moments, respectively, can give [5]. Two-point distributions attain both bounds (for the closed event $D \geq \tau$). This is exactly why the two bounds ignore the structure of a sum of independent terms, and why they are loose for such a sum.

- **Chernoff.** The exponential (Cramér–Chernoff) bound reads

$$\Pr(D > \tau) \;\leq\; \inf_{s>0} e^{-s\tau} M_D(s) \;=\; \inf_{s>0} \exp\Big(-s\tau + \textstyle\sum_t \ln M_{\Delta D_t}(s) \Big), \tag{66}$$

and it is valid for every fixed $s > 0$ [10].

Table 7: Ways to summarise the empirical per-mission damage-increment distribution and propagate it to the accumulated damage $D = \sum_t \Delta D_t$ for the constraint $\Pr(D > \tau) \le \varepsilon$. "Conservative" means a guaranteed upper bound on the failure probability (Route I) or $\le_{\mathrm{st}}$ dominance (Route II); "$\widetilde{D}$" means the method yields an explicit surrogate *law* for the sum, as opposed to a one-sided bound. All inputs are *per mission*; the schedule combines them through the assignment counts n_{ij} by Equation (62). Only the closed form of Bernstein's inequality merges the supports into one constant b_ℓ, the largest of them. CGF: cumulant generating function; var: variance.

Method	**Inputs**	**Cons.?**	**$\widetilde{D}$?**	**Pros**	**Cons**	**Ref.**
Route I — bound the tail of D directly (no surrogate law)						
Markov	mean	Yes	No	needs only the mean; always valid	extremely loose	[5]
Cantelli	mean, var	Yes	No	only two moments; tight *as* a two-moment bound	ignores independence; loose in the tail	[5]
Hoeffding	per-mission mean, support b_j	Yes	No	exploits independence; each mission keeps its own support; budget linear in the counts	ignores the variance	[17, Thm. 2]
Bernstein	per-mission mean, var, support b_j	Yes	No	exploits independence; few parameters; moment inputs linear in the counts	one shared support b_ℓ, the largest; one-sided only	[7]
Chernoff	per-mission CGF $\psi_j(s)$ (or mean + support)	Yes	No	tightest in the tail; count-weighted; valid for any fixed $s > 0$	needs the CGF; empirical CGF is an estimate, not a bound	[10]
Route II — dominate each increment, then convolve (explicit $\widetilde{D}$)						
Moment-matched Gamma / IG	mean, var	No	Yes	explicit law; closed under +; ordering tables apply	matches moments but does not dominate (survival curves cross)	[35, 42]
Dominating shared-rate Gamma	moments + empirical survival	Yes	Yes	conservative *and* explicit; serves reliability and loop; shape linear in the schedule	shared rate + dominance margin add slack; commits to Gamma	[32]

- **Bernstein.** Bernstein's inequality is the special case of (66) for independent summands bounded in $[0, b]$:

$$\Pr(D - \mathbb{E}[D] > a) \le \exp\Big(- \frac{a^2/2}{\mathrm{Var}[D] + b\,a/3}\Big) \tag{67}$$

  with $a = \tau - \mathbb{E}[D]$ [7].

Unlike Markov and Cantelli, these bounds use the logarithm of the MGF, which is additive by (62). So they exploit independence. The derivation of the Markov, Cantelli and Chernoff bounds is in Appendix A.2, page 77.

Per-mission inputs keep the differences between missions. Suppose that all summands share one worst-case support

$$b = \max_j b_{\ell j}. \tag{68}$$

Then Bernstein's inequality treats every mission as the most damaging one, and it loses the differences between missions that the schedule exploits. This loss comes from the closed form of Bernstein's inequality, not from Route I. With inputs kept per term, each bound keeps the support and the tail of every mission.

- **Hoeffding.** Hoeffding's inequality reads

$$\Pr(D - \mathbb{E}[D] > a) \le \exp\big(- 2a^2 / \sum_j n_{ij} b_{\ell j}^2\big) \tag{69}$$

  with $a = \tau - \mathbb{E}[D]$, where mission j enters only through its own $b_{\ell j}^2$.
- **Chernoff.** The Chernoff bound (66) becomes the count-weighted reliability constraint

$$\Pr(D > \tau) \;\le\; \inf_{s>0} \exp\Big(- s\tau + \sum_j n_{ij}\, \psi_{\ell j}(s)\Big), \tag{70}$$

  in which each mission contributes its own cumulant generating function $\psi_{\ell j}$.

For a fixed s the exponent is linear in the assignment counts, so (70) is a linear reliability constraint. The infimum couples s to the counts. For a fixed schedule it is a one-dimensional convex problem. Its minimiser $s^\star$ solves

$$\sum_j n_{ij}\, \psi'_{\ell j}(s) = \tau : \tag{71}$$

the exponentially tilted mean of D (its mean under the law reweighted by e^{sD}) equals τ. But $s^\star$ depends on the counts. So one either fixes a conservative s (every $s > 0$ gives a valid bound) or sweeps a small grid of values.

Bernstein's inequality is the only Route I bound that does not allow this per-mission treatment. The reason is the same coupling, seen from the other side: its closed form needs a *single* support constant b_ℓ in the correction term. Only the variance stays per mission.

Between replacements the damage is a weighted sum, with weights in $[0, 1]$, of the terms of the initial or replacement damage and of the increments. Bernstein's inequality needs a bound on every centred term. So the constant is

$$b_\ell = \max\{b_\ell^{\mathrm{start}},\, b_\ell^{\mathrm{new}},\, \max_j b_{\ell j}\}, \tag{72}$$

where b_ℓ^{start} and b_ℓ^{new} bound the terms of the initial and of the replacement damage, and are 0 for a known damage.

Explicitly, with $\kappa = \ln(1/\varepsilon)$, the reliability constraint $\Pr(D > \tau) \le \varepsilon$ becomes

$$\sum_j n_{ij}\, \mu_{\ell j} \;+\; \frac{\kappa\, b_\ell}{3} \;+\; \sqrt{\Big(\tfrac{\kappa\, b_\ell}{3}\Big)^2 + 2\kappa \sum_j n_{ij}\, v_{\ell j}} \;\le\; \tau, \tag{73}$$

a constraint whose left-hand side is *concave* in the assignment counts. An equivalent form is

$$(\kappa b_\ell/3)^2 + 2\kappa \sum_j n_{ij} v_{\ell j} \le \Big(\tau - \kappa b_\ell/3 - \sum_j n_{ij}\mu_{\ell j}\Big)^2, \tag{74}$$

together with the side row

$$\sum_j n_{ij}\mu_{\ell j} \le \tau - \kappa b_\ell/3, \tag{75}$$

which keeps the bracket non-negative.

The feasible set of (73) is reverse-convex (its complement is convex); it cannot be written as a second-order cone. In the classical chance constraint of a second-order cone program (SOCP), the variance is a convex quadratic function of the decisions. Here, independence across days makes the variance affine, and this reverses the curvature.

For a mixed-integer linear program (MILP) this curvature helps. One can replace the square root in (73) by a fixed tangent, or by the pointwise minimum of several tangents. This minimum is piecewise-linear (PWL), and an equality with a special ordered set of type 2 (SOS2: a set of variables of which at most two consecutive ones are non-zero) encodes it exactly. Both replacements imply (73), so they stay conservative: a fixed tangent gives one linear constraint in the counts, and the minimum of tangents gives linear constraints together with the SOS2 set.

The closed form (73), which has no s, uses one support constant; with a support per mission, the bound keeps the minimisation over s. The derivation of these properties of (73) is in Appendix A.2, page 77.

Guaranteed bound versus estimate. For (70) to be a true bound, the per-mission ingredient must be an *upper* bound on $\psi_{\ell j}$. Two per-mission choices are enough, and both keep the support local.

- **Mean-and-support bound.** For an increment whose range is $[0, b_{\ell j}]$, so that it starts at 0, the first is
$$\psi_{\ell j}(s) \le \ln\big(1 + (\mu_{\ell j}/b_{\ell j})(e^{s b_{\ell j}} - 1)\big), \tag{76}$$
which the two-point law on $\{0, b_{\ell j}\}$ attains.
- **Empirical cumulant generating function.** The second is
$$\widehat{\psi}_{\ell j}(s) = \ln\Big(\frac{1}{m}\sum_{r=1}^{m} e^{s\Delta D_r^{(j)}}\Big), \tag{77}$$
built from the m recorded executions of mission j. It is an *estimate*, not a bound, so it must carry a one-sided confidence margin.

Repeating a mission tightens both ingredients: more executions sharpen $\widehat{\psi}_{\ell j}$, $\mu_{\ell j}$ and $b_{\ell j}$, and they shrink the margin. In this precise sense, a mission served many times has a sharper description of its tail.

The choice among the Route I bounds depends on how far into the tail the constraint lies. A reliability constraint with small ε lies several standard deviations above $\mathbb{E}[D]$, and there the Chernoff bound is much tighter than the Cantelli bound. A numerical experiment compared the bounds on sums of $k \in \{30, 100\}$ independent increments with one law on $[0, 1]$. It used five laws: the uniform law, Beta$(2, 2)$, Beta$(0.5, 5)$, and the two-point laws on $\{0, 1\}$ with $\Pr(1) = 0.5$ and $\Pr(1) = 0.05$. For each case the threshold τ gives an exact tail $\Pr(D > \tau)$ close to 10^{-4}, and each bound was divided by this exact tail.

- **Chernoff.** With the exact cumulant generating function and the best s, the Chernoff bound exceeded the tail by a factor between 4 and 12, so it stayed within about one order of magnitude. With only the mean and the width of the range (the mean-and-support bound), it exceeded the tail of the three continuous laws by a factor between 1000 and 3200.

- **Cantelli.** The Cantelli bound exceeded the tail by a factor between 330 and 710, more than two orders of magnitude, for both values of k.
- **Bernstein.** With one support constant $b = 1$, Bernstein's bound exceeded the tail by a factor between 15 and 560. It was tighter than Cantelli in nine of the ten cases; the exception was Beta$(0.5, 5)$ with $k = 30$, whose increments are small compared with the width 1. It became tighter as k grew: for the uniform law its factor fell from 232 to 83.
- **Hoeffding and Markov.** With width 1, Hoeffding's bound exceeded the tail by a factor between 8 and 6500. It was close to Chernoff only for the two-point law with $\Pr(1) = 0.5$, whose variance is the largest that the width allows; in the other eight cases it was looser than Cantelli. Markov's bound exceeded the tail by a factor between 2200 and 8600.[1]

So Cantelli, the simplest bound, can compete with Bernstein only for short accumulation windows and shallow tails; otherwise it is a coarse fallback that needs only the mean and the variance.

Route II gives an explicit surrogate law, not a bound.

- **Moment-matched Gamma or inverse Gaussian.** Matching only the first two moments of a family closed under convolution is convenient: Gamma, with shape and rate

$$\alpha = \mathbb{E}[\Delta D]^2 / \operatorname{Var}[\Delta D] \quad \text{and} \quad \beta = \mathbb{E}[\Delta D] / \operatorname{Var}[\Delta D], \tag{78}$$

  or inverse Gaussian. The sum then stays in the family under the closure conditions of Table 5. But this choice is *not* conservative: a law with matched moments in general crosses the empirical survival function and does not dominate it.
- **Dominating shared-rate Gamma.** Using more of the empirical information than the moments restores conservatism. Fix a common rate $\beta^\star$, and, for each mission, increase the shape α_j until the survival function of the surrogate dominates the empirical one:

$$\overline{\mathrm{Ga}}(x; \alpha_j, \beta^\star) \geq \widehat{\overline{F}}_j(x) \quad \text{for all } x. \tag{79}$$

  This is possible exactly when some recorded increment is 0. If every record is positive, $\widehat{\overline{F}}_j(x) = 1$ below the smallest record, while $\overline{\mathrm{Ga}}(x; \alpha, \beta^\star) < 1$ for every $x > 0$, so no Gamma law dominates. If a share $p_0 > 0$ of the records is 0, then $\widehat{\overline{F}}_j(x) \leq 1 - p_0$ for $x > 0$, and a large α_j dominates, because $\overline{\mathrm{Ga}}(x_{\max}; \alpha, \beta^\star) \to 1$ as $\alpha \to \infty$, with $x_{\max}$ the largest record. The accumulated surrogate is then $\widetilde{D} = \mathrm{Ga}(A, \beta^\star)$, by the closure under a shared rate, and $D \leq_{\mathrm{st}} \widetilde{D}$ by Proposition 2.1. Its law has a cumulative shape

$$A = \sum_{t,j} x_{ijt}\, \alpha_j \tag{80}$$

  that is linear in the schedule. The reliability constraint reduces to the scalar condition $A \leq A^\star(\varepsilon)$. In the absence of repairs, the loop constraint reduces to the rows $A_{H_1} \leq y^A$ and $A_T \leq y^A$ on the cumulative shape, with a start y^A that is free above A_{H_1}. The cost is the slack that the shared rate and the dominance margin add.

In summary, the formulation certifies the reliability constraint of a component with measured increments by one of five Route-I bounds, chosen per component: Markov, Cantelli, Hoeffding, Bernstein or Chernoff. The loop constraint acts on the statistics that the chosen bound tracks (Section 2.1.5), so it needs no surrogate law. Route II is not part of the formulation.

One trade-off cannot be removed.

[1] This is a numerical illustration, not a proof. The script `tail_experiment.py` and its output are in the folder `archive/merge-2026/analysis/scripts/scoping_checks` of the repository. The exact tail comes from the Irwin–Hall formula (the exact law of a sum of uniform variables), from the binomial law, or from a lattice convolution with a relative error below 0.5%; no sampling is used. For the two-point laws the sum takes integer values, and the bounds are compared with $\Pr(D \geq \tau)$. The output also covers tails close to 10^{-2}, 10^{-6} and 10^{-8}: near 10^{-2} Bernstein was looser than Cantelli in four of the ten cases, and near 10^{-6} and 10^{-8} it was tighter in all cases. Asymptotically the gap is structural: Cantelli decays only polynomially, as $\operatorname{Var}[D]/(\tau - \mathbb{E}[D])^2$, whereas (66) decays exponentially in the deviation.

- **Route-I bounds.** The bounds assume nothing about the shape of the law, but they give only a tail bound.
- **Dominating Gamma.** The dominating Gamma gives a law that propagates through the dynamics, but it commits to one family and adds some conservatism.

The five bounds of the formulation. The rainflow model of the formulation is *distribution-free*. The law of a mission increment is measured from recorded load histories by rainflow counting (Palmgren–Miner damage). It enters the model only through summary statistics: its mean and, as the bound requires, its variance, the width of its range or its cumulant generating function at a fixed point s. No parametric family is assumed.

Each bound tracks the mean μ_k of the damage on day k and at most one more statistic. The second statistic is the variance v_k, the *Hoeffding budget* R_k (the sum of the squared range widths of the terms of the damage), or the cumulant generating function $K_k = \ln \mathbb{E}\left[e^{sD_k}\right]$ at a fixed point $s > 0$.

Table 8 lists the rows, with $\kappa = \ln(1/\varepsilon)$ and $c = \kappa\, b_\ell/3$. Each row is the corresponding tail bound of this section, solved for the target ε and written in the tracked statistics. If the tracked values satisfy a row and are at least the true statistics of the damage, the bound gives $\Pr(D_k > \tau) \le \varepsilon$. The rows rest on four modelling assumptions.

- **A0 (non-negative damage).** Every increment, the initial damage and every replacement draw are non-negative.
- **A1 (independence).** The mission increments are mutually independent, and independent of the initial damage and of every replacement draw. Each replacement installs a new draw, independent of everything else. Every bound except Markov needs A1.
- **A2 (bounded ranges; Hoeffding and Bernstein).** Each mission increment lies in an interval of known width. The initial damage is a known part plus independent terms, each in an interval; its Hoeffding budget is the sum of their squared widths, and its support is the width of the range of their sum. The replacement draw is described in the same way. A known damage has budget and support 0.
- **A3 (repetition).** The row of the bound holds on every day $k = 1, \dots, T$, and after day T the operating phase repeats with period H_2, schedule and increment laws included. This is the setting of Theorem 1.4.

Table 8: Reliability rows of the five Route-I bounds, on every day k of the horizon. μ_k: tracked mean; v_k: tracked variance; R_k: tracked Hoeffding budget; K_k: tracked cumulant generating function at the fixed point s.

Bound	**Tracked statistics**	**Reliability row**
Markov	μ	$\mu_k \le \varepsilon\, \tau$
Cantelli	μ, v	$(1-\varepsilon)\, v_k \le \varepsilon\, (\tau - \mu_k)^2, \quad \mu_k \le \tau$
Hoeffding	μ, R	$(\tau - \mu_k)^2 \ge (\kappa/2)\, R_k, \quad \mu_k \le \tau$
Bernstein	μ, v	$(\tau - c - \mu_k)^2 \ge c^2 + 2\kappa\, v_k, \quad \mu_k \le \tau - c$
Chernoff	μ, K	$K_k - s\,\tau \le \ln \varepsilon$

Width of a range. The support of an increment X is the width b of an interval $[l, l+b]$ that contains X almost surely. The width has this one meaning in every bound, and a width of 0 means that the increment is known. Hoeffding's inequality uses the widths. Bernstein's inequality needs an upper bound of the centred term, and the width is one.

A width w is consistent with a variance v only if $v \le w^2/4$ (Popoviciu's inequality). The width is the only information on the range, and the range may start at any point $l \ge 0$. The inequality $\operatorname{Var} X \le w^2/4$ holds for every such position, so it is a necessary condition for the

pair (variance, width). The derivations of the bound on the centred term and of Popoviciu's inequality are in Appendix A.2, page 78.

For example, increments in an interval of width 0.2 have variance at most 0.01; a two-point law on $\{0, 0.2\}$ with probability $1/2$ each has mean 0.1 and reaches 0.01.

A condition of the form $v \le \mu\,(b-\mu)$ assumes that the range starts at 0, so it is not a necessary condition when only the width is known.

Day maps of the statistics. Between replacements the damage is

$$D_k = \sum_t w_t X_t : \tag{81}$$

the X_t are the known part, the independent terms of the initial damage or of the replacement draw, and the mission increments. Each weight $w_t \in [0,1]$ is a product of factors $1-\rho$, and a term of weight 0, possible when $\rho = 1$, is dropped.

Under ARD_1, write $D^- = G + \Delta$, with G the damage just after the last intervention and Δ the weighted increments added since. G and Δ are independent by A1, and a repair gives $D^+ = G + (1-\rho)\,\Delta$. Before the first intervention, G is the part of the initial damage that the starting anchors describe (Section 2.3.4).

For a damage

$$D = \sum_t w_t X_t, \tag{82}$$

write

$$m(D) = \mathbb{E}D, \qquad V(D) = \operatorname{Var} D, \qquad R(D) = \sum_t w_t^2 r_t, \qquad K(D) = \ln \mathbb{E}\, e^{sD}, \tag{83}$$

with r_t the squared width of X_t.

Proposition 2.2 (Day maps of the statistics). *Let $\theta \in \{m, V, R\}$, with factor $a_\theta = 1-\rho$ for m and $a_\theta = (1-\rho)^2$ for V and R. Under A1:*

1. *a mission day adds the statistic of the increment:*

$$\theta(D+X) = \theta(D) + \theta(X) \quad \textit{and} \quad K(D+X) = K(D) + K(X); \tag{84}$$

2. *under ARD_∞:*

$$\theta\big((1-\rho)D\big) = a_\theta\,\theta(D) \quad \textit{and} \quad K\big((1-\rho)D\big) \le (1-\rho)\,K(D); \tag{85}$$

3. *under ARD_1:*

$$\theta(D^+) = a_\theta\,\theta(D^-) + (1-a_\theta)\,\theta(G) \quad \textit{and} \quad K(D^+) \le (1-\rho)\,K(D^-) + \rho\,K(G). \tag{86}$$

The proof is in Appendix A.2, page 78.

So each tracked statistic follows the repair map $a\,\theta + (1-a)\,g$ of Section 2.3.4, with $a = 1-\rho$ for μ and K, and $a = (1-\rho)^2$ for v and R. Its anchor is its value just after the last intervention.

The maps of μ, v and R are exact; the map of K is an upper bound, which is what the domination argument of Section 1.3.1 needs. Hence each true statistic of D_k, written m, V, R^{t} and K^{t}, is at most the tracked statistic of the same name (Proposition 1.3).

Derivation and validity of the rows. Each row of Table 8 gives $\Pr(D_k > \tau) \le \varepsilon$ when the tracked statistics satisfy it and are at least the true statistics of D_k. The derivation also gives the following properties of the rows.

- **Markov.** The row uses the mean only and does not need A1 or A2.

- **Cantelli.** The row is equivalent to

$$\mu_k + \kappa_C \sqrt{v_k} \le \tau \quad \text{with} \quad \kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}. \tag{87}$$

- **Bernstein.** The support constant b_ℓ of (73) is a constant, not a decision variable; the replacement support enters it only when replacements are allowed.
- **Chernoff.** The row bounds K by $s\tau - \kappa$, its natural range, and $K \ge 0$. So a known zero damage satisfies the row only if $s\tau \ge \kappa$, and a replacement day satisfies it only if the replacement value of K is at most $s\tau - \kappa$.

The derivation is in Appendix A.2, page 78.

The Bernstein side row. The quadratic Bernstein row alone also admits the point $\mu_k = \tau$, $v_k = 0$: a known damage equal to τ. That point is safe; for $c > 0$ it is separated from the rest of the feasible set. The side row $\mu_k \le \tau - c$ removes it and keeps the branch

$$\tau - \mu_k \ge c + \sqrt{c^2 + 2\kappa v_k} \tag{88}$$

of (73). At $\mu_k = v_k = 0$ the row needs $\tau \ge 2c$, and a positive variance is feasible only if

$$\tau > 2c = 2\kappa b_\ell / 3. \tag{89}$$

Initial and replacement values of the Chernoff statistic. The statistic K_k is the cumulant generating function of the whole damage, initial damage included.

- **Known damage.** A known initial damage μ_0 gives the initial value $K_0 = s\,\mu_0$, and a known replacement damage μ^{new} gives the reset value $s\,\mu^{\text{new}}$.
- **Random damage.** A random initial or replacement damage needs its own cumulant generating function at s, which is larger.
- **Lower bounds.** The cumulant generating function of every increment at s is at least s times its mean. Also the initial value and the replacement value of K are at least s times the corresponding mean. An input check enforces these inequalities (Section 5.2).

The derivation, from Jensen's inequality, is in Appendix A.2, page 80.

A start at $K_0 = 0$ ignores the initial damage and can accept an unsafe schedule. Example: $\tau = 1$, $\varepsilon = 0.1$, $s = 5$, a known initial damage $\mu_0 = 0.6$, and one mission that adds 0 or 0.5 with probability 1/2 each. The cumulant generating function of the mission at s is $\ln(0.5 + 0.5\,e^{2.5}) = 1.886$. The true failure probability is 0.5.

- With $K_0 = 0$ the row reads

$$1.886 - 5 = -3.11 \le \ln 0.1 = -2.30 \tag{90}$$

  and accepts the schedule.
- With $K_0 = s\,\mu_0 = 3$ the row reads

$$4.886 - 5 = -0.11 > -2.30 \tag{91}$$

  and rejects it.

In the same way, the Hoeffding budget of a known damage is 0, because its range has width 0, and a random initial or replacement damage needs its own budget. So the defaults of the formulation describe a known initial damage and a known replacement damage: zero variance, zero support, zero budget, and $K = s$ times the mean.

2.1.5 The loop constraint under the Route-I bounds

The probability comparison (15) asks for $\Pr(D_T > \tau) \leq \Pr(D_{H_1} > \tau)$, so that repeating the schedule keeps the failure probability below ε without end. Under Route I we never observe these probabilities. We have only an upper bound $U_k \geq \Pr(D_k > \tau)$ from one of the bounds of Table 8, for example the count-weighted Chernoff bound (70) or the Bernstein constraint (73).

We write $U^{(n)}$ for the bound on day $H_1 + nH_2$, the end of the n-th repetition of the operating phase, so $U^{(0)} = U_{H_1}$ and $U^{(1)} = U_T$. A natural translation replaces both sides by their bounds and imposes $U^{(1)} \leq U^{(0)}$. This condition is *not* sufficient for repeatability, and the reason needs a precise look.

Why $U^{(1)} \leq U^{(0)}$ does not chain. The bound U is a many-to-one function of the parameters of the state, and its level sets are not invariant under the dynamics. So the relation

$$U^{(2)} \;>\; U^{(1)} \;=\; U^{(0)} \tag{92}$$

is possible: the certified bound can rise slowly over the repetitions, and the inequality over one repetition does not carry over by induction. The derivation is in Appendix A.2, page 80.

A concrete instance[2] has $\tau = 10$, $b_\ell = 3$, $\rho = 0.5$, and an operating phase of one ARD_∞ repair day followed by mission days with

$$\sum_j n_{ij}\mu_{\ell j} = 3, \qquad \sum_j n_{ij}v_{\ell j} = 3.99, \qquad \big(\mathbb{E}[D_{H_1}], \mathrm{Var}[D_{H_1}]\big) = (4, 9). \tag{93}$$

- **Cantelli.** The bound gives $U^{(0)} = 0.2000$ and $U^{(1)} = 0.1997 \leq U^{(0)}$, but $U^{(2)} = 0.2151 > U^{(0)}$, and $U^{(n)} \to 0.2495$.
- **Bernstein.** A similar instance (variance 3.15 added over the operating phase) also fails at the second repetition: $U^{(0)} = 0.3012$, $U^{(1)} = 0.3006$ and $U^{(2)} = 0.3247$.
- **Loop constraint.** The loop constraint (19) rejects both, since $\mathbb{E}[D_T] = 5 > 4$.

A fixed margin $\delta > 0$, $U^{(1)} \leq U^{(0)} - \delta$, does not fix this. Every such margin rejects a schedule whose bound repeats exactly, $U^{(1)} = U^{(0)}$. A small margin still admits the rise: for Cantelli with $U^{(0)} = 0.2$, instances that satisfy it and still fail exist for every $\delta < 0.088$. The defect is that the level sets are not invariant, not a constant gap.

The certificate stays valid in every repetition. This worsening is not hidden: it can be detected. Each statistic that Route I tracks follows a monotone recursion under the dynamics (12): the mean, the variance, the Hoeffding budget and the cumulant generating function at a fixed point s.

- **Increment.** Increments add to the statistic.
- **Imperfect repair.** An imperfect repair maps it from θ to $a\,\theta + (1-a)\,g$, with its anchor g and a factor $a \in [0, 1]$ (Section 2.3.4).
- **Replacement.** A replacement resets it.

The recursions are exact for the mean, the variance and the Hoeffding budget, and an upper bound for the cumulant generating function (see below).

Consequently U_k is a valid bound on *every* day of the endless schedule, $\Pr(D_k > \tau) \leq U_k$. The true probability can exceed ε only if U_k itself does. Repeatability therefore needs a loop condition that forces $U_k \leq \varepsilon$ on every such day. So the condition must act on a statistic that both determines U and propagates monotonically, not on the scalar U, which loses information.

[2] The script `loop_routeI_instance.py` in the folder `archive/merge-2026/analysis/scripts/scoping_checks` of the repository reproduces these numbers.

The loop on the moments (the natural condition). For the Cantelli and Bernstein bounds, the loop constraint (19) acts on the tracked moments,

$$\mathbb{E}[D_T] \;\le\; \mathbb{E}[D_{H_1}] \qquad \text{and} \qquad \mathrm{Var}[D_T] \;\le\; \mathrm{Var}[D_{H_1}], \tag{94}$$

and, under ARD_1, on their anchors g^μ and g^v in the same way. The moments evolve as monotone recursions, and componentwise domination chains. So the certified risk does not increase across repetitions,

$$U^{(n)} \;\le\; U^{(0)} \;\le\; \varepsilon, \tag{95}$$

and the schedule is repeatable. The derivation is in Appendix A.2, page 80.

Both inequalities in (94) are linear in the tracked moments $(\mathbb{E}[D_k], \mathrm{Var}[D_k])$. The model carries these moments as auxiliary variables, and big-M rows or indicator constraints encode their recursion linearly. So the loop adds one linear constraint per tracked moment and per anchor. These constraints are strictly stronger than the single relation $U^{(1)} \le U^{(0)}$, and this stronger form is exactly what makes the induction valid.

The loop on the cumulant generating function under imperfect repair. If the Chernoff bound with a fixed s certifies the reliability instead, the loop acts on the tracked cumulant generating function,

$$K_T \;\le\; K_{H_1}. \tag{96}$$

Without imperfect repair, (96) carries over by induction. An imperfect (ARD) repair does not keep this scalar exact. The tracked statistic then follows

$$K \mapsto (1-\rho)\,K \quad \text{under } \mathrm{ARD}_\infty, \qquad K \mapsto (1-\rho)\,K + \rho\, g^K \quad \text{under } \mathrm{ARD}_1, \tag{97}$$

with the anchor g^K. These maps are non-decreasing in (K, g^K), and the tracked K stays an upper bound on the true cumulant generating function, which is all that the Chernoff certificate needs. The derivation is in Appendix A.2, page 81.

Theorem 1.4 then applies: the loop row on K, and on its anchor under ARD_1, chains under both repair laws. The cost is slack: after a repair, the tracked K exceeds the true value. The moments need no such bound, because a repair acts on each of them through a fixed power of $1-\rho$, which keeps their recursion exact.

Summary. Under Route I the loop constraint must act on statistics that determine the bound and that propagate by themselves through the dynamics, never on the scalar bound U. The formulation imposes the loop constraint (19) on the statistics of the chosen bound; Table 9 lists them for each bound. Under ARD_1 the anchors of these statistics enter the loop too.

The loop is valid for every bound and for both repair laws. In every case the guarantee is that the *certified* risk does not worsen. Hence $\Pr(D_k > \tau) \le \varepsilon$ on every day of the endless schedule. This is the practical meaning of repeatability for a model based on bounds.

Table 9: Statistics on which the Route-I loop constraint acts, for each bound, and the recursion that tracks them under the dynamics.

Bound	**Statistics in the loop constraint**	**Recursion of the statistics**
Markov	the mean	exact
Cantelli, Bernstein	the mean and the variance (94)	exact
Hoeffding	the mean and the Hoeffding budget	exact
Chernoff	the mean and the cumulant generating function (96)	exact for the mean; an upper bound for the cumulant generating function

2.1.6 Two refinements of the Route-I constraints: schedule-dependent support and piecewise-linear tightening

Two identifiable sources of slack remain in the Route-I constraints. Standard mixed-integer constructions can remove both, at a cost in binaries and rows that can be quantified.

- **Schedule-dependent support.** The first one, a schedule-dependent support, is not part of the formulation, which uses the constant support of Section 2.1.4.
- **Piecewise-linear tightening.** The formulation includes the second one, the piecewise-linear tightening, as the form `pwl` of its reliability rows (Section 2.1.7).

Neither refinement changes the loop argument of Section 2.1.5. The moment condition (94) acts on the exactly tracked moments, and it does not depend on how tightly the certificate is enforced on them.[3]

Schedule-dependent Bernstein support.

- **Slack.** The closed form (73) carries the single constant

$$b_\ell \;=\; \max\{b_\ell^{\text{start}},\, b_\ell^{\text{new}},\, \max_j b_{\ell j}\}. \tag{98}$$

  This constant includes the supports b_ℓ^{start}, b_ℓ^{new} of the initial and replacement draws. So a component pays the support of the most damaging mission even if the schedule never assigns that mission to it. The effect is not small, because the correction $\kappa b_\ell/3$ enters twice at zero variance: even the empty schedule satisfies (73) only if $2\kappa b_\ell/3 \le \tau$.
- **Example.** Take $\varepsilon = 0.01$ ($\kappa = \ln 100 \approx 4.605$), $\tau = 1$, two missions with supports $b_{\ell 1} = 0.02$ and $b_{\ell 2} = 0.30$, initial and replacement supports 0.02, and moments per day of mission 1 $\mu_{\ell 1} = 0.01$, $v_{\ell 1} = 4 \cdot 10^{-6}$. Consider a component that never serves mission 2.
  - The constraint with the constant b_ℓ ($\kappa b_\ell/3 = 0.4605$) leaves a mean budget of 0.079 and admits at most $n_{i1} = 7$ days of mission 1.
  - The same constraint evaluated with the supports actually present ($\kappa b_\ell/3 = 0.0307$) leaves 0.939 and admits $n_{i1} = 90$.

  The gap is a factor of 12.9, and it comes entirely from the unused $b_{\ell 2}$. If $b_{\ell 2}$ rises to 0.35, then

$$2\kappa b_\ell/3 \;=\; 1.075 \;>\; \tau, \tag{99}$$

  and the model with the constant b_ℓ becomes infeasible for a component that never serves mission 2.
- **Tightened constant.** The tightened constant is

$$b_\ell(\zeta) \;=\; \max\Big\{b_\ell^{\text{start}},\, b_\ell^{\text{new}},\, \max_{j:\,\zeta_{ij}=1} b_{\ell j}\Big\}. \tag{100}$$

  Here $\zeta_{ij} \in \{0,1\}$ marks the missions that member i actually serves ($\zeta_{ij} \ge x_{ijt}$ for all t; only this direction of the forcing is needed). The constant $b_\ell(\zeta)$ bounds every summand of the state, and Bernstein's inequality applies without change, on every day and in every repetition of the schedule.
- **Rows.** A monotonicity observation makes $b_\ell(\zeta)$ implementable without a nonlinear tracking of the maximum: the Bernstein row written for a larger constant implies every row written for a smaller one. Consequently, impose the row with the per-mission constant c_j *conditionally on* $\zeta_{ij} = 1$, together with one unconditional row at c_0, where

$$c_j \;=\; \kappa \max\{b_{\ell j}, b_\ell^{\text{start}}, b_\ell^{\text{new}}\}/3, \qquad c_0 \;=\; \kappa \max\{b_\ell^{\text{start}}, b_\ell^{\text{new}}\}/3. \tag{101}$$

[3] Every monotonicity and equivalence claim in this subsection, both worked examples and the comparison between the union and the intersection of the tangent regions are checked, symbolically or numerically, by the scripts `refine_schedule_support.py` and `refine_pwl_tightening.py`. The scripts and their outputs are in the folder `archive/merge-2026/analysis/scripts/scoping_checks` of the repository.

These rows are exactly equivalent to the single row with the schedule-dependent maximum. The derivation of this equivalence and of the validity of $b_\ell(\zeta)$ is in Appendix A.2, page 81.

- **Cost.** The cost is one binary per pair of member and mission ($F \cdot M$ in total), their linking rows and the conditional certificate rows of each mission. All of them are linear in the new variables. The rows of the tangent surrogate attach to ζ_{ij} by indicator constraints or by big-M rows. Written as $2\kappa v \le \dots$, these rows have the uniform valid constant $2\tau(\tau - c_0)$ for every mission with $2c_j < \tau$; a mission with $2c_j \ge \tau$ gets $\zeta_{ij} = 0$ and no row. The exact quadratic rows must use big-M, because Gurobi's indicator constraints enforce a linear constraint [15].
- **Side effects.** Two side effects are worth recording.
  1. Big-M constants derived from the largest admissible variance must be recomputed from c_0. The feasible region has grown, and a bound taken from the old constant b_ℓ would cut it.
  2. The feasibility precondition $2\kappa b_\ell/3 < \tau$ becomes local: only c_0 must satisfy it. A mission with $2c_j > \tau$ simply forces $\zeta_{ij} = 0$; at $2c_j = \tau$ its row admits only $\mu = v = 0$. So that mission is never assigned to that component, and the whole model does not become inconsistent.
- **Effect left out.** By choice, the refinement does *not* capture one further effect. The repair weights shrink the support of each summand further, to $w_t\, b_{\ell j}$. Exploiting this needs a running maximum over products of repair factors and mission indicators that depend on the schedule, a truly nonlinear refinement. The variable ζ, defined over the whole window, captures the first-order gap: missions that are never served.

Piecewise-linear tightening of the tangent surrogates.

- **Slack of one tangent.** The linear surrogate with one fixed tangent (the device described after (73)) has a conservatism that grows quadratically with the mean. Cantelli's bound (65) is at most ε if and only if

$$\mathbb{E}[D] + \kappa_C \sqrt{\mathrm{Var}[D]} \;\le\; \tau, \qquad \text{with} \qquad \kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}. \tag{102}$$

  Write (μ, v) for the tracked moments. The exact feasible set is

$$\kappa_C^2\, v \;\le\; f(\mu) := (\tau - \mu)^2, \tag{103}$$

  and the tangent at $\mu_r = 0$ gives the linear row $\kappa_C^2\, v \le \tau^2 - 2\tau\mu$. Since $f(\mu) - (\tau^2 - 2\tau\mu) = \mu^2$, the surrogate loses exactly μ^2/κ_C^2 of admissible variance. It reaches zero at $\mu = \tau/2$, and it excludes *every* point with $\mu > \tau/2$, a deterministic component included. The exact set still admits

$$v \;=\; (\tau - \mu)^2/\kappa_C^2 \;>\; 0 \tag{104}$$

  up to $\mu = \tau$.
- **Several tangents: union, not intersection.** The refinement takes tangents at several points μ_r. Each row

$$\kappa_C^2\, v \;\le\; T_r(\mu) := (\tau - \mu_r)(\tau + \mu_r - 2\mu) \tag{105}$$

  is conservative by itself, because tangents lie below the convex parabola. The way the rows are combined is the critical point. Imposing all rows *together* gives their intersection

$$\kappa_C^2 v \;\le\; \min_r T_r(\mu), \tag{106}$$

  which is *more* conservative than any single tangent. The tightening is the *union* of the single-tangent regions, a disjunction:

$$\kappa_C^2 v \;\le\; \max_r T_r(\mu). \tag{107}$$

- **Encodings.** Two encodings give this union.
  (i) One selector binary per tangent, with indicator (or big-M) rows:

$$\sum_r \lambda_r = 1, \qquad \lambda_r = 1 \;\Rightarrow\; \kappa_C^2 v \le T_r(\mu). \tag{108}$$

  (ii) An auxiliary variable y, tied by a piecewise-linear *equality* (the SOS2 λ-formulation) to the pointwise minimum of the tangents to $\sqrt{v}$ at

$$v_r \;=\; (\tau - \mu_r)^2/\kappa_C^2, \tag{109}$$

  followed by the single linear row $\mu + \kappa_C\, y \le \tau$.
  The two encodings describe the same set: the tangent at μ_r on the side of μ and the tangent at v_r on the side of v bound the same half-plane

$$\kappa_C^2 v \;\le\; (\tau - \mu_r)(\tau + \mu_r - 2\mu). \tag{110}$$

- **Shortcuts that fail.** Two natural shortcuts both fail. Requiring $y \ge$ every tangent gives their pointwise *maximum*: a region strictly inside the single-tangent region, worse than no refinement. Requiring $y \le$ every tangent lets y fall below $\sqrt{v}$, and this destroys the certificate. Likewise, the equality must interpolate the vertices of the tangent envelope, not points on the curve. Consecutive tangents meet at

$$v \;=\; \sqrt{v_r v_{r+1}} \qquad \text{with value} \qquad \big(\sqrt{v_r} + \sqrt{v_{r+1}}\big)/2, \tag{111}$$

  above the curve. The chords between points on the curve underestimate the concave square root.
- **Recovery.** The recovery is fast, and it can be quantified exactly because f is quadratic. Where the union admits $v \ge 0$, the remaining gap of the union is

$$\min_r\, (\mu - \mu_r)^2. \tag{112}$$

  This gap is quadratic in the distance to the nearest tangent point, with worst case $(\Delta/2)^2$ for a spacing Δ. So doubling the number of tangents divides it by four. The derivation is in Appendix A.2, page 82. With $\tau = 1$, $\varepsilon = 0.01$ ($\kappa_C^2 = 99$) and tangents at $\mu_r \in \{0, 0.25, 0.5, 0.75\}$, the admissible variance recovers as follows ($v \times 10^3$):

μ	0	0.2	0.4	0.5	0.6
exact $(\tau-\mu)^2/\kappa_C^2$	10.10	6.46	3.64	2.53	1.62
single tangent ($\mu_r = 0$)	10.10	6.06	2.02	0	infeasible
four tangents (union)	10.10	6.44	3.54	2.53	1.52

  The union recovers 94% of the area of the feasible region that the single tangent loses.
- **Bernstein and the schedule-dependent support.** The construction applies without change to Bernstein. The parabola $(\tau - \kappa b_\ell/3 - \mu)^2$ follows the same law for the gap, and the tangents to the concave $\sqrt{(\kappa b_\ell/3)^2 + 2\kappa v}$ lie above it everywhere. The construction also combines soundly with the schedule-dependent support above: each conditional row carries its own mission constant and stays conservative for it.

2.1.7 Linear forms of the reliability rows

The formulation offers three forms for each quadratic reliability row. These rows are the Gaussian and Wiener row $\kappa_G^2\, v \le (\tau-\mu)^2$ with $\kappa_G = \Phi^{-1}(1-\varepsilon)$ (Section 2.2.2), and the Cantelli, Hoeffding and Bernstein rows of Table 8. Each has the form

$$\chi \;\le\; f(\mu), \qquad f(\mu) = c_2\,(\tau - \mu)^2 + c_1\,(\tau - \mu), \qquad \mu \le \bar{\mu}_{\text{side}}, \tag{113}$$

Table 10: Constants of the quadratic reliability rows (113).

Row	χ	c_2	c_1	$\bar{\mu}_{\text{side}}$
Gaussian, Wiener	v	$1/\kappa_G^2$	0	τ
Rainflow, Cantelli	v	$1/\kappa_C^2 = \varepsilon/(1-\varepsilon)$	0	τ
Rainflow, Hoeffding	R	$2/\kappa$	0	τ
Rainflow, Bernstein	v	$1/(2\kappa)$	$-b_\ell/3$	$\tau - c$

where χ is the variance or the Hoeffding budget, with the constants of Table 10. The third relation is the side row; it selects the branch of the parabola on which the row is a valid certificate. Since $c_2 > 0$, f is convex in μ, with $f'(\mu) = -2c_2(\tau - \mu) - c_1$.

Every form keeps the side row.

- `exact`: the row (113) itself. The feasible set $\{(\mu, \chi) : \chi \le f(\mu)\}$ lies below a convex function, so it is not convex: the row is a nonconvex quadratic constraint. Example (Gaussian row): $(\mu, v) = (\tau, 0)$ and $(0, \tau^2/\kappa_G^2)$ satisfy it, and their midpoint $(\tau/2, \tau^2/(2\kappa_G^2))$ does not, since $\tau^2/2 > \tau^2/4$. The solver Gurobi handles such rows through its nonconvex quadratic support.

- `tangent`: one linear row per day, the tangent of f at a point $\mu_r \in [0, \tau)$ chosen per component; the default is $\mu_r = \tau/2$. At $\mu_r = 0$ the Gaussian row becomes $\kappa_G^2\, v + 2\tau\mu \le \tau^2$. The point $\mu_r = \tau$ is excluded: for $c_1 = 0$ the tangent at τ is $\chi \le 0$.

- `pwl`: the interval $[0, \tau]$ is split into P equal segments, by default $P = 8$. One binary per segment and day selects the segment that contains the mean, and the tangent at the midpoint of that segment caps χ. This is encoding (i) of the piecewise-linear tightening (Section 2.1.6), with each selector tied to its segment. With $P = 1$ the form is the single tangent at $\tau/2$, with no binaries.

Proposition 2.3 (Safety of the linear forms)**.** *For every μ_r and every μ,*

$$f(\mu_r) + f'(\mu_r)\,(\mu - \mu_r) \;=\; f(\mu) - c_2\,(\mu - \mu_r)^2 \;\le\; f(\mu). \tag{114}$$

Therefore every point that satisfies the `tangent` or the `pwl` form satisfies the `exact` form, and both linear forms lie inside the certificate set $\mathcal{C}_{i\ell}$ of Section 1.3.1.

The proof is in Appendix A.2, page 82.

Accuracy.

- **The `tangent` form.** The `tangent` form loses $c_2(\mu - \mu_r)^2$ in χ: it is exact at $\mu = \mu_r$ and loosest far from it. A tangent point near the operating mean of the component is the best choice.
- **The `pwl` form.** In the `pwl` form the loss is

$$c_2(\mu - \hat{\mu}_p)^2 \;\le\; c_2\,\tau^2/(4P^2), \tag{115}$$

with $\hat{\mu}_p$ the midpoint of the segment that contains μ. That midpoint is the nearest tangent point to μ. So the `pwl` form accepts exactly the points with

$$\chi \;\le\; \max_p\,[f(\hat{\mu}_p) + f'(\hat{\mu}_p)(\mu - \hat{\mu}_p)], \tag{116}$$

the union of the P single-tangent regions and the largest set that the P tangents allow. The cost is P binaries per component and day.

Example 2.4 (Cantelli, $\varepsilon = 0.1$, $\kappa_C^2 = 9$, $\tau = 1$, $v = 0.004$)**.** The largest admissible mean is $1 - \sqrt{9 \cdot 0.004} \approx 0.810$ with `exact`; 0.482 with `tangent` at 0 ($9v + 2\mu \le 1$); 0.714 with `tangent` at $0.5\,\tau$ ($9v \le 0.75 - \mu$); and 0.810 with `pwl` and $P = 8$. The tangent at 0 never admits a mean above $\tau/2$, even with zero variance.

The tangent point of a Bernstein row. For Bernstein with $b_\ell > 0$, the tangent point must satisfy $\mu_r < \sqrt{\tau(\tau - 2c)}$, with $c = \kappa b_\ell/3$, and the `pwl` form needs $\tau/(2P) < \sqrt{\tau(\tau - 2c)}$ for its first segment. The derivation is in Appendix A.2, page 82.

2.1.8 Random fatigue life in the Palmgren–Miner rule

The rainflow pipeline of Section 2.1.4 measures the law of the per-mission increment $\Delta D^{(j)}$ directly and nonparametrically. A complementary pipeline keeps the rainflow *counts* deterministic and makes the fatigue life N_f in Miner's rule (51) the random object. It has two steps.

1. Fit a scatter family of Table 6 to constant-amplitude test data or to records of failures in service. Basquin's law (50) ties the location to the stress amplitude. For the log-normal,
$$\ln N_f \;\sim\; \mathcal{N}(\mu(\sigma_a), \sigma^2), \qquad \text{with} \qquad \mu(\sigma_a) \;=\; \ln C - m \ln \sigma_a \tag{117}$$
and a σ that does not depend on the stress. For the Weibull, the scale is $\lambda(\sigma_a) = C\sigma_a^{-m}$, with a common shape k. Run-outs (tests stopped before failure) enter the likelihood as censored observations.
2. *Derive* the increment from the rainflow histogram $\{n_c^{(j)}\}$ of the mission as
$$\Delta D^{(j)} \;=\; \sum_c n_c^{(j)}/N_f(\sigma_c). \tag{118}$$

Both pipelines follow the same idea: collect data, then approximate a distribution. They differ in the random object and in the assumptions they commit to.

- **Empirical route.** It is distribution-free, and it captures the variability of the load and of the material together.
- **Random-N_f route.** It is parametric. It pools information across stress levels, extrapolates to unobserved amplitudes through (50), and separates the load spectrum (the counts) from the material scatter (the law of N_f).

Inverse moments of the scatter families. Every quantity that the framework uses later is a moment of $1/N_f$. For the three families of Table 6 and $r > 0$,
$$\mathbb{E}\big[N_f^{-r}\big] \;=\; \begin{cases} \exp\big(-r\mu + r^2\sigma^2/2\big), & \ln N_f \sim \mathcal{N}(\mu, \sigma^2) \quad \text{(finite for all } r\text{)}, \\ \lambda^{-r}\,\Gamma(1 - r/k), & N_f \sim \mathrm{Wei}(k, \lambda) \quad \text{(finite if and only if } r < k\text{)}, \\ e^{-r\mu}\,\Gamma(1 - r\beta), & \ln N_f \sim \mathrm{Gumbel}_{\min}(\mu, \beta) \quad \text{(finite if and only if } r\beta < 1\text{)}. \end{cases} \tag{119}$$
The derivation is in Appendix A.2, page 82. In particular, for the three families,
$$\mathrm{Var}[1/N_f] \;=\; \begin{cases} e^{-2\mu+\sigma^2}(e^{\sigma^2} - 1), & \text{log-normal}, \\ \lambda^{-2}\big[\Gamma(1 - 2/k) - \Gamma(1 - 1/k)^2\big], & \text{Weibull}, \\ e^{-2\mu}\big[\Gamma(1 - 2\beta) - \Gamma(1 - \beta)^2\big], & \text{Gumbel-minimum}. \end{cases} \tag{120}$$
These values match the increment moments already given for the log-normal below Table 6.

Remark 2.5 (Two tail behaviours, one real restriction). The Gumbel-minimum log-life is *exactly* a Weibull life:
$$\Pr(N_f > x) \;=\; \exp\big(-e^{(\ln x - \mu)/\beta}\big) \;=\; \exp\big(-(x/e^\mu)^{1/\beta}\big), \tag{121}$$
that is, $\mathrm{Wei}(k = 1/\beta,\ \lambda = e^\mu)$. The existence conditions in (119) correspond to each other. So the three families of Table 6 give two tail behaviours for $1/N_f$: log-normal (all polynomial moments finite) and Fréchet tails with index k (moments finite only below order k).

The second behaviour is a real restriction, not a formality. The mean of the increment requires $k > 1$, and the variance, the input of the Cantelli certificate (65), requires $k > 2$. So the fitted shape must be checked before the method is used. For $k \in (1, 2]$ only the Markov bound remains, and for $k \leq 1$ the untruncated model has an infinite mean increment and cannot enter the framework at all.

Moreover, for *all three* untruncated families, $1/N_f$ has unbounded support and an infinite moment generating function. So the exponential certificates (Chernoff (70), Bernstein (73), Hoeffding) are not admissible. Only a lower bound $N_{\min} > 0$ on the life, defended on physical grounds, restores them (a truncated family). This bound gives the per-mission support $b_{\ell j} = n_j/N_{\min}$, and the moments must then be recomputed under the truncated law (closed forms with Gaussian tails and incomplete gamma functions).

What "random N_f" means: two interpretations. The marginal law of N_f does not determine the model. The joint law of the scatter *across days* must also be chosen, and the two extreme choices lead to different mathematics. Table 11 compares the two interpretations.

Table 11: The two interpretations of a random fatigue life N_f in the Palmgren–Miner rule.

	(I) Block-independent scatter	**(II) Unit-specific scatter**
Scatter across days	each execution of mission j draws a new, independent life	the component carries *one* realisation of the scatter for its entire life
Stochastic state	the moments of the Route-I pipeline, with the per-mission inputs (122)	a deterministic scalar $M_{i\ell k}$ with linear dynamics
Chance constraint	Cantelli certificate (65)	the exact linear inequality (125)
Loop constraint	moment loop (94)	the linear condition (129)

(I) Block-independent scatter. Interpretation (I) suits the case in which the variability from cycle to cycle and from day to day dominates, and the component does not carry one fixed N_f. Assume a single dominant stress level per mission. (For a spectrum in bins, the means add over the bins; the variance depends on the assumed dependence within the mission, and it is computed from (119) in the same way.) Then the per-mission inputs are

$$\mu_{\ell j} \;=\; n_j\,\mathbb{E}\big[N_f^{-1}\big], \qquad v_{\ell j} \;=\; n_j^2\,\mathrm{Var}\,\big[N_f^{-1}\big]. \tag{122}$$

Independence across days is exactly the hypothesis of Equation (62). So the Cantelli certificate (65) and the moment loop (94) apply *without change*, with parametric instead of empirical moments per mission. The rainflow route of Section 2.1.4 already uses this interpretation. The random-N_f model is then only another input pipeline for the same formulation, subject to the admissibility conditions of Remark 2.5.

(II) Unit-specific scatter. The fitted families have a scale or a shape that does not depend on the stress, so the realisation of the scatter is a single multiplicative factor,

$$N_{f,j} \;=\; A\,\bar{N}_j, \qquad A > 0 \text{ drawn once at installation,} \tag{123}$$

with $\bar{N}_j$ the median life (log-normal: e^{μ_j}) or the characteristic life (Weibull: λ_j; Gumbel: e^{μ_j}), and $A \sim F_A$ the unit factor ($\ln A \sim \mathcal{N}(0, \sigma^2)$, $A \sim \mathrm{Wei}(k, 1)$ or $A = e^{\beta G}$, respectively). This is exactly the "random critical damage sum" of source (ii) above. Dividing the Miner sum by A is the same as comparing a deterministic sum with a random capacity.

- **Damage.** The damage becomes $D_k = M_k/A$, where M_k is the *deterministic* Miner state. Its dynamics mirror (12): a mission day adds

$$\sum_j x_{ijk}\, n_j/\bar{N}_j, \tag{124}$$

an ARD repair maps $M \mapsto (1-\rho)M$, and a replacement resets M to zero *and draws a new* A (a new physical component carries a new factor with the same law).

- **Reliability constraint.** For a continuous F_A the reliability constraint is then *exact*, deterministic and linear in the assignment counts:

$$\Pr(D_k \geq 1) \;=\; \Pr(A \leq M_k) \;=\; F_A(M_k) \;\leq\; \varepsilon \quad\Longleftrightarrow\quad M_k \;\leq\; q_\varepsilon \;:=\; F_A^{-1}(\varepsilon), \tag{125}$$

with the closed-form quantiles

$$q_\varepsilon \;=\; \begin{cases} e^{\sigma\Phi^{-1}(\varepsilon)} & \text{(log-normal)}, \\ (-\ln(1-\varepsilon))^{1/k} & \text{(Weibull)}, \\ (-\ln(1-\varepsilon))^{\beta} & \text{(Gumbel-minimum)}. \end{cases} \tag{126}$$

In fact no family is needed beyond one scalar: an empirical ε-quantile of A with a confidence margin serves equally well. Without repair since the last replacement, and with n_{ij} counted from it, rewriting $M_k \leq q_\varepsilon$ as

$$\sum_j n_{ij}\, n_j/(q_\varepsilon \bar{N}_j) \;\leq\; 1 \tag{127}$$

gives back a classical design practice: deterministic Miner on the ε-percentile lives

$$N_j^{(\varepsilon)} \;=\; q_\varepsilon \bar{N}_j, \tag{128}$$

which are exactly the ε-quantiles of $N_{f,j}$. Here this practice is an *exact* chance constraint rather than a heuristic, in the per-day sense of (40g).

- **Loop constraint.** The loop constraint simplifies in the same way. F_A is strictly increasing on $(0,\infty)$, so the probability comparison (15) is equivalent to the single linear condition

$$M_T \;\leq\; M_{H_1}. \tag{129}$$

Unlike the scalar bound of Section 2.1.5, which loses information, this condition *chains*. So imposing (125) on all $k = 1,\ldots,T$ extends the per-day reliability to every repetition, replacements (which draw a new A) included. The derivation is in Appendix A.2, page 83.

Remark 2.6 (The two interpretations are not two levels of conservatism). Interpretations (I) and (II) can share the *same* marginal law per day and differ only in the dependence across days: zero or perfect correlation of the scatter. Yet they give different answers. The means coincide, while the variances satisfy

$$\mathrm{Var}_{\mathrm{II}} \,/\, \mathrm{Var}_{\mathrm{I}} \;=\; \Big(\sum_t m_t\Big)^2 \Big/ \sum_t m_t^2. \tag{130}$$

By Cauchy–Schwarz this ratio is at most the number of days, with equality for equal daily increments m_t. So under (II) the dispersion never averages out.

In the small-ε regime, (II) has by far the heavier tail. For example, take a log-normal factor with $\sigma = 0.5$ and fifty days with equal increments, with Miner sum $M = 0.6$, the median of the damage under (II). Under (I), each day draws its own factor with the same law. The failure probability $\Pr(D \geq 1)$ is then 0.153 under (II) and $2.5 \cdot 10^{-7}$ under (I). The order reverses only at high damage. In the same example, the two probabilities are equal (0.39) at $M = 0.87$. For a larger M, (I) gives the larger probability. So neither interpretation is conservative in all cases.[4]

The choice is a question of identification.

[4] The scripts and their outputs are in the folder `archive/merge-2026/analysis/scripts/scoping_checks/miner` of the repository. They check the closed forms (119), the divergent sample variance at Weibull $k \leq 2$, the exactness of (125) along a schedule with repair and replacement, the chaining of (129), the variance ratio and the example. The value under (I) comes from a discretised convolution, which bounds it between $2.45 \cdot 10^{-7}$ and $2.46 \cdot 10^{-7}$.

- **Unit-specific (II).** Scatter that comes from the variability between specimens (material batch, manufacturing) is unit-specific (II).
- **Block-independent (I).** Scatter that comes from the randomness of the load history within a mission is block-independent (I).

Rainflow-counted data recorded in service on a fleet mix both. A decomposition of the variance of the recorded increments into a part between units and a part within units estimates the split. When both sources matter, one writes $N_f = A\,\widetilde{N}_f$, with a unit factor and block-independent residuals.

Conservative combinations exist. One example is ε-splitting:

$$\Pr(\text{fail}) \;\le\; \delta + \Pr(\text{fail} \mid A = a_\delta), \tag{131}$$

with a_δ the δ-quantile of A. After this step the increments are again independent across days, with moments scaled by a_δ^{-1} and a_δ^{-2}, and the Route-I method runs on the remaining budget $\varepsilon - \delta$. The calibration of such combinations is left as future work.

Implications for the abstract formulation.

- **Interpretation (I).** Interpretation (I) requires no change to Problem (40). It supplies the per-mission moments (122) to the same Route-I pipeline that uses the moments measured by rainflow counting, and it keeps the certificates and the loop condition of that pipeline.
- **Interpretation (II).** Interpretation (II) defines a strictly simpler model family (Table 11). There is no moment tracking, no slack from a tail bound and no big-M linearisation of a variance recursion. One quantile q_ε per component holds the entire stochastic description. So interpretation (II) is a natural candidate for a dedicated model type (a deterministic equivalent) beside the models based on processes.

The caveat of Remark 2.6 applies: which interpretation matches a given component is an empirical question, not a preference of the formulation.

2.2 Stochastic orderings

A comparison of the damage of a component at two days of the schedule needs a notion of "one random variable is larger than another". Several stochastic orders formalise this idea, with increasing strength [32, 40]. Let X and Y be two non-negative random variables with CDFs F_X, F_Y, survival functions $\bar{F}_X = 1 - F_X$, $\bar{F}_Y = 1 - F_Y$, and densities f_X, f_Y (when they exist).

Definition 2.7 (Usual stochastic order)**.** $X \le_{\text{st}} Y$ if $F_X(t) \ge F_Y(t)$ for all $t \in \mathbb{R}$, or equivalently $\Pr(X > t) \le \Pr(Y > t)$ for all t.

Definition 2.8 (Hazard rate order)**.** $X \le_{\text{hr}} Y$ if $\bar{F}_Y(t)/\bar{F}_X(t)$ is non-decreasing in t over $\{t : \bar{F}_X(t) > 0\}$. Equivalently, the hazard rates satisfy $r_X(t) \ge r_Y(t)$ for all t, where $r(t) = f(t)/\bar{F}(t)$.

Definition 2.9 (Likelihood ratio order)**.** $X \le_{\text{lr}} Y$ if $f_Y(t)/f_X(t)$ is non-decreasing in t over the union of their supports.

These three orders are progressively stronger [32, Ch. 1]. The implications are strict, and none of the converses holds in general:

$$X \le_{\text{lr}} Y \implies X \le_{\text{hr}} Y \implies X \le_{\text{st}} Y \implies \mathbb{E}[X] \le \mathbb{E}[Y]. \tag{132}$$

Remark 2.10 (Connection to the loop constraint). The probability comparison (15) requires

$$\Pr(D_{i\ell T} > \tau) \le \Pr(D_{i\ell H_1} > \tau), \tag{133}$$

which is a *pointwise* comparison of the survival functions at the single threshold τ. This is strictly weaker than

$$D_{i\ell T} \le_{\text{st}} D_{i\ell H_1}, \tag{134}$$

which demands the inequality at every point. So the usual stochastic order is a convenient sufficient condition for (15), but not a necessary one. The loop constraint (19) of the formulation compares tracked statistics instead (Section 1.3).

2.2.1 Stochastic ordering within each degradation family

For the fleet management problem, the relevant comparison is between two damages of the same component that belong to the same parametric family. An example is the damage at the end of the transitory phase and the damage at the end of the horizon.

Tables 12–14 give, for each family of Table 4, the necessary and sufficient conditions for each of the three orders.

- An entry marked "—" means that the author knows no simple closed-form characterisation.
- For the inverse Gaussian rows, the $\leq_{\text{lr}}$ conditions remain a valid sufficient criterion by (132).
- For the compound Poisson rows, see the discussion below.

Table 12: Conditions for $X \leq_{\text{lr}} Y$ (likelihood ratio order) within each family. Conditions are necessary and sufficient except where noted.

Family	X	Y	**Condition**
Gamma	$\text{Ga}(\alpha_1, \beta_1)$	$\text{Ga}(\alpha_2, \beta_2)$	$\alpha_1 \leq \alpha_2$ and $\beta_1 \geq \beta_2$
Gaussian	$\mathcal{N}(\mu_1, \sigma_1^2)$	$\mathcal{N}(\mu_2, \sigma_2^2)$	$\mu_1 \leq \mu_2$ and $\sigma_1 = \sigma_2$
Inv. Gaussian	$\text{IG}(\mu_1, \lambda_1)$	$\text{IG}(\mu_2, \lambda_2)$	$\lambda_2 \geq \lambda_1$ and $\lambda_1/\mu_1^2 \geq \lambda_2/\mu_2^2$
Comp. Poisson	$\text{CP}(\nu_1, F_{Y_1})$	$\text{CP}(\nu_2, F_{Y_2})$	— (can fail even for $F_{Y_1}{=}F_{Y_2}$, $\nu_1 \leq \nu_2$; holds, for example, for deterministic jumps; see text)

Table 13: Conditions for $X \leq_{\text{hr}} Y$ (hazard rate order) within each family.

Family	X	Y	**Condition**
Gamma	$\text{Ga}(\alpha_1, \beta_1)$	$\text{Ga}(\alpha_2, \beta_2)$	$\alpha_1 \leq \alpha_2$ and $\beta_1 \geq \beta_2$
Gaussian	$\mathcal{N}(\mu_1, \sigma_1^2)$	$\mathcal{N}(\mu_2, \sigma_2^2)$	$\mu_1 \leq \mu_2$ and $\sigma_1 = \sigma_2$
Inv. Gaussian	$\text{IG}(\mu_1, \lambda_1)$	$\text{IG}(\mu_2, \lambda_2)$	— (sufficient: $\leq_{\text{lr}}$ conditions)
Comp. Poisson	$\text{CP}(\nu_1, F_{Y_1})$	$\text{CP}(\nu_2, F_{Y_2})$	— (no general criterion; the $\leq_{\text{lr}}$ route is unavailable, see text)

Table 14: Conditions for $X \leq_{\text{st}} Y$ (usual stochastic order) within each family.

Family	X	Y	**Condition**
Gamma	$\text{Ga}(\alpha_1, \beta_1)$	$\text{Ga}(\alpha_2, \beta_2)$	$\alpha_1 \leq \alpha_2$ and $\beta_1 \geq \beta_2$
Gaussian	$\mathcal{N}(\mu_1, \sigma_1^2)$	$\mathcal{N}(\mu_2, \sigma_2^2)$	$\mu_1 \leq \mu_2$ and $\sigma_1 = \sigma_2$
Inv. Gaussian	$\text{IG}(\mu_1, \lambda_1)$	$\text{IG}(\mu_2, \lambda_2)$	— (sufficient: $\leq_{\text{lr}}$ conditions)
Comp. Poisson	$\text{CP}(\nu_1, F_{Y_1})$	$\text{CP}(\nu_2, F_{Y_2})$	for $F_{Y_1} = F_{Y_2}$, $Y \geq 0$, $\Pr(Y > 0) > 0$: $\nu_1 \leq \nu_2$

Gamma. All three orders coincide for the Gamma family [32, Sec. 1.C]. Under the closure condition (same β), they all reduce to $\alpha_1 \le \alpha_2$. The derivation is in Appendix A.3, page 83.

Gaussian. All three orders coincide: $\sigma_1 = \sigma_2$ and $\mu_1 \le \mu_2$. The derivation is in Appendix A.3, page 83.

Inverse Gaussian. For $X \sim \mathrm{IG}(\mu_1, \lambda_1)$ and $Y \sim \mathrm{IG}(\mu_2, \lambda_2)$, $X \le_{\mathrm{lr}} Y$ holds if and only if

$$\lambda_2 \ge \lambda_1 \quad \text{and} \quad \lambda_1/\mu_1^2 \ge \lambda_2/\mu_2^2. \tag{135}$$

The derivation is in Appendix A.3, page 83. Under the closure condition (same μ^2/λ), these simplify to $\mu_1 \le \mu_2$. For the general two-parameter case, the standard references give no closed-form necessary and sufficient conditions for $\le_{\mathrm{hr}}$ and $\le_{\mathrm{st}}$. The $\le_{\mathrm{lr}}$ conditions serve as a sufficient criterion.

Compound Poisson. Take the same jump-size distribution F_Y for both laws, with $Y \ge 0$ and $\Pr(Y > 0) > 0$. Then

$$\mathrm{CP}(\nu_1, F_Y) \le_{\mathrm{st}} \mathrm{CP}(\nu_2, F_Y) \tag{136}$$

if and only if $\nu_1 \le \nu_2$. The derivation is in Appendix A.3, page 83.

The stronger orders do *not* follow in general. Take Y uniform on $\{1, 10\}$, $\nu_1 = 0.1$ and $\nu_2 = 0.2$. The likelihood ratio drops from $\approx$463 at $x = 9$ (reachable only by nine unit jumps) to $\approx$1.8 at $x = 10$ (reachable by a single large jump), so $\le_{\mathrm{lr}}$ fails. The ratio of the survival functions is not monotone either, so $\le_{\mathrm{hr}}$ fails as well. Both orders hold in special cases, such as deterministic jump sizes, where the process is a scaled Poisson process.

When the jump-size distributions differ, the ordering conditions depend on the specific form of F_Y, and no universal closed-form characterisation is available.

Implications for the fleet problem. Under the closure conditions of Table 5, the cumulative damage $D_{i\ell k}$ depends on the schedule only through cumulative parameters.

- **One-parameter families.** The comparison of two damages is scalar: for Gamma (shared rate β), the total shape $\sum_t \alpha_t$; for IG (shared dispersion μ^2/λ), the total mean $\sum_t \mu_t$.
- **Gaussian family.** The state carries *two* cumulative parameters,

$$\Big(\sum_t \mu_t,\ \sum_t \sigma_t^2\Big). \tag{137}$$

  The variance depends on the schedule: it accumulates with the mission days served, and only an intervention reduces it. The usual stochastic order between Gaussian laws requires equal variances (Table 14). So the Gaussian comparison uses the single-threshold condition (161) or the componentwise domination of the mean and the variance, not the mean alone.

The formulation does not rely on closure for the Gamma and inverse Gaussian models, for two reasons.

- The terms of the damage need not share the closure parameter. The missions of one component may have different Gamma rates, and an inverse Gaussian term of mean a in the formulation has the dispersion $\mu^2/\lambda = \eta\, a$, which grows with a (the model is stated below).
- A repair acts on the damage path, and the repaired damage leaves the family with the shared parameter, in both cases. Indeed,

$$(1-\rho)\,\mathrm{Ga}(A, \beta) = \mathrm{Ga}\big(A,\ \beta/(1-\rho)\big) \tag{138}$$

has another rate, and

$$c\,\mathrm{IG}(\mu,\lambda) = \mathrm{IG}(c\mu,\, c\lambda) \tag{139}$$

multiplies the dispersion μ^2/λ by c.

For these two models the formulation therefore tracks the mean only. It bounds the failure probability over a whole class of laws that contains every possible law of the damage, and it imposes a cap on the tracked mean. The inverse Gaussian cap uses a further order.

Definition 2.11 (Increasing convex order)**.** For integrable non-negative X and Y, $X \le_{\mathrm{icx}} Y$ if $\mathbb{E}(X-d)_+ \le \mathbb{E}(Y-d)_+$ for every $d \ge 0$.

The quantity $\mathbb{E}(X-d)_+$ is the stop-loss transform of X at d. This order does not compare tails directly: $X \le_{\mathrm{icx}} Y$ allows $\Pr(X>\tau) > \Pr(Y>\tau)$. For example, the constant $X=1$ and the variable Y equal to 0 or 2 with probability $1/2$ each satisfy $X \le_{\mathrm{icx}} Y$, while

$$\Pr(X > 0.9) = 1 > \Pr(Y > 0.9) = 1/2. \tag{140}$$

Inverse Gaussian: a cap on the mean. The inverse Gaussian law $\mathrm{IG}(\mu,\lambda)$ with mean $\mu>0$ and shape $\lambda>0$ has the density

$$f(x;\mu,\lambda) = \sqrt{\frac{\lambda}{2\pi x^3}}\,\exp\Big(-\frac{\lambda\,(x-\mu)^2}{2\mu^2 x}\Big), \qquad x>0, \tag{141}$$

and the variance μ^3/λ; for $c>0$,

$$c\,\mathrm{IG}(\mu,\lambda) = \mathrm{IG}(c\mu, c\lambda). \tag{142}$$

The formulation gives each inverse Gaussian component one parameter $\eta>0$. A term of mean a has the law $\mathrm{IG}(a, a/\eta) = a\,Z$, with $Z \sim \mathrm{IG}(1, 1/\eta)$. Its density is

$$\sqrt{a/(2\pi\eta x^3)}\,\exp\big(-(x-a)^2/(2\eta a x)\big), \tag{143}$$

and η is its squared coefficient of variation. Every term has this law with its own mean: each mission increment, the initial damage and each replacement draw. A known damage a is the constant term $a\cdot 1$. The terms may depend on each other in any way.

Between replacements, under either repair law, the damage is

$$D_k = \sum_t w_t\, a_t\, Z_t, \tag{144}$$

where each Z_t has the law of Z or equals 1, and each weight $w_t \in [0,1]$ is a product of factors $1-\rho$.

- **Weight zero.** A term of weight 0, possible when $\rho = 1$, is dropped.
- **Repair laws.** ARD_∞ multiplies every weight by $1-\rho$, and ARD_1 multiplies only the weights of the terms added since the last intervention.
- **Anchor below the initial mean.** When the anchor before the first intervention is below the initial mean μ_0, the initial damage $\mu_0 Z_0$ is read as the two terms $g\,Z_0$ and $(\mu_0 - g)\,Z_0$, with g the anchor.
- **Mean.** The mean $\sum_t w_t a_t$ is at most the tracked mean μ_k (Proposition 1.3).

Proposition 2.12 (Stop-loss order)**.** *Let*

$$D = \sum_t w_t\, a_t\, Z_t \tag{145}$$

with weights $w_t \in [0,1]$, means $a_t \ge 0$ and $\sum_t w_t a_t \le \mu$, where each Z_t has the law of Z or equals 1, under any joint law. Then $D \le_{\mathrm{icx}} \mu Z$.

The proof is in Appendix A.3, page 84.

Let q be the $(1-\varepsilon)$-quantile of Z, so $\Pr(Z > q) = \varepsilon$, and let $\bar{q} = \mathbb{E}[Z \mid Z > q]$, the conditional value at risk (CVaR) of Z at the level $1 - \varepsilon$.

Theorem 2.13 (CVaR cap)**.** *Let* $D \geq 0$ *satisfy* $D \leq_{\text{icx}} \mu Z$ *with* $0 \leq \mu \leq \tau/\bar{q}$*. Then* $\Pr(D > \tau) \leq \varepsilon$*.*

The proof is in Appendix A.3, page 84.

So the formulation tracks the mean of an inverse Gaussian component and imposes the linear row

$$\mu_k \leq \bar{\mu} = \tau/\bar{q} \tag{146}$$

on every day, under both repair laws. By Proposition 2.12 and Theorem 2.13, the row gives $\Pr(D_k > \tau) \leq \varepsilon$, whatever the dependence between the terms.

The quantile q follows from the distribution function (166) of Z, with $\mu = 1$ and $\lambda = 1/\eta$. The tail mean has the closed form $\mathbb{E}[Z;\, Z > q] = F_Z(1/q)$, so

$$\bar{\mu} = \tau\, \varepsilon / F_Z(1/q). \tag{147}$$

The derivation is in Appendix A.3, page 84. Table 15 gives values of the cap.

Table 15: The CVaR cap $\bar{\mu}/\tau$ of the inverse Gaussian model, for three reliability levels and five values of η.

	$\eta = 0.05$	0.25	1	4	10
$\varepsilon = 0.001$	0.491	0.242	0.101	0.0370	0.0190
$\varepsilon = 0.01$	0.570	0.322	0.155	0.0679	0.0401
$\varepsilon = 0.1$	0.693	0.477	0.299	0.183	0.141

No larger cap is safe for every law below μZ in the order of Definition 2.11. The derivation is in Appendix A.3, page 84. The schedules produce only some of the laws below μZ, so the row can reject some safe schedules.

The quantile cap τ/q, at which the single law μZ meets the target exactly, is not safe for sums. Example: $\eta = 10$ and $\varepsilon = 0.1$, so $\tau/q = 0.473\,\tau$. Three independent missions of equal mean, with total mean τ/q, give $\Pr(D > \tau) = 0.118$. The CVaR cap is $0.141\,\tau$ there. At $\varepsilon = 0.01$ and η from 0.05 to 10, the CVaR cap $\bar{\mu}$ is 0.93 to 0.59 times τ/q.

Computation of the inverse Gaussian cap. The cap $\bar{\mu}$ is computed once per component, before the model is built. Its numerical choices have the following reasons.

- **Distribution function.** F_Z is a sum of two positive terms, and $\ln F_Z$ is computed from their logarithms. So the sum loses no digits, and the factor $e^{2/\eta}$ of (166) does not overflow when η is small.
- **Survival function.** $1 - F_Z$ is a difference of two positive terms, and $\ln(1 - F_Z)$ is computed from their logarithms. The subtraction $1 - F_Z$ would lose the relative accuracy when ε is small, because F_Z is then close to 1 at the quantile.
- **Quantile.** The quantile is the root of $\ln \Pr(Z > e^u) = \ln \varepsilon$ in $u = \ln q$. The root search `brentq` of SciPy (a scientific library of the Python language) finds it on an interval $[-50, \ln b]$. By Cantelli's inequality (65) with $\operatorname{Var} Z = \eta$, the right end satisfies $\Pr(Z > b) \leq \varepsilon/2$. So the function changes sign on the interval. The derivation is in Appendix A.3, page 84.
- **Tail mean.** The closed form $\mathbb{E}[Z;\, Z > q] = F_Z(1/q)$ needs no numerical integration, and it is evaluated in logarithms too.

- **Accuracy of the quantile.** The root search is more accurate than the quantile function `invgauss.isf` of SciPy. The table gives the relative error of the survival function at each computed quantile, against a survival function computed with 60 significant digits.

	`invgauss.isf`	root search
$\eta = 10^{-6}$, $\varepsilon = 0.5$	$1.6 \cdot 10^{-6}$	$2 \cdot 10^{-13}$
$\eta = 10^{-3}$, $\varepsilon = 10^{-12}$	$1.4 \cdot 10^{-5}$	$5 \cdot 10^{-14}$

- **Checks.** On a grid of 11 values of η from 10^{-3} to 10^{3} and 9 values of ε from 10^{-6} to 0.9, the quantile agrees with `invgauss.isf` to a relative $2.2 \cdot 10^{-12}$. On the same grid the closed-form tail mean agrees with a numerical integral to $4 \cdot 10^{-14}$. For η from 10^{-6} to 10^{6} and ε from 10^{-12} to 0.999, the computation never fails, and every cap lies in $(0, \tau/q]$. A Monte Carlo sample of $2 \cdot 10^{6}$ draws of Z gives the conditional tail mean 2.61699 against the computed $\bar{q} = 2.61568$ at $\eta = 0.5$ and $\varepsilon = 0.1$. At $\eta = 2$ and $\varepsilon = 0.05$ it gives 5.76704 against 5.76507.

Gamma: a sharp cap on the mean. The formulation lets each mission of a Gamma component have its own rate: an increment of mean $m > 0$ and rate β has the law $\mathrm{Ga}(\beta m, \beta)$. The initial damage and each replacement draw are Gamma terms with their own rates, or known constants. All terms are independent.

A repair acts on the damage path by the law of the component: ARD_∞ gives $D^+ = (1-\rho)\, D^-$, and ARD_1 gives

$$D^+ = D^{\mathrm{last}} + (1 - \rho)\,(D^- - D^{\mathrm{last}}). \tag{148}$$

After a repair the damage is, in general, not a Gamma variable: it is a weighted sum of independent Gamma terms and known constants. The cap below holds for every such sum. The day map of the tracked mean is exact for the true mean $\mathbb{E}[D_k]$, because expectation is linear.

Let β^* be the smallest rate among the terms of the component with a positive mean, over the missions and the days of both phases, and $t = \beta^* \tau$. The formulation computes β^* from the data; no input gives it.

- A mission with mean 0 adds no term, and a known damage is a constant, which does not enter β^*.
- The rate of the replacement draw enters only when replacements are allowed.
- If no term has a rate, the damage is a known constant, and the cap below reads $\mu_k \le \tau$.

Measured in units of $1/\beta^*$, every law below has a rate of at least 1. Write $Q(a, x) = \Pr(\mathrm{Ga}(a, 1) > x)$ for the regularised upper incomplete gamma function.

Definition 2.14 (Class of weighted Gamma sums)**.** For $m \ge 0$, the class $\mathcal{G}(m)$ holds the laws of the random variables

$$S = \delta + \sum_{i=1}^{n} c_i\, Y_i \tag{149}$$

with $Y_i \sim \mathrm{Ga}(a_i, 1)$ independent, $c_i \in (0, 1]$, $a_i > 0$, $\delta \ge 0$, $n \ge 0$, and mean $\delta + \sum_i c_i a_i = m$. The term cY with $Y \sim \mathrm{Ga}(m'/c, 1)$ has mean m'; as $c \to 0$ it tends to the constant m'.

Lemma 2.15 (True states belong to the class)**.** *On every day k of a schedule, under either repair law, the law of $\beta^* D_k$ belongs to $\mathcal{G}(\beta^*\, \mathbb{E}[D_k])$.*

The proof is in Appendix A.3, page 84.

For $0 < m < t$, let $W(m, t)$ be the supremum of $\Pr(S > t)$ over the random variables S whose law is in $\mathcal{G}(m)$. The one-term element $c\,\mathrm{Ga}(m/c, 1)$ has the tail $Q(m/c,\, t/c)$ at t.

An element of $\mathcal{G}(m)$ is described by its *atoms*: the pairs (m_i, c_i) of a term mean $m_i = c_i a_i$ and a scale c_i, and the constant δ, read as an atom at scale $c = 0$. Terms with equal scales merge into one term, because a sum of independent Gamma variables of one rate is a Gamma variable.

Lemma 2.16 (First-order rates). *Let the law of S be an element of $\mathcal{G}(m)$ with at least one term, with density f_S. Since $\delta \le m < t$, f_S is continuous at t. For $c \in (0,1]$ let*

$$\omega_S(c) = \frac{1}{c}\int_0^\infty \Pr(t-y<S\le t)\,\frac{e^{-y/c}}{y}\,\mathrm{d}y, \qquad \omega_S(0) = f_S(t). \tag{150}$$

(a) Adding mean Δm at scale c (a term $c\,\mathrm{Ga}(\Delta m/c,1)$, or the constant Δm when $c=0$) raises $\Pr(S>t)$ at the rate $\omega_S(c)$ as $\Delta m \to 0$. Removing mean from an atom at scale c lowers it at the same rate. (b) Moving the scale of an atom (m_i, c_i), $c_i \in (0,1)$, at fixed mean m_i changes $\Pr(S>t)$ at the rate $m_i\,\partial\omega_S/\partial c\,(c_i)$.

The proof is in Appendix A.3, page 85.

Theorem 2.17 (One term is the worst case). *For $0 < m < t$,*

$$W(m,t) \;=\; \max_{c\in(0,1]} Q\!\left(\frac{m}{c}, \frac{t}{c}\right). \tag{151}$$

The proof is in Appendix A.3, page 85.

Lemma 2.18 (Monotone in the mean). *$W(m,t)$ is non-decreasing in $m \in (0,t)$.*

The proof is in Appendix A.3, page 87.

For $A \in [0,t)$ let

$$w(A) = \max_{c\in(0,1]} Q(A/c, t/c). \tag{152}$$

- w is continuous and non-decreasing in A.
- $w(A) < 1/2$ for every $A < t$.
- $w(A) \to 1/2$ as $A \to t$.

The derivation is in Appendix A.3, page 87. So, for $\varepsilon < 1/2$, the *sharp cap* $A_1(t,\varepsilon)$ is the largest $A \in [0,t)$ with $w(A) \le \varepsilon$, and it lies below t. For $\varepsilon \ge 1/2$ no largest value exists, because every $A < t$ qualifies. The cap is then a value below t that the computation fixes.

The formulation tracks the mean of a Gamma component and imposes, on every day and under both repair laws, the linear row

$$\mu_k \;\le\; \frac{A_1(\beta^*\tau,\,\varepsilon)}{\beta^*}. \tag{153}$$

Proposition 2.19 (The sharp cap is safe). *If the tracked mean μ_k satisfies (153) and $\mu_k \ge \mathbb{E}[D_k]$, then $\Pr(D_k > \tau) \le \varepsilon$, for any mix of rates.*

The proof is in Appendix A.3, page 87.

Computation of the cap. The cap A_1 is computed once per component, before the model is built, by bisection on A with an inner maximum over c. Three facts justify the computation, and $w(A_1) \le \varepsilon$ holds at the returned value. The derivation is in Appendix A.3, page 87.

The numerical choices of the computation have the following reasons.

- **One-rate cap.** A bisection on the first argument of Q finds the exact one-rate cap $\hat{A}$, the root of $Q(\hat{A}, t) = \varepsilon$. Its interval keeps $Q \le \varepsilon$ at the lower end and $Q > \varepsilon$ at the upper end, and the bisection returns the lower end. So $Q(\hat{A}, t) \le \varepsilon$, and the numerical error lies on the safe side.
- **Grid and its extensions.** The maximum over c starts from a logarithmic grid of 401 values of c. When the best grid point is the smallest value of the grid, the maximum may lie below the grid. A new grid of 401 values then covers the next factor 100 below, at most 10 times. After that the computation stops with an error: a maximum below the last grid would be underestimated, and the cap would be too large.

- **Golden-section search.** A golden-section search shrinks an interval around a single peak by the factor 0.618 per step. The function of c has a single peak, by the second of the three facts. So 60 steps between the neighbours of the best grid point reduce the interval by the factor $0.618^{60} \approx 3 \cdot 10^{-13}$. The search returns the largest value found, so it never returns less than the best grid value.
- **Tolerances and cost.** The bisection for $\hat{A}$ stops at a relative width 10^{-12}, and the bisection on A at 10^{-9}. Each step of the bisection on A computes w once. One cap takes about 30 ms and about 16 evaluations of w.
- **Levels of at least one half.** For $\varepsilon \geq 1/2$ the test $w(A) \leq \varepsilon$ holds without computing w, because $w(A) < 1/2$ for every $A < t$. Near $A = t$ the maximum over c moves towards $c \to 0$, where Q approaches $1/2$, so the grid would need many extensions. The bisection returns a value close to t: $A_1/t = 0.999999999069$ at $\varepsilon = 0.5$.
- **Cache.** The cap is computed once for each pair (t, ε) in a run and then reused. The input check of the replacement state and the row of the model use the same cap, and components with the same pair share it. So every use after the first costs nothing.
- **Accuracy.** The table compares the computation with independent ones. The brute force evaluates 20,001 values of $s = t/c$, refines the best one and bisects on A to 10^{-13}.

Value	Reference	ε	t	Largest difference
$\hat{A}$	`brentq`, tolerance 10^{-15}	10^{-6} to 0.99	10^{-6} to 10^{4}	$8.8 \cdot 10^{-13}$ (relative)
A_1	brute force	0.001 to 0.3	10^{-4} to 1000	$6.7 \cdot 10^{-10}$
A_1	brute force	10^{-6} and 10^{-8}		$2.2 \cdot 10^{-10}$
A_1	brute force	0.3 to 0.499	10^{-6} to 10^{4}	$7.4 \cdot 10^{-10}$

  In the last row no grid reaches its last extension. At nine pairs with $A_1 < \hat{A}$, $w(A_1)/\varepsilon = 1.000000000$ and $w\big(A_1(1 + 10^{-6})\big)/\varepsilon = 1.000001$. So the computed cap is safe, and a cap larger by a relative 10^{-6} is not.

The sharp cap and the exact one-rate cap. $\hat{A}$ is the exact cap of a single Gamma law of rate β^*, and $A_1 \leq \hat{A}$, because $c = 1$ is in the maximum. Let $t_1(\varepsilon)$ be the value of t at which $\partial_c Q(\hat{A}/c, t/c)$ at $c = 1$ changes sign: $t_1 = 0.436$, 0.448 and 0.614 for $\varepsilon = 0.001$, 0.01 and 0.1.

- **Above t_1.** On a grid of 200 values of t in $[t_1, 10^5]$, no $c < 1$ gives a larger tail than $c = 1$, so $A_1 = \hat{A}$ there. This is a numerical check, not a proof; the computation of the cap does not rely on it, because it tests $w(\hat{A}) \leq \varepsilon$ before it uses $\hat{A}$.
- **Below t_1.** $A_1 < \hat{A}$.

For $\varepsilon = 0.01$:

$t = \beta^*\tau$	0.01	0.1	0.3	1	10
$A_1/\hat{A}$	0.142	0.641	0.961	1	1

When every term has the same rate and no repair occurs, the true state is a single Gamma law and $\hat{A}$ is exact for it. With several rates, or with repairs, the true state is a weighted sum, and below t_1 the exact one-rate cap is not safe.

Example: $\varepsilon = 0.01$ and $\beta^*\tau = 0.1$, so $\hat{A} = 0.0055012$. A new part serves one mission of mean $\hat{A}/\beta^*$ and rate $4.56\,\beta^*$. Its mean meets the exact one-rate cap, but its tail is $1.565\,\varepsilon$. The sharp cap there is $A_1 = 0.641\,\hat{A}$.

Below the threshold. For $t \leq t_1(\varepsilon)$,

$$A_1(t, \varepsilon) = r^*(\varepsilon)\, t. \tag{154}$$

The derivation is in Appendix A.3, page 87. The mean cap is then $A_1/\beta^* = r^*(\varepsilon)\,\tau$, whatever the rates: $r^* = 0.00355$, 0.0353 and 0.331 for $\varepsilon = 0.001$, 0.01 and 0.1.

The choice of β^*. The mean cap

$$A_1(\beta^*\tau, \varepsilon)/\beta^* = \tau\, A_1(t, \varepsilon)/t \tag{155}$$

is non-decreasing in β^*. The derivation is in Appendix A.3, page 87. Lemma 2.15 holds for every β^* at most the smallest rate. So the smallest rate gives the largest safe cap of this form.

The class $\mathcal{G}(m)$ also contains laws that no schedule produces, so below t_1 the row can reject some safe schedules. The row keeps the mean of each mission. β^* only bounds how dispersed each term may be.

2.2.2 Loop constraint for each degradation model

The loop constraint (19) is one rule for every model: each tracked statistic, and its anchor under ARD_1, must be no larger at the end of the horizon than at the end of the transitory phase. Table 16 lists the tracked statistics and the reliability row of each model of the formulation. Every model accepts both repair laws, and Theorem 1.4 applies to each of them. We drop the (i, ℓ) subscripts for clarity.

Table 16: Tracked statistics and reliability rows of the five degradation models of the formulation, on every day k of the horizon. Under ARD_1 each tracked statistic also has an anchor, except the Wiener variance. The loop constraint (19) covers every tracked statistic and every anchor. $\kappa_G = \Phi^{-1}(1-\varepsilon)$, with $\varepsilon < 0.5$.

Model	Tracked statistics	Reliability row
Gaussian	mean μ, variance v	$\kappa_G^2\, v_k \le (\tau - \mu_k)^2, \quad \mu_k \le \tau$
Wiener	mean μ, variance v	as for the Gaussian model
Inverse Gaussian	mean μ	$\mu_k \le \tau/\bar{q}$ (Theorem 2.13)
Gamma	mean μ	$\mu_k \le A_1(\beta^*\tau, \varepsilon)/\beta^*$, Eq. (153)
Rainflow	mean μ and the statistic of the bound	Table 8

Gamma. With one shared rate β and no repair, the damage is $\mathrm{Ga}(A_k, \beta)$, where A_k is the cumulative shape on day k. Its survival function is $\Pr(D > \tau) = Q(A, \beta\tau)$, with Q the regularised upper incomplete gamma function of Section 2.2. Since $Q(a, x)$ is strictly increasing in a for fixed $x > 0$, a loop on the shape would then be a scalar inequality.

One caveat concerns repairs. The *pathwise* ARD repair $D \mapsto (1-\rho)D$ leaves the shared-rate sub-family, since

$$(1-\rho)\,\mathrm{Ga}(A, \beta) = \mathrm{Ga}\big(A,\ \beta/(1-\rho)\big). \tag{156}$$

A repair modelled in parameter space, as the shape reduction $A \mapsto (1-\rho)A$ at fixed rate β, matches ARD in expectation,

$$\mathbb{E}[D^+] = (1-\rho)\,\mathbb{E}[D^-]. \tag{157}$$

But it is a different operator on the paths, and it can accept an unsafe schedule.

Example: $\tau = 1$, $\varepsilon = 0.01$, $\rho = 0.5$ and one rate $\beta = 0.01$, so the exact one-rate cap is $\hat{A} = 0.0024877$. A new part serves a mission of shape $\hat{A}$ on day 1, is repaired on day 2, and serves a mission of shape $0.5\,\hat{A}$ on day 3. The shapes in parameter space are $\hat{A}$, $0.5\,\hat{A}$ and $\hat{A}$, so every cap row holds. But the true tail on day 3 is 0.0133, that is, $1.33\,\varepsilon$.

The formulation therefore repairs the damage path, tracks the mean, and certifies reliability by the sharp cap (153), which is safe for every weighted sum of Gamma terms (Proposition 2.19). Its loop constraint acts on the mean and, under ARD_1, on the anchor of the mean. Facts (M1)–(M3) of Section 1.3.1 hold. The derivation is in Appendix A.3, page 88.

The Gaussian family is closed under both readings of the repair,

$$(1-\rho)\,\mathcal{N}(\mu, \sigma^2) = \mathcal{N}\big((1-\rho)\mu,\ (1-\rho)^2\sigma^2\big). \tag{158}$$

Gaussian. Let

$$D_{H_1} \sim \mathcal{N}(\mu_{H_1}, \sigma_{H_1}^2) \quad \text{and} \quad D_T \sim \mathcal{N}(\mu_T, \sigma_T^2). \tag{159}$$

The survival function is

$$\Pr(D > \tau) = \Phi\big((\mu - \tau)/\sigma\big), \tag{160}$$

where Φ is the standard normal CDF. Since Φ is strictly increasing, the probability comparison (15) is equivalent to

$$\frac{\mu_T - \tau}{\sigma_T} \leq \frac{\mu_{H_1} - \tau}{\sigma_{H_1}}, \tag{161}$$

or equivalently

$$(\tau - \mu_T)/\sigma_T \geq (\tau - \mu_{H_1})/\sigma_{H_1}. \tag{162}$$

Note that $\sigma_T = \sigma_{H_1}$ does *not* follow from serving the same missions in both phases. Without interventions the variance accumulates,

$$\sigma_T^2 = \sigma_{H_1}^2 + \sum_{k=H_1+1}^{T} v_k^{\text{inc}} > \sigma_{H_1}^2, \tag{163}$$

where v_k^{inc} is the variance of the increment of day k (0 on a day without a mission). Equality requires that the repairs or replacements of the operating phase (which scale the variance by $(1-\rho)^2$ or reset it) remove exactly the variance that the mission days add. In that case (161) reduces to $\mu_T \leq \mu_{H_1}$.

In general, condition (161) couples the mean and the standard deviation: a sufficiently lower mean can compensate a higher variance at T. Condition (161) constrains only the single scalar $(\mu - \tau)/\sigma$ of a two-parameter state. So, by itself, it does not make the schedule repeatable (see Section 2.1.5).

The componentwise condition $\mu_T \leq \mu_{H_1}$ and $\sigma_T^2 \leq \sigma_{H_1}^2$ restores repeatability; it is the analogue of (94) for the exact law. It implies (161) whenever $\mu_{H_1} < \tau$, and it chains because both moments propagate monotonically under the dynamics. This componentwise condition is the loop constraint (19) of the Gaussian model, together with the rows on the two anchors under ARD_1.

In the formulation, a mission increment of a Gaussian component is $\mathcal{N}(\mu^{\text{inc}}, v^{\text{inc}})$, the initial damage is $\mathcal{N}(\mu_0, v_0)$, and each replacement draw is $\mathcal{N}(\mu^{\text{new}}, v^{\text{new}})$. A known damage has variance 0. These terms are mutually independent (assumption N1). A repair acts on the damage path by the law of the component.

The Gaussian law gives negative damage a small probability. The input checks $\mu \geq 3\sqrt{v}$ for the initial damage and for each increment keep this probability at most $\Phi(-3) = 0.135\,\%$ for each of these terms.

Proposition 2.20 (The state stays Gaussian)**.** *Under N1, on every day of a schedule $D_k \sim \mathcal{N}(m_k, V_k)$, where (m_k, V_k) follow the day maps of the mean and the variance, with factors $1-\rho$ and $(1-\rho)^2$, and with anchors equal to the mean and the variance just after the last intervention.*

The proof is in Appendix A.3, page 88.

So ARD_1 is exact for this model. Its variance map uses the factor $(1-\rho)^2$ and the variance anchor. For $D \sim \mathcal{N}(m, V)$,

$$\Pr(D > \tau) = 1 - \Phi\big((\tau - m)/\sqrt{V}\big) \leq \varepsilon \tag{164}$$

exactly when $m + \kappa_G\sqrt{V} \leq \tau$, with $\kappa_G = \Phi^{-1}(1-\varepsilon)$. For $\varepsilon < 0.5$, so that $\kappa_G > 0$, and with $\tau - \mu \geq 0$, this is equivalent to the squared row of Table 16.

Validity. Let $(m_k, V_k) \leq (\mu_k, v_k)$ be the true moments (Proposition 2.20 and Proposition 1.3), and let (μ_k, v_k) satisfy the row. Then $\Pr(D_k > \tau) \leq \varepsilon$. The derivation is in Appendix A.3, page 88.

Remark 2.21 (The level ε and the range $V_{\max}$). $V_{\max} = \tau^2/\kappa_G^2$ is the largest variance that the Gaussian row admits, at $\mu = 0$. So it is the natural range of the variance, and the big-M constant of its rows (Section 5.1).

$V_{\max}/\tau^2$ is 0.18 at $\varepsilon = 0.01$, 0.61 at 0.1, 3.6 at 0.3 and about 1600 at 0.49. It grows without bound as $\varepsilon \to 0.5$, and at $\varepsilon \geq 0.5$ the squared row does not describe the reliability requirement, because $\kappa_G \leq 0$. Hence the condition $\varepsilon < 0.5$ for Gaussian and Wiener components. The range is large only close to 0.5.

The other models and bounds have no limit below $\varepsilon = 1$: for example, the variance cap $\tau^2\varepsilon/(1-\varepsilon)$ of the Cantelli row is finite for every $\varepsilon < 1$. The variance ranges of the rainflow rows behave like $V_{\max}$ at the other end: the Cantelli and Bernstein variance ranges and the Hoeffding range $2\tau^2/\kappa$ grow without bound as $\varepsilon \to 1$.

Facts (M1)–(M3) of Section 1.3.1 hold. The derivation is in Appendix A.3, page 88.

Wiener. The damage is a Wiener process with drift, observed once a day. A mission adds a Gaussian increment with a mean that depends on the mission and with the variance σ^2, the same for every mission. The reason is that σ describes the material of the component, not the route. The initial damage is $\mathcal{N}(\mu_0, v_0)$ with $v_0 \geq 0$, and each replacement draw is $\mathcal{N}(\mu^{\text{new}}, v^{\text{new}})$. These terms are mutually independent (assumption W1).

In the formulation, a repair acts on the mean only; this is a modelling choice. The repair shifts the damage by a known amount: $D^+ = D^- - \rho\, m^-$ under ARD_∞ and $D^+ = D^- - \rho\,(m^- - g^\mu)$ under ARD_1, with $m^- = \mathbb{E}D^-$. The law stays Gaussian, the variance is unchanged, and only a replacement lowers it.

Under W1 the damage is $\mathcal{N}(m_k, V_k)$ on every day, with (m_k, V_k) given by these maps: a mission adds an independent Gaussian term, and a repair or a replacement acts as stated.

The reliability row is the Gaussian one, with the same validity argument, forms and remark on $V_{\max}$. The rows of the Wiener model are safe also when a repair acts on the damage path, as in the Gaussian model. The derivation is in Appendix A.3, page 88.

Facts (M1)–(M3) hold as for the Gaussian model, with the factor 1 for the variance, which has no anchor. The loop row on the variance forces a replacement in the operating phase whenever the member serves a mission there. Without a replacement,

$$v_T \geq v_{H_1} + n\,\sigma^2 > v_{H_1} \tag{165}$$

after $n \geq 1$ mission days. No input check enforces this. When replacements are not allowed, the member of a Wiener component cannot serve a mission in the operating phase.

Inverse Gaussian. The CDF of $\text{IG}(\mu, \lambda)$ admits the closed form

$$F_{\text{IG}}(x;\, \mu, \lambda) \;=\; \Phi\!\left(\sqrt{\frac{\lambda}{x}}\Big(\frac{x}{\mu} - 1\Big)\right) \;+\; e^{2\lambda/\mu}\,\Phi\!\left(-\sqrt{\frac{\lambda}{x}}\Big(\frac{x}{\mu} + 1\Big)\right). \tag{166}$$

For

$$D_{H_1} \sim \text{IG}(\mu_{H_1},\, \lambda_{H_1}) \quad \text{and} \quad D_T \sim \text{IG}(\mu_T,\, \lambda_T), \tag{167}$$

the probability comparison (15) reads

$$F_{\text{IG}}(\tau;\, \mu_T, \lambda_T) \;\geq\; F_{\text{IG}}(\tau;\, \mu_{H_1}, \lambda_{H_1}). \tag{168}$$

This is an analytical but nonlinear inequality in the four parameters $(\mu_{H_1}, \lambda_{H_1}, \mu_T, \lambda_T)$. Under the closure condition (same μ^2/λ), the $\leq_{\text{lr}}$ results of Table 12 guarantee that $\mu_T \leq \mu_{H_1}$ is a sufficient condition.

The formulation does not use this comparison, because its damage is not an inverse Gaussian variable in general (Section 2.2). It tracks the mean, imposes the CVaR cap of Theorem 2.13, and loops on the mean and, under ARD_1, on the anchor of the mean. Facts (M1)–(M3) of Section 1.3.1 hold. The derivation is in Appendix A.3, page 89.

Rainflow. The tracked statistics and the rows are those of Section 2.1.4. Facts (M1)–(M3) of Section 1.3.1 hold. The derivation is in Appendix A.3, page 89. The loop constraint covers the mean, the second statistic of the bound and, under ARD_1, their anchors.

Compound Poisson. When $D \sim \mathrm{CP}(\nu, F_Y)$, the survival function does not admit a universal closed form. Conditioning on the number of shocks gives the series

$$\Pr(D > \tau) \;=\; \sum_{n=0}^{\infty} \frac{e^{-\nu}\,\nu^n}{n!}\;\Pr\Big(\sum_{k=1}^{n} Y_k > \tau\Big), \tag{169}$$

which can be evaluated numerically (it converges rapidly) or bounded.

Upper bounds when no closed form is available. Sometimes the survival function $\Pr(D > \tau)$ has no closed form. This happens for the compound Poisson process, and when the closure condition fails and the marginal of the sum is not in a standard family. The *reliability* constraint can then still be enforced through conservative upper bounds on the failure probability:

- Markov ($\Pr(D > \tau) \le \mathbb{E}[D]/\tau$);
- Cantelli (65);
- Chernoff (66).

Section 2.1.4 develops these bounds in per-mission form, together with the Bernstein and Hoeffding refinements, their relative tightness in the small-ε regime and their MILP encodings. For the compound Poisson process the Chernoff bound is particularly convenient, since $M_D(s) = \exp\big(\nu\,(M_Y(s) - 1)\big)$ has a closed form whenever M_Y has one.

The *loop* constraint, by contrast, must not be translated into the same bounds. Imposing $U^{(1)} \le U^{(0)}$ on a scalar bound U does not make the schedule repeatable. The loop acts instead on the tracked statistics of the bound, for example through the componentwise moment condition (94), whose rows are linear in the tracked statistics (Section 2.1.5).

2.3 Maintenance strategies

Maintenance actions on a repairable component lie between two extremes. *Minimal repair* restores the component to its state just before failure ("as bad as old", ABAO). *Perfect repair*, or *replacement*, resets it to the condition of a new part ("as good as new", AGAN). Real interventions almost always lie between these two extremes, and together they are called *imperfect maintenance* [28]. This section surveys the principal modelling families.

2.3.1 Taxonomy of repair actions

Table 17 classifies five canonical maintenance actions by their effect on the component state. Here D^- and D^+ denote the damage immediately before and after maintenance, and D^{new} is the damage of a new component.

2.3.2 Virtual age models

Kijima [20] formalised imperfect repair through the concept of *virtual age*. After the n-th repair, the component behaves as if it had the age V_n instead of its calendar age. The failure intensity at calendar time t is $\lambda(t - T_n + V_n)$, where T_n is the time of the n-th repair and $\lambda(\cdot)$ is the baseline intensity. Two variants are standard:

- **Kijima I.** The repair acts only on the virtual age accumulated since the previous repair:

$$V_n \;=\; V_{n-1} + (1 - \rho)\,(T_n - T_{n-1}), \tag{170}$$

  where $\rho \in [0, 1]$ is the repair efficiency. Only the time since the previous repair becomes partly younger ("memory-one").

Table 17: Taxonomy of maintenance actions, from least to most restorative.

Action	Effect on damage D	Model
No intervention	$D^+ = D^-$	Damage unchanged
Minimal repair (ABAO)	$D^+ = D^-$	Failure intensity restored to pre-failure level; damage history preserved [8]
Imperfect repair	$D^- > D^+ > D^{\text{new}}$	Partial reduction of damage or virtual age (see below)
Perfect repair (AGAN)	$D^+ = D^{\text{new}}$	Component returned to like-new state
Replacement	$D^+ = D^{\text{new}}$	Physical component substituted; equivalent to perfect repair

- **Kijima II.** The repair acts on the entire accumulated virtual age:

$$V_n \;=\; (1-\rho)\,(V_{n-1} + T_n - T_{n-1}). \tag{171}$$

  The whole history becomes younger ("infinite memory").

In both cases $\rho = 0$ gives minimal repair (ABAO) and $\rho = 1$ gives perfect repair (AGAN).

2.3.3 Reduction-based models

Doyen and Gaudoin [12] classify imperfect repair models by the *quantity* that the repair reduces. The classes ARA (arithmetic reduction of age) reduce the virtual age, and the classes ARI (arithmetic reduction of intensity) reduce the failure intensity. Each class has a memory: the largest number of previous failure times that enter the failure intensity [12, p. 8]. The memory-one variant carries the subscript 1, and the infinite-memory variant the subscript ∞.

No class of Doyen and Gaudoin acts on a degradation level. Mercier and Castro [22] define a reduction that acts on the degradation level itself. Table 18 compares the reduction classes.

Table 18: Reduction classes of imperfect repair (arithmetic reduction of age, ARA; of deterioration, ARD; of intensity, ARI): the reduced quantity, the laws and their sources.

Class	Reduced quantity	Laws	Source
ARA	virtual age V	ARA_1 (172) ARA_∞ (173)	[12, eq. (22), p. 12] for ARA_1 and [12, eq. (20), p. 11] for ARA_∞
ARD	degradation level D	ARD_1 (175) ARD_∞ (176)	ARD_1: the "arithmetic reduction of deterioration of order 1" of Mercier and Castro [22, §3.1, eq. (4)]; ARD_∞: neither source names it, and its name is an analogy
ARI	failure intensity λ	ARI_1 (177) ARI_∞ (178)	Doyen and Gaudoin [12]

Arithmetic Reduction of Age (ARA). The repair removes a fraction ρ of the virtual age V:

$$\text{ARA}_1: \quad V^+ = V^- - \rho\,(V^- - V^{\text{last}}), \tag{172}$$

$$\text{ARA}_\infty: \quad V^+ = (1-\rho)\,V^-, \tag{173}$$

where V^{last} is the virtual age just after the previous repair. ARA_1 reduces only the age accumulated since the previous repair, and ARA_∞ reduces the whole virtual age. With

$$V^- = V^{\text{last}} + (T_n - T_{n-1}), \tag{174}$$

ARA_1 coincides with Kijima I and ARA_∞ with Kijima II.

Arithmetic Reduction of Deterioration (ARD). The repair removes a fraction ρ of the degradation level D:

$$\mathrm{ARD}_1: \quad D^+ = D^- - \rho\,(D^- - D^{\mathrm{last}}), \tag{175}$$

$$\mathrm{ARD}_\infty: \quad D^+ = (1-\rho)\,D^-. \tag{176}$$

ARD_1 removes the fraction ρ of the degradation accumulated since the last maintenance. It acts on the degradation path, and the age is unchanged. The reference level D^{last} is the level just after the last intervention, repair or replacement. Before the first intervention, it is the initial level. ARD_∞ applies the rule of ARA_∞ to the whole degradation level, as ARD_1 applies the rule of ARA_1 to the degradation accumulated since the last intervention.

Arithmetic Reduction of Intensity (ARI). Instead of reducing the virtual age, the repair directly reduces the failure intensity:

$$\mathrm{ARI}_1: \quad \lambda^+ = \lambda^- - \rho\,(\lambda^- - \lambda^{\mathrm{last}}), \tag{177}$$

$$\mathrm{ARI}_\infty: \quad \lambda^+ = (1-\rho)\,\lambda^-, \tag{178}$$

where λ^{last} is the failure intensity just after the previous repair. ARI models are natural when maintenance targets the failure rate (for example, condition-based interventions) rather than the underlying degradation level. They are in general not equivalent to ARA or ARD models. A reduction in intensity corresponds to a unique reduction in virtual age only if the baseline intensity is strictly monotone [12].

2.3.4 Connection to the fleet management formulation

The abstract repair operator $\mathcal{R}(D^-, \rho)$ of Section 1 covers the ARD models and replacement. In particular:

- ARD_∞: $\mathcal{R}(D^-, \rho) = (1-\rho)\,D^-$, the default repair law of the formulation;
- ARD_1:
$$\mathcal{R}(D^-, \rho) = D^- - \rho\,(D^- - D^{\mathrm{last}}), \tag{179}$$
requiring the additional state variable D^{last};
- Replacement / AGAN: $\mathcal{R}(D^-, \rho) = D^{\mathrm{new}}$ (captured by the separate replacement variable $x^r_{i\ell k}$ in the formulation).

The formulation offers both repair laws for every degradation model, chosen per component. The choice of the repair law affects the damage dynamics (12), the definition of the repair-cost variable $z_{i\ell k}$ and the tractability of the resulting optimisation problem.

ARI models do not fit directly into the formulation of Section 1, which tracks the damage. They would need a reformulation in terms of the failure intensity instead of the damage. No law of the formulation acts on a virtual age.

Anchors. Under ARD_1 the formulation tracks, for each statistic of the damage, an *anchor*: the value of the statistic just after the last intervention, repair or replacement. On an intervention day the anchor takes the new value of its statistic. On any other day it keeps its value.

Example: mean 0.6 and anchor 0.2 before a repair with $\rho = 0.5$. The repair gives the mean $0.6 - 0.5\,(0.6 - 0.2) = 0.4$ and the anchor 0.4. On the next day the gap between the mean and the anchor is 0, and it grows again with each mission.

Before the first intervention of the horizon, the anchor is the value just after the last intervention before the horizon. When that value is unknown, the formulation takes the initial value of the statistic. Each statistic is non-decreasing between interventions (fact (M1) of Section 1.3.1), so the true value lies between the value of a new part and the initial value. A larger anchor gives a weaker repair, so the initial value never overstates what a repair achieves.

Example: initial mean 0.5, $\rho = 0.5$, and a repair on day 1 before any mission. With the anchor 0 the mean drops to 0.25. With the anchor 0.5 it stays at 0.5, and the repair removes only damage accumulated within the horizon.

Repair maps of the tracked statistics. A model tracks moments or other summaries of the damage D, not D itself. A repair maps a tracked statistic θ with anchor g to $a\,\theta + (1-a)\,g$, with a factor $a \in [0,1]$ and $g = 0$ under ARD_∞.

- **Mean.** $a = 1 - \rho$.
- **Variance and Hoeffding budget.** $a = (1-\rho)^2$.
- **Cumulant generating function of the Chernoff bound.** $a = 1 - \rho$ gives an upper bound (Proposition 2.2).
- **Wiener variance.** It is the exception: a repair leaves it unchanged ($a = 1$), and it has no anchor (Section 2.2.2).

The derivation of the factors of the mean and the variance is in Appendix A.3, page 89. These maps are non-decreasing in the statistic and in its anchor, as fact (M1) and Lemma 1.2 require.

Laws per model. Every model of the formulation accepts both laws. Table 19 gives the reason for each model. Under ARD_1 with $\rho < 1$, the loop constraint requires a replacement in the operating phase of a component that gains damage there (Proposition 1.8).

Table 19: Repair laws accepted by each degradation model, with the reason.

Model	ARD_∞	ARD_1	Reason or note
Gaussian	yes	yes	both laws keep the state Gaussian, with exact moments (Proposition 2.20)
Inverse Gaussian	yes	yes	the cap holds for any repair weights and any dependence between the terms (Proposition 2.12 and Theorem 2.13)
Wiener	yes	yes	on the mean only; the variance changes only by replacement
Gamma	yes	yes	both laws keep the state a weighted sum of Gamma terms (Lemma 2.15), and the sharp cap holds for every such sum (Proposition 2.19)
Rainflow, every bound	yes	yes	under both laws, Chernoff uses a conservative recursion for the cumulant generating function (Proposition 2.2)

Maintenance under the Palmgren–Miner damage model. The remaining-life framework of Section 2.1.3 represents damage as the normalised fraction

$$D_{\mathrm{PM}} = \sum_i n_i / N_{f,i} \geq 0, \tag{180}$$

with failure at $D_{\mathrm{PM}} \geq 1$. D_{PM} adds up over loading blocks. So it fits the same damage-tracking structure as the stochastic processes of Table 4: each mission day adds an increment $\Delta D_{\mathrm{PM}} = n_j / N_{f,j}$. This increment is deterministic if the S–N parameters are fixed, and random if the scatter is modelled [25]. The repair operator $\mathcal{R}$ and the replacement therefore apply directly:

- *Replacement* resets the fatigue fraction to zero: $D^{+}_{\mathrm{PM}} = 0$ (equivalently $D^{\mathrm{new}} = 0$). A new component, with a cycle count of zero, takes the place of the physical component.
- ARD_{∞}:
$$D^{+}_{\mathrm{PM}} = (1-\rho)\, D^{-}_{\mathrm{PM}}. \tag{181}$$
Physically, this models an intervention that partly restores the fatigue life without resetting the whole loading history, for example a surface treatment, shot peening or a local crack repair.
- ARD_{1}:
$$D^{+}_{\mathrm{PM}} = D^{-}_{\mathrm{PM}} - \rho\,(D^{-}_{\mathrm{PM}} - D^{\mathrm{last}}_{\mathrm{PM}}). \tag{182}$$
The repair reverses part of the damage accumulated since the last intervention (repair or replacement), and only that damage.

Unlike the stochastic process models, D_{PM} is normalised so that failure occurs at 1. So the threshold is fixed at $\tau = 1$, and the reliability constraint becomes

$$\Pr(D_{\mathrm{PM}} \geq 1) \leq \varepsilon. \tag{183}$$

In the deterministic variant of Miner's rule this is a hard feasibility check. In the stochastic extension it becomes a chance constraint analogous to (40g).

Closure under summation holds trivially: D_{PM} is a sum of real-valued increments. When the increments are independent, as under the block-independent scatter (I) of Section 2.1.8, the marginal on any day is the convolution of the individual increments, whatever their distributional family. Under the unit-specific scatter (II) the increments share the factor A, and the marginal is the law of M_k/A.

- In the deterministic variant of Miner's rule, the comparison of two schedules reduces to the comparison of two real numbers, and no stochastic order is needed.
- In the stochastic extension (scatter in $N_{f,i}$), the parametric ordering conditions of Section 2.2 apply only if the assumption on the scatter places the increments $n_i/N_{f,i}$ in one of the families of Table 4. Otherwise the general definitions of the orders still hold, but they must be verified case by case.

3 Focus on electric vehicles

A battery-electric road vehicle, such as a transit bus or a passenger car, is a complex repairable system. Its parts degrade at very different rates and through very different physical mechanisms. Not every part needs an explicit place in the fleet-management model of Section 1. We keep a component if it satisfies two criteria:

(C1) **Usage-driven degradation.** Its condition worsens measurably with *how* the vehicle is used. So the mission assigned on a given day produces a damage increment $\Delta D^{(j)}_{i\ell k}$ that can be quantified (Section 1). A part that ages only with calendar time, whatever the mission served, carries no information for the schedule. Its damage need not enter the model, which tracks the damage that the missions cause.

(C2) **Operational significance.** Its failure or its preventive renewal strongly affects at least one of the three quantities that the schedule balances: *safety*, *availability* or *life-cycle maintenance cost*.

We choose *not* to require that the component has a degradation model within the families of Section 2.2. A component can be worth considering even when the abstract model of Section 1 does not yet capture it. Such cases are listed here with the modelling extension that they would need, rather than dropped without a note.

The components that meet these criteria form two groups.

- The *electrochemical and electrical powertrain* (traction battery, electric motor, power electronics, charging interface) degrades through electrochemical ageing and through thermal and electrical cycling. Energy throughput, charge and discharge rates and temperature drive this degradation.
- The *mechanical running gear* (tires, friction brakes, wheel bearings, suspension, reduction gear) degrades through wear and fatigue. Distance travelled, vehicle mass, road quality and the profile of acceleration and braking drive this degradation.

Table 20 lists the main candidates in each group, the dominant degradation mechanism, the operational quantity that drives it, and the reason why the component is operationally significant.

Two facts of the domain shape the relative importance of these components.

1. **The traction battery dominates.** A recent predictive-maintenance study of electric buses [19, Secs. 2, 4.2] centres entirely on the battery pack as the main maintenance-critical component. The study models the end of life of the pack through the accumulated charge throughput. It shows that optimising the driving profile alone can cut the modelled capacity loss by 25%. This is direct evidence that battery damage can be steered by exactly the kind of mission-assignment decisions that our framework optimises.
2. **Regenerative braking shifts the wear burden to the tires.** Regenerative braking absorbs most of the service braking, and it reduces the friction-brake wear emissions by an estimated 60–90% [14, Sec. 6.1]. The relative wear burden therefore shifts to the tires, and the higher mass of electric vehicles increases tire wear. On motorways, the savings from regenerative braking do not compensate the added tire-wear particle emissions [14, Sec. 6.1].

Power-electronic and electronic components. Beyond the battery, the traction chain contains three electrical sub-families, and they fit the framework to different degrees.

- **Power semiconductors.** Junction-temperature-swing fatigue governs the *power semiconductors* of the inverter and of the DC–DC and on-board-charger *converters*. They fit well into the remaining-life family (treated in detail below).
- **Converter capacitors.** The *DC-link capacitors* of the converters stand out among the components that fail in the field [37, Sec. I].
  - **Mechanisms.** Small electrolytic capacitors wear out mainly by vaporisation of the electrolyte, and large ones mainly by an electrochemical reaction of the oxide layer [37, Sec. III-A]. Film capacitors lose capacitance through self-healing breakdowns of the film and through corrosion by moisture [37, Sec. III-A, Table II]. When the ripple current is high, as in electric vehicles, film capacitors can give a cheaper and smaller DC link [37, Sec. II].
  - **End of life.** In both types the capacitance falls or the equivalent series resistance (ESR) rises until an end-of-life limit, for example a 20% loss of capacitance for an electrolytic capacitor [37, Table III].
  - **Lifetime model.** The standard model is an Arrhenius law in the temperature and a power law in the voltage [37, Eqs. (1)–(2)]. The ripple current is not a variable of this model: it dissipates power in the ESR and so raises the hot-spot temperature [37, Table I, Secs. IV-B and V, Fig. 10]. Humidity is a further critical stressor of film capacitors [37, Table II].
  - **Mission.** The mission acts through the load of the converter, which sets the ripple current and raises the local ambient temperature of the capacitor [38, Sec. IV, Fig. 7].
  - **Fit to the framework.** Wear-out predictions under changing stress usually add the damage linearly, by the Palmgren–Miner rule [38, Sec. III-D]. The accuracy of this linear rule is still an open question [38, Sec. IV-C]. So capacitor wear-out is a monotone accumulation driven by usage, and it fits the framework.
- **Printed circuit boards.** The *control and power printed circuit boards* (gate-driver,

Table 20: Candidate electric-vehicle components for degradation-aware modelling. Relevance flags: **S** safety-critical, **A** availability driver (its failure or renewal takes the vehicle out of service), **C** maintenance-cost driver. The "mission driver" column names the operational quantity that determines the per-mission damage increment $\Delta D_{i\ell k}^{(j)}$. Acronyms: SEI, solid electrolyte interphase; DOD, depth of discharge; C-rate, charge or discharge current divided by the capacity; IGBT, insulated-gate bipolar transistor; SiC, silicon carbide; DC, direct current; PCB, printed circuit board; BMS, battery management system; HVAC, heating, ventilation and air conditioning.

Component	**Dominant degradation mechanism**	**Mission driver**	**Rel.**
Electrochemical / electrical powertrain			
Traction battery pack	Capacity fade and impedance growth: SEI growth, lithium plating, loss of active material (cycle + calendar ageing)	Energy throughput, DOD, C-rate, temperature	A, C
Traction motor	Winding-insulation thermal ageing; bearing wear	Thermal/load cycles, current	A, C
Power semiconductors (IGBT / SiC modules in the inverter and converters)	Thermal-cycling fatigue (bond-wire lift-off, solder cracking)	Power throughput, thermal cycles	A, C
Converter capacitors (DC link)	Electrolyte vaporisation (electrolytic); self-healing breakdowns and moisture corrosion (film)	Ripple current (through the temperature), ambient temperature, voltage, humidity	A, C
Control / power PCBs (gate drivers, BMS, converter boards)	Solder-joint and via thermo-mechanical fatigue; sudden overstress failures	Thermal cycles (power on/off), vibration	A
Charging interface (inlet, connector)	Contact wear and contact-resistance growth over mating cycles	Number of charging events	A
Mechanical: running gear and chassis			
Tires	Tread abrasion (friction work at the contact patch)	Distance, vehicle mass, driving style, road surface	S, C
Brake friction elements (pads, discs)	Abrasive and thermal wear (mitigated by regenerative braking)	Number and energy of friction-braking events	S, C
Wheel bearings	Rolling-contact fatigue spalling	Distance, load	S, A
Suspension (springs, dampers, bushings)	Fatigue; loss of damping	Load cycles, road quality	A
Reduction gear / drive shafts	Wear, contact fatigue, lubricant ageing	Transmitted torque, distance	A
Auxiliaries (HVAC, battery thermal management, doors)	Wear, cycle-driven failures	Operating hours, actuation cycles	A

battery-management and converter boards) are the borderline case.

- **Wear-out.** Temperature cycles, such as power on/off cycles or day–night cycles, fatigue the solder joints and the plated through-holes of a board [41, Table 1]. Vibration also fatigues the solder joints [41, Table 1]. With each cycle the damage accumulates, until a crack grows and the joint fails [41, Table 1]. The power on/off cycles and the vibration depend on usage, so the boards satisfy (C1). Field data from photovoltaic and wind-turbine converters rank control boards high among the failure sources [38, Sec. II-A], so the boards also satisfy (C2).
- **Sudden failures.** Not every failure of a board comes from accumulated damage. Sudden failures come from design defects, manufacturing issues, single-event effects, overstress or misuse [38, Sec. II]. A mechanical shock, for example, loads the solder joints suddenly [41, Table 1]. For the board failures seen in the field, the causes and the dominant stressors are still open [38, Sec. II-A]. The damage-accumulation dynamics of Section 1 do not fully capture this mix of wear-out and sudden failures.
- **Faithful model.** It would need either a compound-Poisson shock term (Section 2.2) or a formulation in the failure intensity instead of the damage formulation of Equation (12).

We list PCBs here, with this caveat, so that the limitation is recorded and not hidden by the choice of model.

Modelling shortlist. For a first concrete instance of the abstract framework, the most informative components meet two conditions. They are safety- or cost-critical, *and* mature, monotone degradation models describe them and fit the closure and ordering analysis of Section 2.2. These components are the *traction battery pack*, the *tires*, the *brake friction elements*, the *wheel bearings*, the *traction-motor insulation* and the *inverter and converter power modules*.

Each of them maps onto a single damage variable $D_{i\ell k}$ per component. The operational driver in the third column of Table 20 governs its mission-dependent increment $\Delta D_{i\ell k}^{(j)}$. For example, a long highway mission produces large increments of tire wear and of deep-discharge battery damage. A stop-intensive urban mission instead multiplies the braking events and the charge–discharge micro-cycles. This dependence on the mission is exactly the coupling that the scheduling problem of Section 1 exploits.

Worked example: traction-battery capacity fade. The battery pack is the most expensive component of an electric vehicle and the focus of its predictive maintenance [19, Secs. 2, 4.2]. So it serves as the reference instance of the abstract increment $\Delta D_{i\ell k}^{(j)}$.

- **Damage and threshold.** Its damage is the relative capacity fade (one minus the state of health), and the failure threshold τ is the largest admissible fade (the end-of-life criterion).
- **Mechanisms.** Capacity is lost through three families of side reactions: growth of the SEI (solid electrolyte interphase) film, lithium plating and gas generation. Their contributions split into loss of lithium inventory and loss of active material [21, Secs. 2.1, 4.1, Eqs. 28–29].
- **Semi-empirical models.** At the fleet level, semi-empirical models abstract this process and split the fade into a *calendar* term and a *cycling* term.
  - The calendar term grows with an Arrhenius factor $a\exp(-E_a/RT)$ in temperature and with a power law in time and state of charge.
  - The cycling term grows with the number of cycles, modulated by stress factors for temperature, C-rate and depth of discharge [18, Eqs. 1–3].

  An even simpler engineering abstraction links the end of life directly to the accumulated charge throughput [19, Sec. 4.2].
- **Cycling term.** For our framework the decisive structure is the following.
  - The cycling fade is monotone and adds up over days.
  - The daily increment is a function of the energy served on the mission, of its depth

of discharge and of its temperature profile. The energy includes the recharge that restores it, so fast charging (a high C-rate) makes the same mission more damaging.
  - So the cycling fade is a monotone, mission-dependent accumulation process; the Gamma and inverse Gaussian processes of Section 2.2 are two models of this kind.
  - The time-varying increment parameters that the formulation already allows ($\Delta D_{i\ell k}^{(j)}$ may depend on k) absorb the non-stationary (power-law) dependence on time within the horizon. Beyond the horizon the endless schedule repeats the increment laws of the operating phase with period H_2 (Section 1.2), so a dependence on time that does not repeat is not represented there.
- **Calendar term.** The calendar term, by contrast, accrues also on idle days and depot days. So the dynamics of Section 1, which leave the state unchanged on such days, do *not* capture it. Over short horizons the calendar term can be neglected. Over long horizons the dynamics (12) must be extended with a small drift that does not depend on the activity and is added to every case, idle days included.
- **Maintenance.** Maintenance also takes a specific form. Replacing a subset of battery modules removes a corresponding fraction of the damage of the pack. This matches the mean map of the ARD_∞ operator $\mathcal{R}$, with ρ the fraction of modules renewed, but not its action on the damage path. With independent modules the variance falls by the factor $1-\rho$, not $(1-\rho)^2$, so a model that tracks the variance cannot use the variance map of Section 2.3.4 for this action. A full replacement of the pack is the AGAN action x^r.

3.1 Degradation models for the shortlisted components

The other shortlisted components degrade through the same two structural mechanisms as the battery. The first is a monotone *wear or accumulation* driven by usage, and the second is the *consumption of fatigue life* under cyclic loads. So these components map onto the same two model families of Section 2.1. Table 21 summarises, for each component, the governing model, the operational quantity that drives the per-mission increment, and the abstract family that the model instantiates.

Tires. The frictional work spent at the tire–road contact patch governs the loss of tread.

- **Model.** The classical theory of Schallamach and Turner assumes that abrasion is proportional to the energy dissipated by friction [30, Eq. 7]. It derives that, at small slip, wear grows approximately with the *square* of the slip [30, Eq. 19]. So the wear rate rises steeply with the tangential forces of traction, braking and cornering.
- **Factors.** The main factors are the tire characteristics (compound, construction), the road-surface characteristics and the vehicle operation [14, Sec. 2.1.2]. The higher kerb mass (the mass of the empty vehicle) of electric vehicles raises these contact forces. The effect is large: on motorways, the brake-wear savings of regenerative braking do not compensate the added tire-wear particle emissions [14, Sec. 6.1].
- **Fit to the framework.** Tread depth decreases monotonically from its new value to the legal limit. The damage is the tread depth lost, and the threshold τ is the new depth minus the legal limit. The per-mission increment depends on distance, load, road surface and the aggressiveness of the driving.
- **Renewal.** For bus fleets, retreading (renewing the tread of a used tire) acts as a near-AGAN replacement of the wear state. Tire rotation only redistributes wear across the positions on the vehicle, and it stays outside the per-component model.

Brake friction elements. Friction braking wears the brake pad and the brake disc.

- **Model.** A physics-based route applies Archard's law on a fractal, elastoplastic contact [31, Eqs. 11, 14]. Sha et al. apply it to the brake disc, not to the pad. The model gives the

Table 21: Degradation models for the shortlisted electric-vehicle components and their mapping onto the abstract model families of Section 2.1. "Accumulation" denotes a monotone damage-accumulation process (Gamma / inverse-Gaussian / compound-Poisson); "remaining-life" denotes a cycles-to-failure model accumulated by the Palmgren–Miner rule. SOC: state of charge; LAM: loss of active material; DOD: depth of discharge.

Component	**Degradation model / law**	**Mission driver**	**Abstract family**
Battery pack	Calendar + cycle capacity fade: Arrhenius in T, power law in t/SOC, cycle-count term [18, Eqs. 1–3]; mechanistic SEI / plating / LAM [21, Sec. 2.1]	Energy throughput, DOD, C-rate, temperature	Accumulation (k-dependent)
Tires	Friction-work abrasion [30, Eq. 7]; factors: tire build, road surface, vehicle operation [14, Sec. 2.1.2]	Distance, mass, accel./braking, road surface	Accumulation
Brake friction elements	Archard-type wear per braking event, shown for the brake disc [31, Eqs. 21–22]; regenerative-braking mitigation [14, Sec. 6.1]	Number and energy of friction-braking events	Accumulation
Wheel bearing	Subsurface-fatigue spall propagation, continuum damage [26, Sec. 2.1.3]	Distance, speed, load	Accumulation / remaining-life
Traction-motor insulation	Arrhenius–Dakin thermal-endurance life [11]; Montsinger rule [24]; cumulative-damage extension [43, Eqs. 1–2]	Thermal / load cycles, current	Remaining-life (Miner)
Inverter power module	Coffin–Manson / Norris–Landzberg cycles-to-failure in ΔT_j [2, Eqs. 3–5]; CIPS08 [4, Eq. (1)]	Power throughput, thermal cycles	Remaining-life (Miner)

average wear depth h_{wear} of the disc in one braking [31, Eq. 21]. The life of the disc, as a number of brakings, is then

$$N = H_0/(S_0\, h_{\text{wear}}), \tag{184}$$

with H_0 the maximum admissible average wear depth and S_0 a dimensionless wear safety factor for frictional heat and other external effects [31, Eq. 22].

- **Regenerative braking.** In an electric vehicle the regenerative brake absorbs most service braking, and it cuts friction-brake wear by an estimated 60–90% [14, Sec. 6.1]. The remaining increments concentrate in the events where regeneration is unavailable or insufficient: a battery near full charge, a very low temperature, an emergency stop.
- **Fit to the framework.** The cumulative wear of a friction element remains a sum of non-negative, mission-dependent increments, one per friction-braking event. So it is a monotone accumulation process, whose rate now also depends on the energy-management strategy.

Wheel bearings. Bearing degradation has two stages: a long *initiation* phase of rolling-contact fatigue, and then a rapid *spall propagation* once a surface defect forms (a spall is a flake of material that breaks off the surface).

- **Model.** Ohana et al. [26, Sec. 2.1.3] model the propagation phase with a formulation from continuum damage mechanics. In it, a scalar damage variable accumulates per over-rolling cycle,

$$\mathrm{d}D/\mathrm{d}N = \big(\sigma_{\text{eff}}/[\sigma_r(1-D)]\big)^m. \tag{185}$$

  The growth is driven by the tensile residual stresses that repeated impacts of the rolling elements generate at the edge of the spall. The model is meant as the basis of a physics-based prognostic tool that estimates the remaining useful life once a critical spall size is reached.
- **Fit to the framework.** The underlying driver is the number of load cycles, and hence the running distance and the speed. So the bearing damage is a monotone accumulation process. A Gamma or inverse Gaussian process (initiation) that feeds a remaining-life threshold (propagation) represents it well.

Traction-motor insulation. The winding insulation is the main element that limits the life of the traction motor, and its dominant ageing mechanism is thermal. A high winding temperature accelerates the chemical degradation of the dielectric.

- **Model.** Dakin recognised thermal ageing as a chemical reaction whose rate constant follows the Arrhenius exponential law $K = A\,e^{-B/T}$ in absolute temperature [11, Eq. 5]. This gives the thermal-endurance life $L(T) = a\,\exp(b/T)$ [43, Eq. 1].
- **Montsinger's rule.** The earlier empirical rule of Montsinger states that the ageing rate roughly doubles for each 8 °C rise of temperature [24, Synopsis, p. 776]. This rule halves the life for each fixed rise, and Dakin showed that it is the constant-coefficient approximation of the Arrhenius law, accurate at one temperature only [11, Fig. 1 and discussion of Eq. 5].
- **Fit to the framework.** Under a temperature profile that varies with the mission, the fraction of life consumed in an interval $\mathrm{d}t$ at temperature $T_i(t)$ is $\mathrm{d}t/L[T_i(t)]$. The total consumed fraction is the integral of this loss rate over the profile. This is a cumulative-damage (Palmgren–Miner) extension of the Arrhenius model [43, Eq. 2 and Fig. 10]. So the abstract description coincides with the remaining-life model of Section 2.1.3, with the mission driving the thermal load (current, cooling, ambient conditions).
- **Stochastic processes.** The same review also surveys Wiener- and Gamma-process formulations of insulation degradation [43, Sec. 5], which match directly the stochastic families of Section 2.2.

Inverter power modules. The IGBT modules of the traction inverter fail mainly by thermo-mechanical fatigue: lift-off of the aluminium bond wires and cracking of the solder layer [2, Sec. 1]. The mismatch of the coefficients of thermal expansion under repeated junction-temperature swings ΔT_j drives this fatigue [2, Secs. 1 and 4].

- **Model.** The classical model of the number of cycles to failure is a Coffin–Manson power law [2, Eq. 3]:

$$N_f = N_0 \, (\Delta T_j)^{-n}. \tag{186}$$

  An Arrhenius factor in the mean junction temperature T_{jm} refines it [2, Eq. 4]:

$$N_f = A \, (\Delta T_j)^{-n} \exp\!\big(E_a/(k_b T_{jm})\big). \tag{187}$$

  Here A, $n > 0$ and the activation energy $E_a > 0$ are fitted parameters, and k_b is the Boltzmann constant. Alavi et al. write the exponent as $\alpha = -n$ and report $\alpha = -5.039$ [2, Sec. 4]. The CIPS08 model adds further factors. Bayerer et al. published it at the 5th International Conference on Integrated Power Electronics Systems (CIPS 2008) [4, Eq. (1)]:

$$N_f = K \, \Delta T_j^{\beta_1} e^{\beta_2/(T_j+273)} \, t_{\mathrm{on}}^{\beta_3} \, I^{\beta_4} \, V^{\beta_5} \, d_{\mathrm{w}}^{\beta_6}. \tag{188}$$

  The factors are the temperature swing ΔT_j, the junction temperature T_j in °C, the power-on time t_{on}, the current I per bond foot on the chip, the blocking voltage V of the chip and the bond-wire diameter d_{w} [4, Secs. 2.2 and 3.1, Fig. 2]. Alavi et al. enter V in volts divided by 100 [2, Sec. 4]. Here K is a scale factor, and $\beta_1, \ldots, \beta_6$ are exponents fitted by multiple linear regression on the logarithm of the model [4, Sec. 3.2]. Bayerer et al. give two sets of exponents $\beta = (\beta_1, \ldots, \beta_6)$ [4, Eqs. (5)–(6)]:
  - with T_j the maximum junction temperature of the cycle,

$$\beta = (-3.483,\ 1917,\ -0.438,\ -0.717,\ -0.751,\ -0.564); \tag{189}$$

  - with T_j the minimum (starting) junction temperature, which removes a correlation between the variables [4, Sec. 3.4],

$$\beta = (-4.416,\ 1285,\ -0.463,\ -0.716,\ -0.761,\ -0.5). \tag{190}$$

  In both sets every exponent is negative except $\beta_2 > 0$. The authors note that the relation of the voltage factor to the chip thickness does not apply to modules for traction applications, whose setup and bonding differ [4, Sec. 3.1].
- **Calibration.** The model is empirical: it is limited to the range of the test data and cannot be used for extrapolation [4, Secs. 3.1 and 3.4]. Its predictions, moved from accelerated tests to real use, can be inaccurate [2, Sec. 4]. Its coefficients depend on the device type, mainly β_1 and β_2, so a new device type needs a new calibration [2, Secs. 1 and 4]. A recent fit on 27 power-cycling tests of one discrete IGBT moved the estimates of $(\beta_1, \beta_2, \beta_3)$ to $(-2.910,\ 1083.714,\ -4.521)$ [2, Sec. 5.2, Table 2]. Its 95% confidence bounds contain the first three exponents of both sets of Bayerer et al. [2, Table 2]. All these tests used one power-on time, $t_{\mathrm{on}} = 2\,\mathrm{s}$, so they cannot determine β_3. For a photovoltaic inverter, the same study also modified a second empirical model, SKiM63, by a factor that depends on ΔT_j. This change cut the predicted B10 lifetime (the time at which 10% of the units have failed) from 43.79 to 13.56 years [2, Sec. 5.3 and Fig. 15].
- **Fit to the framework.** Under a realistic mix of missions, the thermal cycles have different amplitudes. The Palmgren–Miner rule sums the accumulated damage,

$$D_{\mathrm{PM}} = \sum_i n_i / N_{f,i}, \tag{191}$$

  with failure at $D_{\mathrm{PM}} = 1$. This places the power module clearly in the remaining-life family of Section 2.1.3, with the mission determining the transmitted power and hence the spectrum of temperature swings.

Common structure. Every shortlisted component reduces to one of the two abstract descriptions of Section 2.1:

- a *monotone accumulation process* (battery cycling fade, tires, brakes, bearings; for bearings, the propagation stage ends at a remaining-life threshold);
- a *remaining-life model accumulated by the Palmgren–Miner rule* (insulation, power modules).

In both, an operational driver that the schedule controls sets the per-mission damage increment $\Delta D^{(j)}_{i\ell k}$. The drivers are energy throughput and charging rate, distance and load, braking profile, or thermal cycling. This is exactly the structure that the abstract formulation of Section 1 assumes. So the electric-vehicle fleet is a faithful concrete instance of the general problem. This analysis also fixes which degradation family describes each component of an electric-vehicle fleet.

Excluded parts. Some components deteriorate almost independently of the mission served: body corrosion, interior fittings, static low-voltage electronics and the 12-volt auxiliary battery. They satisfy criterion (C2) but not (C1). Time-based overhaul outside the optimisation suits them best, so the model does not track their damage. Their fixed maintenance windows can still enter the model as exogenous availability constraints.

The calendar ageing of the traction battery is the one borderline case. The drift extension discussed in the worked example above handles it.

4 Optimization problem for a fleet of electric vehicles

Table 22 lists the additional symbols that appear when the abstract formulation of Section 1 is specialised to a concrete fleet of electric vehicles. All symbols of Table 1 remain valid. The entries below either instantiate abstract quantities or add new ones.

Table 22: Additional notation for the electric-vehicle fleet specialisation (extends Table 1).

Symbol	Description
Cost parameters	
C_M	Cost of a depot day, paid once per member and depot day (scalar)
C_D	Damage-penalty coefficient (scalar)
$C_{R,i\ell}$	Cost of imperfect repair per unit of expected damage removed
$C_{\mathrm{rep},i\ell}$	Cost of a replacement of component ℓ of member i

4.1 Cost function

With the cost parameters of Table 22, the abstract objective of Section 1 takes the form:

$$\min \quad C_M \sum_{k=1}^{T}\sum_{i=1}^{F} x_{i0k} \;+\; \sum_{k=1}^{T}\sum_{i=1}^{F}\sum_{\ell=1}^{L} C_{R,i\ell}\, z_{i\ell k} \;+\; \sum_{k=1}^{T}\sum_{i=1}^{F}\sum_{\ell=1}^{L} C_{\mathrm{rep},i\ell}\, x^{r}_{i\ell k} \\ +\; C_D\, u. \tag{192}$$

The first three terms are weighted sums over the planning horizon. The first term counts depot days, and the second grows with the damage that each repair removes. The third term counts replacements, and the last one penalises the largest aggregate expected damage through u (see (31)).

5 Implementation details

The formulation of this document is implemented in the Python package `fleet_management`. The package builds the mixed-integer program of Section 1 for a given fleet and solves it with the commercial solver Gurobi.

The package covers the five degradation models of Table 16, both repair laws, the five Route-I bounds and the three forms of the reliability rows. It also offers tools around the solve: sweeps over the horizon lengths, warm starts, plots of a schedule, and a check of a result against its input.

The software specification `spec/spec.tex` is the reference for the implementation. It states the formulation in full, with every row, the input and output files and the tools. The code follows it. Where this document and the specification differ in a detail, the specification holds.

The subsections below give the reasons behind the implementation choices of the specification.

5.1 Big-M and indicator forms of the one-sided rows

The specification writes each one-sided row of Section 1.3.1 in one of two forms, chosen per component. In the *big-M form*, each rule of a tracked statistic s holds unless a binary switches it off through a constant M_s. The rows are the following:

$$s_k \geq s_{k-1} + \delta s_k - M_s\,(x_k^m + x_k^r) \qquad \text{(the carry row)}, \tag{193}$$

$$s_k \geq a\,s_{k-1} + (1-a)\,g_{k-1} - M_s\,(1 - x_k^m) \qquad \text{(the repair row)}, \tag{194}$$

$$|s_k - s^{\text{new}}| \leq M_s\,(1 - x_k^r) \qquad \text{(the replacement rows)}, \tag{195}$$

$$|g_k - s_k| \leq M_s\,(1 - x_k^m - x_k^r) \qquad \text{(the anchor rows)}, \tag{196}$$

$$|g_k - g_{k-1}| \leq M_s\,(x_k^m + x_k^r) \qquad \text{(the anchor rows)}. \tag{197}$$

In the *indicator form*, the same inequalities are Gurobi indicator constraints, switched on by the binary of the day's case. At binary points the two forms have the same feasible set.

The constant M_s is the *natural range* of s: an upper bound of s at every point that satisfies the reliability row of the component (for the mean at a Chernoff cell, at every point of the reference trajectory). For a variance, the Hoeffding budget and the Chernoff statistic, the value below is the largest value that the exact row admits at $\mu \geq 0$. For a mean, τ is an upper bound: the row itself, its side row or a smaller cap bounds the mean by τ. The Chernoff row does not contain the mean; there $\mu \leq K/s < \tau$ holds on the reference trajectory, because the input checks keep $K \geq s\mu$ (Jensen's inequality).

- τ for a mean and for its anchor.
- $V_{\max} = \tau^2/\kappa_G^2$ for a Gaussian or Wiener variance (Remark 2.21).
- $\varepsilon\tau^2/(1-\varepsilon)$ for a Cantelli variance.
- $\tau(\tau - 2c)/(2\kappa)$ for a Bernstein variance.
- $2\tau^2/\kappa$ for the Hoeffding budget.
- $s\tau - \kappa$ for the Chernoff statistic.

An anchor has the range of its statistic.

Validity of the big-M constants. Every tracked statistic and every anchor lies in $[0, M_s]$. The initial values, the starting anchors and the replacement values lie in $[0, M_s]$ too, by the input checks (Section 5.2). A relaxed row then cuts no point. The derivation is in Appendix A.4, page 89.

Choice of the default form. The big-M form is the default. Gurobi accepts a binary as integral within its integrality tolerance `IntFeasTol` (default 10^{-5}). A big-M row can therefore leak by at most $2 \cdot 10^{-5} M_s$, because the carry row and the anchor rows contain the sum $x_k^m + x_k^r$

of two binaries. Because M_s is the range of the statistic itself, this is a relative error of at most $2 \cdot 10^{-5}$ per row. The leaks of the carry rows of successive days add up, to at most $2 \cdot 10^{-5} T$ over the horizon. Indicator constraints have no such leak, and Gurobi usually needs more time to solve them. Which form solves this model faster has not been measured.

5.2 Reasons for the input checks

The specification validates the input before it builds the model, and a violation raises an error that names the key and the component. A check of one model runs only at the components of that model. Each item below gives a rule and, after the dash, its reason.

Keys, shapes and entries.

- Every key is known, and every required key is present. — A misspelled or missing key must not change the problem without a message.
- Each value has one of the shapes of its key. — Silent broadcasting by the array library (the automatic extension of an array to a larger shape) could read a value for another component.
- At a component where a key applies, every entry is a finite number, or a string for a selector. — A missing value at a component must not take a default silently.
- At the three keys without a default (the optional variance cap and the rates of the initial and the replacement damage), `null` at a component means that the key is not given there. — These keys can be given at some components of a model and not at others, and an array needs an entry at every component; for example, the optional variance cap at one Gaussian component only.
- An unknown value of a selector is an error at every component, also where the reliability row is linear and the form is ignored. — Reading an unknown value as `exact` would turn a linear model into a nonconvex one without a message.
- Selector values are case sensitive, like key names. — Older inputs name the law ARD_∞ with the value `ARD1` (Section 6); a comparison that ignores the case would read it as `ard1`, the other law, without a message.
- An integer key accepts a number with zero fractional part, such as 3.0, and refuses any other number, such as 2.5, and a flag. — Numerical tools often write integers as floating-point numbers, above all in HDF5 files (a binary file format for numerical data); a flag is not a number, although Python reads it as the integer 0 or 1.

Sizes and selectors.

- $F > M$. — Every mission needs a member on every day; with $F = M$ no member could be in the depot, and no repair could take place.
- $H_1, H_2 \geq 1$. — Each phase has at least one day.
- With a positive C_D and the quadratic penalty, every quadratic reliability row is in its exact form. — The quadratic penalty is a quadratic row, and the linear forms exist to give a linear model (Section 1.4).

Reliability, repair and initial state.

- $\tau > 0$. — The threshold is a positive degradation level.
- $\varepsilon \in (0, 1)$, and $\varepsilon < 0.5$ at Gaussian and Wiener components. — A probability lies in $(0, 1)$; the normal quantile $\kappa_G = \Phi^{-1}(1 - \varepsilon)$ of the Gaussian and Wiener rows is positive only below 0.5; at $\varepsilon \geq 0.5$ the squared row does not describe the requirement (Remark 2.21).
- $\rho \in (0, 1]$. — A repair removes a positive fraction of the level, at most all of it.
- The initial mean satisfies

$$0 \leq \mu_0 < \tau. \tag{198}$$

— A component at its threshold has failed already.

- Initial and replacement variances, supports and budgets are non-negative. — They describe a variance, a width or a squared width.
- The relative tangent point lies in $[0, 1)$, and the number of `pwl` segments is at least 1. — At the point τ the tangent row reduces to $\chi \leq 0$ (Gaussian, Wiener, Cantelli, Hoeffding) or admits no point (Bernstein with $b > 0$), and a value outside $[0, 1]$ has no tangent point in $[0, \tau]$.
- Each starting anchor lies between 0 and the initial value of its statistic. — Each statistic is non-decreasing between interventions (fact (M1) of Section 1.3.1), so its level after the last intervention cannot exceed its initial level.
- With the shortcut for a new part, every replacement value is 0. — The shortcut means a new part, and a non-zero value would be discarded without a message.

Natural ranges and replacement states.

- Every initial value, starting anchor and replacement value lies in the natural range of its statistic (Section 5.1). — The big-M rows take the natural range as their constant, and this constant is valid only for values in the range (Appendix A.4, page 89): with a value outside it, a switched-off row can cut feasible points.
- When replacements are allowed, the replacement state satisfies every row that holds on a replacement day, in the form built at the component (for `pwl`, the cap of the segment that contains the replacement mean), its side row, and the optional variance cap. — The rows hold on the day of a replacement, so a replacement state that violates one of them makes every replacement infeasible (Section 1.2).

Increments.

- The mean of an increment satisfies

$$0 \leq \mu^{\mathrm{inc}} < \tau \tag{199}$$

  for every mission and day. — A mission with mean τ or more fails the component in one day; the mean 0 is allowed, so that a mission that does not wear a component can be written as 0.
- Where the mean of an increment is 0, its variance, support and cumulant generating function are 0. — A non-negative increment with mean 0 is 0 almost surely, so it has no variance, no range and a zero cumulant generating function.
- Variances, supports and cumulant generating functions of increments are non-negative, and a transitory profile comes with its operating profile. — A variance and a width are not negative, and the lookup of the increment laws needs the operating laws after the transitory phase (Section 1.2).

Model-specific checks.

- Gaussian: $\mu \geq 3\sqrt{v}$ for the initial damage, for every increment and, when replacements are allowed, for the replacement draw. — A Gaussian term is then negative with probability at most about 0.13 %, so the damage stays non-negative in practice.
- Inverse Gaussian: $\eta > 0$. — The squared coefficient of variation of the inverse Gaussian terms must be positive.
- Wiener: $\sigma > 0$ and $v_0 \geq 0$. — The diffusion constant is positive; a known initial damage has variance 0 and is a valid input.
- Gamma: every given rate is finite and positive. — The rate of a Gamma law is positive, and the smallest rate sets the cap (Section 2.2); an omitted rate of the initial or replacement damage means a known damage, which has no Gamma law and no rate.

- Optional variance cap: positive, and at least the initial variance and, when replacements are allowed, the replacement variance. — A cap below the replacement variance makes every replacement infeasible, and a cap below the initial variance forces an intervention on day 1.
- Bernstein: every variance is at most a quarter of the squared width of its range, for increments, the initial damage and the replacement draw. — Popoviciu's inequality (Section 2.1.4): a law on a range of width w has variance at most $w^2/4$, so a larger variance contradicts the width.
- Bernstein: $\tau > 2\kappa\, b_\ell/3$. — Otherwise the certificate admits no positive variance: it is infeasible at zero damage, and at equality only the point

$$\mu = v = 0 \tag{200}$$

  survives, so that the big-M rows break.
- Bernstein with a linear form: the tangent point, or the midpoint $\tau/(2P)$ of the first `pwl` segment, lies below $\sqrt{\tau(\tau - 2c)}$. — Otherwise no point with $v > 0$ satisfies the row (Section 2.1.7).
- Chernoff: $s > \ln(1/\varepsilon)/\tau$. — The statistic K is not negative, so the row $K - s\tau \le \ln\varepsilon$ needs $s\tau \ge \ln(1/\varepsilon)$; with a smaller s every schedule is infeasible, and at equality only a component without damage is feasible.
- Chernoff: the cumulant generating function of every increment, the initial value, the starting anchor under ARD_1 and, when replacements are allowed, the replacement value are at least s times the corresponding mean. — Jensen's inequality (Section 2.1.4); the two starting anchors describe the same damage after the last intervention, and the defaults satisfy the inequality.

Run options and costs.

- The evaluation horizon, when given, exceeds H_1. — The projected cost needs at least one operating day beyond the transitory phase; for a smaller horizon the operating cost would have a negative weight (Section 1.4).
- Numerical run options lie in their ranges, and flags are flags. — A value outside its range must raise an error and must not be changed silently.
- All costs are finite and non-negative. — A negative cost would reward a depot day, a repair, a replacement or damage.

Tolerance of the checks.

- A non-strict inequality $a \le b$ passes when it holds up to $10^{-12}\max(1, |a|, |b|)$. — The rounding of floating-point numbers can break an exact comparison on valid data. For example, with $s = 5$ and $\mu_0 = 0.14$, the Chernoff initial value $K_0 = 0.7$ fails $K_0 \ge s\mu_0$ without a tolerance, because $5 \cdot 0.14 = 0.7000000000000001$ in floating point. The rounding error grows with the size of the numbers, so the tolerance is relative; the term 1 keeps the tolerance 10^{-12} near 0.
- A strict inequality has no tolerance. — It excludes its boundary on purpose: for example, a component with $\mu_0 = \tau$ has failed already.
- A negative value of the run option `mip_gap` (the relative gap at which Gurobi stops) that passes through the tolerance is stored as 0. — Gurobi refuses a negative gap. Other values that pass through the tolerance reach the model unchanged, for example an initial variance of -10^{-13}; the feasibility tolerance of Gurobi, 10^{-6}, absorbs them.

5.3 Outputs

- **Solution and status.** The solution arrays are present whenever Gurobi has found at least one solution, and then the status tells how the solve ended. For example, a solution found

before the time limit is reported with the status `time_limit`. With the default MIP gap of 0.1 (MIP means mixed-integer program), the status `optimal` means that the best solution exceeds the best bound by at most 10 % of its own value, that is, by at most about 11 % of the bound.

- **Tracked statistics.** The output carries every tracked statistic and anchor, so that a validator can check every loop row. The dynamics rows are one-sided, so these values, and the costs computed from them, are upper bounds, not exact values (Section 1.3.1).
- **Objective value.** When the variable u has no slack, the objective value of the solve equals the cost of the objective mode computed from the output. Otherwise it is larger by C_D times that slack.
- **Result file.** The extension and the directory of the result file are checked before the model is solved, so that a long solve is not lost.
- **Numbers that are not finite.** YAML and JSON files (two plain-text data formats) write NaN (not a number), $+\infty$ and $-\infty$ as `null`. Standard JSON has no notation for these values, and the two text formats follow one rule. HDF5 files store floating-point numbers in binary form and keep these values.
- **Objective without a value.** HDF5 has no `null`, so an HDF5 file does not store the objective of a solve without a solution. Every reader treats a missing key as `null`, so the file carries the same information as a YAML or JSON file.
- **One reader.** One function, `read_result`, reads a result file of every format for every tool: the schedule plot, the validators and the dashboards. These tools need the same reading rules, for example NaN at a `null` entry of an array, so the rules have one implementation. The function sits in the module of the writers, so the rules for writing and for reading a result file are in one place.

5.4 Horizon sweeps

A single solve uses one horizon pair $H = [H_1, H_2]$. A *horizon sweep* solves one scenario for several pairs and compares the solved cases. It ranks the cases by the operating average

$$\bar{J}_{\mathrm{op}} = J_{\mathrm{op}}/H_2, \tag{201}$$

the cost per day of the repeating regime, which is comparable across values of H_2 (Section 1.4). A sweep over both horizons with an evaluation horizon E ranks by the projected cost

$$P_E = J_{\mathrm{init}} + \frac{E - H_1}{H_2}\, J_{\mathrm{op}}, \tag{202}$$

which is the cost of the transitory phase, then of the operating phase repeated until day E. This criterion also prices the transitory phase, so it compares different values of H_1.

- **What a proof certifies.** A case counts as proven when Gurobi reports `optimal` at the MIP gap requested for that case. Such a case is optimal within that gap, not exactly optimal, and the sweep report records the requested gap of each case. The status certifies the objective that the solve minimised: $\bar{J}_{\mathrm{op}} + C_D\, u$ or $P_E + C_D\, u$, while the ranking value is $\bar{J}_{\mathrm{op}}$ or P_E alone. So the sweep ranks by cost and uses the damage penalty only inside each solve.
- **Early stop.** With the default threshold, a case qualifies for the early stop when Gurobi has met the gap requested for it. A case that does not qualify, for example one that stopped at the time limit above its requested gap, is recorded, but it cannot trigger a stop.
- **Parallel cases.** Each running solve uses one Gurobi licence token, so a licence with one token allows one case at a time. The parallel cases do not use each other's results. So the parallel option is refused together with warm starts between cases, which need the preceding results. It is also refused together with the early stop, which needs the preceding cases before it starts the following ones.

- **Report.** The report is a mapping of plain types, written by a safe writer of YAML (a plain-text data format), so it contains no arrays and no program objects. If a job ends before the run is complete, the checkpoint file is a valid report of the cases evaluated so far.

5.5 Warm starts

A warm start changes the time of a solve, not the model: it does not change the feasible set or the objective.

- **Day mapping.** A start taken from a previous result maps its days to the days of the new problem.
  - The transitory days keep their index.
  - The operating days repeat the operating phase of the source cyclically, so a shorter operating phase can seed a longer one.
  - When the new transitory phase is longer than the source's, its extra days all take the first operating day of the source.

  When the source has the same sizes F, M, L, the mapped schedule copies whole days of the source, so it keeps the one-activity rows, the mission coverage and the depot-day rows of every day. It need not satisfy the dynamics or the loop rows of the new horizons.
- **Invalid source.** A source whose horizons are not integers of at least 1, or whose arrays do not have the days of these horizons on their last axis, raises an error. The day mapping needs the horizons of the source, and values taken from the new problem in their place would map the days wrongly without a message.
- **Start.** A start sets values of the binaries only. Gurobi tries to complete it by itself, with limited effort, and discards a start that it cannot make feasible. The greedy heuristic returns its start also when its loop pass counts failures, and it requests the completion of the specification. That completion can then end with the status `infeasible`, with an irreducible infeasible subset of rows.
- **Completion report.** In the mode `hint` the completion never runs, although the greedy heuristic requests it. The report then gives the status `not_requested`, the status of every completion that does not run, so no further status is needed. Its flag `requested` still shows the request of the source.
- **Hints.** Gurobi uses hints to guide its heuristics and its branching during the whole solve. A hint needs no feasible point, so it also helps when the mapped schedule does not fit the horizons of the problem and Gurobi discards it as a start.
- **Relaxation start.** The relaxed solution need not be a feasible point of the original model. It gives integer values only, and the original model must find its own first solution.
- **Greedy heuristic.** The heuristic builds a schedule that tries to satisfy the rows of the model. It does not try to give a low cost. After its construction pass the missions are fixed, so the components are independent except through the depot days of a member.
  - **Rows in the excess.** The excess of a component counts the violations of the reliability row, of its side row and of the optional variance cap. The tracked values are at least the reference values (Proposition 1.3), and each of these rows sets upper limits on them. So a schedule whose reference trajectory breaks one of these rows has no feasible completion.
  - **Reference trajectory.** The excess is computed on the reference trajectory. This test is stricter than the one-sided rows, because the reference trajectory of a feasible schedule can violate the loop row (Section 1.3.3). In a Gaussian test case the heuristic counts one failure, and the same schedule is feasible for the rows of the model.
  - **Wiener components.** A free day is an operating day on which the member flies no mission and the component has no intervention. At a Wiener component the heuristic tries a repair and a replacement on every free day, and it keeps the intervention with

the smaller excess. At the components of the other models it tries repairs first, and a replacement only when no repair lowers the excess. The Wiener variance falls only at a replacement (Section 1.3.4), so a repair placed first can take the last free day of a component that needs a replacement. Example: a Wiener test case with $H = [2, 2]$. A repair on day 4 removes the excess of the mean and takes the only free day; the excess of the variance remains, and each of the two members counts one failure. A replacement on that day removes both excesses.

– **Report.** The report gives the largest ratio of a reference mean to its threshold on the final trajectory of each component. So the ratio describes the schedule that the heuristic returns.

5.6 Plots

- **Readers.** The schedule plot reads its file with the reader of the result files (Section 5.3), and the sweep plot reads its reports with the input reader. A file whose data are not a mapping raises the type error of the reader, so every way of reading a file has one error for this case.
- **Result without models.** A result file without the key of the models is plotted, and each row shows the model "unknown". Older result files lack this key, and every solve writes it.
- **Counts from the Gurobi log.** The summary panel of the schedule plot takes the presolved numbers of rows and variables from the last line "Presolved" of the Gurobi log. The completion of a warm start and the relaxation start solve their own copies of the model before the main solve, and they write to the same log. So the last such line belongs to the main solve.
- **Sweep without cases.** A sweep report whose list of cases is empty raises an error. There is nothing to plot, and a sweep writes its first report only after its first case.

5.7 Validation tools

- **Objective check.** The saved damage penalty u is the maximum of its rows. The solver objective contains the variable u, which can lie above this maximum in a solution that is not optimal. So the check is one-sided: it tests that the reported objective is not below the cost of the reported solution, and it does not test optimality.
- **Loop checks.** They use the saved tracked states, which satisfy the loop rows, and not the reference trajectory: the reference trajectory of a feasible schedule can violate the loop row (Section 1.3.3).
- **Scope of the structural validator.** It checks the binaries, the assignment and depot-day rows, the loop rows of every tracked statistic and anchor, and the objective. It does not check the dynamics rows or the reliability rows; the dashboards show the reliability rows as margins.
- **Fixed list of checks.** The report always lists the eight loop checks, with the violation 0 where the result holds no number for the statistic. So every report has the same 14 checks, or 13 for a result without the array of idle days.
- **Incomplete results.** A result with the schedule but without another solution array, the damage penalty or the objective raises an error. No solve writes such a result, and without the error the report would hold NaN, which the JSON writer refuses.
- **Margins in the dashboards.** A margin of at least the negative tolerance means that the row holds. The exact form is used also when the solve used a linear form, so the margin shows the true slack. At a row with a side row, the margin is the smaller of the two slacks, because the row holds if and only if both relations hold. The slack of the quadratic relation alone can be positive when the side row fails: a Gaussian component with $\mu = 1.2$, $\tau = 1$ and $v = 0$ has $(\tau - \mu)^2 = 0.04 > 0$. The two slacks have different units. The margins of

components with different models and bounds are not comparable in size. Only their signs are comparable.

- **Model of the result.** The dashboards and the Gamma validators raise an error when the model of a component in the result differs from the input. The structural validator does not compare the models. The margins and the Gamma validators need the parameters that the input gives for the model of each component, and the structural validator needs no model.
- **Tolerance of the dashboards.** The tolerance is at least 10^{-12}, because the Gamma validators that the dashboards run need a positive tolerance.
- **Start of the dashboards.** The entry script draws the page only when Python runs it as its main program. It imports Streamlit (a Python library for web dashboards) only inside the functions that draw. The command `streamlit run` executes the script as the main program, so the page is drawn. An import of the script, by the package or by a test, draws nothing and needs no Streamlit, which is an optional part of the installation.
- **Repair cost in the replay Gamma validator.** The check of the repair cost uses the saved means and anchors, not the reference values. This is the row of the model, and at binary points the saved difference between the mean and the anchor is at least the reference difference, under both repair laws. So the check on the saved values also implies the check on the reference values.
- **Exact Gamma validator.** It computes the tail $\Pr(D_k > \tau)$ of the sum of the terms with the Moschopoulos series.
  - **Mixture.** The series writes the law of a sum of independent Gamma variables with different rates as a mixture of Gamma laws with non-negative weights. When all rates are equal, the tail is the Gamma tail.
  - **Certified bound.** The partial sum of the series is a lower bound of the tail, so adding the mixture weight that remains gives a certified upper bound.
  - **Saved means.** The saved means may be larger than the exact ones, because they are upper bounds.
  - **Initial damage under ARD_1.** The initial damage is split at the starting anchor into a part that no repair reduces and a part that the repairs reduce. Otherwise a repair could bring the damage below its level after the last intervention, which the law ARD_1 excludes.
  - **Cost.** The number of terms of the series grows with the ratio of the largest to the smallest rate. Frequent repairs under ARD_∞ make this ratio large, because each repair multiplies the rates of the repaired terms by $1/(1-\rho)$. Example: one term repaired 2, 4, 6, 8 and 10 times at $\rho = 0.5$, beside a new term, needs 151, 725, about 3,000, about 12,400 and 49,884 terms. One tail then takes about 2 s after 8 repairs and about 31 s after 10. From 11 repairs the series needs more than 100,000 terms, the default limit, and the validator stops with an error.
  - **Precision.** The series stops when the mixture weight that remains falls below the convolution tolerance, 10^{-12} by default. Its sums use the extended precision of NumPy, the numerical library of Python: 80 bits on Linux, and the usual 64 bits on Windows. With 80 bits the series can reach a convolution tolerance near 10^{-19}. With 64 bits the remaining weight is understated by $2 \cdot 10^{-15}$ to $9 \cdot 10^{-15}$ after 2, 4 and 6 repairs in the example above. This is about 1 % of the default convolution tolerance, and far below the default tolerance 10^{-8} with which a day passes. The convolution tolerance has no lower limit: a value that the series cannot reach ends in an error.
- **Monte Carlo validator.** Its result is an empirical stress test and not a proof: zero failures in n runs does not show that the true failure rate is below $1/n$. It raises no error for a violation of the schedule rows. The mode `both` runs the replay and the Monte Carlo validators together, and it reports the schedule through the replay validator.

- **Settings of the Gamma validators.** An unknown key in the input block of these settings raises an error. The input checks test only that the block is a mapping, so a misspelled key, such as `repetition` for `repetitions`, would otherwise be ignored without a message.

5.8 Package and inputs

- **Modules.** The package has one module per task. The shared Gurobi model and the registry of component builders are in one module, and each degradation model has its own module.
- **Modules without Gurobi.** The inverse Gaussian and Gamma modules compute their caps with NumPy and SciPy only, and they import the module of the shared Gurobi model only when Gurobi is installed. So each cap lives in the module of its model. The input check of the replacement state calls the same cap function as the row, also on a computer without Gurobi. The rainflow module loads in the same way, so the constants of its reliability rows can be tested without Gurobi. Without Gurobi these modules register no builder.
- **One route for every fleet.** Every fleet, whatever its models, goes through the same route: no model has a separate solver. Every builder defines the tracked mean on every day for each of its components, because the damage penalty and the output read it. After the solve, a shared extraction reads the variables, and then the extraction of each model runs.
- **Cluster scripts.** The cluster scripts call only the public functions of the package, so the behaviour that the specification defines does not depend on them.
- **Plots and dashboards.** The package has one schedule plot and one sweep plot. The schedule plot colours each component by its tracked mean divided by the failure threshold. The tracked means are upper bounds of the expected damage (Section 1.3.1). The dashboards are Streamlit pages (Streamlit is a Python library for web dashboards).
- **Run options.** Run options are keys of the input. They do not change the mathematical problem, except the objective mode and the evaluation horizon, which choose the objective. A horizon sweep is not an input option: the sweep functions read the same input and solve it for several horizons.
- **Input parser.** The input parser rejects old and unknown keys, because each of them names a quantity that has another name, or another meaning, in the specification. It cannot detect an array whose mission axis and component axis are swapped when $M = L$.

The input example of the specification has $F = 3$ members, $M = 2$ missions and $L = 2$ components. Its six components use the Gaussian, inverse Gaussian, Wiener and Gamma models, and the rainflow model with the Bernstein bound and with the Chernoff bound. It passes every input check.

- **Inverse Gaussian component.** It uses ARD_1 with the starting anchor

$$0.06 \le \mu_0 = 0.10. \tag{203}$$

- **Gamma component.** It has the starting anchor

$$0.05 \le \mu_0 = 0.08 \tag{204}$$

  and no rate for the initial damage, so its initial damage is known.
- **Chernoff component.** It has

$$K_0 = 5 \cdot 0.14 = 0.70 \tag{205}$$

  by default, so its starting anchor 0.50 is allowed.
- **Bernstein component.** It has the support constant

$$b = \max(0.05, 0.03, 0.05) = 0.05, \tag{206}$$

so

$$2\kappa b/3 = 0.077 < \tau = 1, \tag{207}$$

with $\kappa = \ln 10$.

5.9 Open questions

These are the design questions that the formulation leaves open. The planned work of the project is listed in its to-do file, `Things to change.txt`.

- **Hierarchical failure-probability bounds.** Every reliability row bounds the failure probability of one component:

$$\Pr(D_{i\ell k} > \tau_{i\ell}) \le \varepsilon_{i\ell} \quad \text{for all } i, \ell, k. \tag{208}$$

  A user may want to state the requirement at a coarser level.

  – **Member level:**

$$\Pr(\exists\, \ell : D_{i\ell k} > \tau_{i\ell}) \le \varepsilon_i \quad \text{for all } i, k, \tag{209}$$

    the probability that *any* component of member i fails. Under independence of the components this reads

$$1 - \prod_{\ell} (1 - \Pr(D_{i\ell k} > \tau_{i\ell})) \le \varepsilon_i, \tag{210}$$

    a nonlinear row that couples the components of a member.

  – **Fleet level:**

$$\Pr(\exists\, i, \ell : D_{i\ell k} > \tau_{i\ell}) \le \varepsilon \quad \text{for all } k, \tag{211}$$

    the probability that *any* component of the fleet fails. Under independence this row couples all members.

  Both levels need either approximations by the union bound, which are conservative and linear, or rows in product form, which are exact and nonlinear. The union bound is $\Pr(A \cup B) \le \Pr(A) + \Pr(B)$. The per-component levels give both through the union bound: $\varepsilon_{i\ell} \le \varepsilon_i / L$ for every ℓ gives the member level, and $\varepsilon_{i\ell} \le \varepsilon/(FL)$ for every component gives the fleet level.

- **Sampling error of the empirical inputs.** Rainflow counting yields *estimates* of the per-mission mean, variance, support width and cumulant generating function, not exact values. A guaranteed reliability bound needs one-sided confidence margins on these inputs.

- **Gamma cap over the reachable states.** The sharp cap of Section 2.2 is the largest cap that is safe for every member of the class $\mathcal{G}(m)$. The states that a schedule can reach form a smaller set, because the tracked mean also satisfies the cap on every earlier day. For a component with one rate and $\beta^* \tau$ below the threshold $t_1(\varepsilon)$, a larger safe cap may exist. Its value is not known.

6 Changelog of the software specification

This section records the versions of the software specification `spec/spec.tex`. Versions 0.3 to 0.5 use older names and an older horizon. Their law ARD_1 is the law ARD_∞ of this document, and their law ARA_1 is the law ARD_1 of this document. Their horizon has $2H$ days, and the second H days are the schedule to repeat.

Version 0.3 (panel review).

- The ARA_1 maintenance was removed from the Gaussian model, because no source in the literature supports its extension to the variance.
- The output `v` has the shape $F \times L \times 2H$, with NaN (not a number) at the components without a variance.
- The bound on ε became tighter: from $(0, 0.5)$ to $(0, 0.01]$.
- A verified proposition, with a proof by the monotone likelihood ratio (MLR), replaced the open item on first-order stochastic dominance (FSD) of the inverse Gaussian (IG) model.
- The remark on tightness became an argument with three mechanisms.

Version 0.4 (rainflow extension).

- The rainflow degradation model was added. It is distribution-free, and its inputs are the empirical mean, variance and support of each mission, from rainflow counting. A Cantelli or a Bernstein tail bound certifies the reliability; the new option `tail_bound` selects the bound, with the default `"cantelli"`. The specification gives both an exact nonconvex quadratic formulation and a conservative linear tangent formulation. The loop constraint acts on both tracked moments. The argument is that the componentwise domination of the moments chains across repetitions, while the scalar comparison of the bounds, $U_{2H} \leq U_H$, does not. The model accepts ARD_1 only, as the Gaussian model did.
- The value `"socp"` of the key `formulation` was renamed `"exact"`. `"socp"` remained as a deprecated alias, with a `UserWarning`.
- The exact reliability constraint of the Gaussian and Wiener models had been called a rotated second-order cone (SOC); this was corrected. The variance is affine in the decisions, so the squared form is a nonconvex quadratic constraint, handled by the `NonConvex=2` support of Gurobi. The `"quadratic"` damage penalty remains a truly convex SOC constraint.
- The wording of the Wiener loop constraint was corrected after the accepted review item A-02: the constraint bounds the moments; it is not first-order stochastic dominance.
- The change log of version 0.3 above describes the revision that the title page had called "v0.2 (revised)".
- A panel review after the draft (markers RF-042–RF-045 and corrections made in place) made further changes.
  - It fixed the base case of the induction over the loop ($m \geq 2$, with the loop row as the case $m = 1$).
  - It fixed the wording at the boundary of the precondition $\tau > 2\kappa b/3$.
  - It fixed the centring step in the application of Bernstein's inequality, which holds a fortiori (it follows from a stronger fact).
  - It fixed the statement of the problem class for fleets whose components are all inverse Gaussian or Gamma.
  - It extended the consistency checks based on the Bhatia–Davis inequality (a bound on the variance from the mean and the range) to the initial and replacement states.
  - It extended the list of checked array shapes to b^{inc}, b^0 and b^{new}.
  - It moved `tail_bound` to the input tables of the models.
  - It added the fifth border style of the plot.
  - It corrected the value of ε in the YAML example, which did not respect the bound $(0, 0.01]$ of the specification.

Version 0.5 (removal of the loop penalty).

- The soft penalty of the loop constraint, $\mathcal{L}_{\mathrm{loop}}$, and its coefficient C_P were removed from every place: the objective, the penalty formulas of the models, the total penalty sum, the input tables, the YAML example and the statement of the full problem. The hard loop constraints of all five models did not change.

- The remark on the tightness of the big-M rows had named three mechanisms. It was revised to name the two cost mechanisms that remain, $C_D\,u$ and $C_R\,z$.
- An argument on robustness to slack was added. The tracked states are componentwise upper bounds on the true moments, and each one-sided dynamics row dominates the exact monotone map of one day. So the hard loop constraint gives $\Pi(s_H) \le s_H$ for the exact map Π of one repetition of the second half (Section 1.3.2; that version wrote Φ for this map). This inequality is the premise of the induction in the repeatability argument. The monotonicity of the reliability certificates then keeps every future repetition certified, even when the tracked states carry slack. The only role of the penalty had been to break ties in favour of larger sustainability margins.
- A later review (markers REV-001–REV-022) closed a gap of the loop constraint under ARA_1: the loop did not cover the anchors. It added the hard rows

$$\mu_{2H}^{\text{last}} \le \mu_H^{\text{last}} \qquad \text{(Wiener with } \mathrm{ARA}_1\text{)}, \tag{212}$$

$$\alpha_{2H}^{\text{last}} \le \alpha_H^{\text{last}} \qquad \text{(Gamma with } \mathrm{ARA}_1\text{)}. \tag{213}$$

  A derivation of monotonicity on the full state shows that these rows restore the chaining premise $\Pi(s_H) \le s_H$. The review extended the remark on robustness to slack in the same way.
- The same review made further changes.
  - It stated that an idle day is a row of zeros in the assignment, so the term with C_M is a real decision cost, not a constant.
  - It corrected the day index of the anchor in the ARA_1 repair-cost row ($k \to k-1$).
  - It removed the old duplicate parameter `alpha_target`, a former name of τ.
  - It made consistent which keys are optional: `tail_bound` and the input fields that have a default.
  - It added the derived constants to the notation table.
  - It expanded the two items on future work of the rainflow model, with pointers to the Route-I refinements of this document (Section 2.1.6).
  - It recorded two questions on the convention of the ARA_1 anchor as open items: whether the anchor stores the state before or after the repair, and the missing reset of the anchor at a replacement.

Version 1.0 (merged package).

- The specification describes the package that merges the branches `lore` and `main`, on the code base of `main`. It holds only the codebase and its mathematical constructions; the reasons, the proofs and the sources are in this scoping document.
- The five degradation models are cell builders in one registry; a cell is component ℓ of member i.
- The repair laws are ARD_∞, which reduces the whole degradation, and ARD_1, which reduces the degradation since the last intervention.
- The horizon has a transitory phase and an operating phase, $H = [H_1, H_2]$, and the loop row bounds the full tracked state, anchors included.
- Every dynamics row is one-sided, in big-M or indicator form.
- The rainflow model offers five tail bounds, and the quadratic reliability rows have an exact, a tangent and a piecewise-linear form.
- The inverse Gaussian row is a cap from a conditional tail mean. The Gamma row is a cap on the tracked mean that holds for any mix of rates.
- The reliability level, the tangent point, the repair cost and the replacement cost are set per cell.
- The specification covers the horizon sweeps, the warm starts, the schedule plot and the validators.

- The input keys follow one table, and unknown or old keys raise an error.
- The input checks give every non-strict inequality the relative tolerance $10^{-12}\max(1, |a|, |b|)$, and they compare selector values with their case. At a key without a default, `null` means that the key is not given at that cell. YAML and JSON result files write every number that is not finite as `null`.
- The specification fixes the numerical methods of the two caps: a root search in $\ln q$ for the inverse Gaussian quantile, and the grid, the golden-section search and the two bisections of the Gamma cap A_1.
- One function, `read_result`, reads the result files for every tool. The greedy heuristic tries a repair and a replacement on every free day of a Wiener cell.

A Proofs and derivations

A.1 Overview of the problem

Derivation that $\mathcal{C}_{i\ell}$ is a closed down-set. Every reliability row, written in chance-bound form, reads "a continuous function of the tracked statistics, non-decreasing in each of them, is at most a constant" (Table 2). So the set $\mathcal{C}_{i\ell}$ is closed, and it is a down-set. □

Proof of Lemma 1.2. Let $\xi \le \xi'$ componentwise. Each coordinate of π_k has one of the forms $s + \delta s_k$, g, s, $a\,s$, $a\,s + (1-a)\,g$ or s^{new}. By (M1), $a \ge 0$ and $1 - a \ge 0$. Therefore each coordinate of $\pi_k(\xi)$ is at most the same coordinate of $\pi_k(\xi')$. □

Proof of Proposition 1.3. At binary points the rows of day k read: $s_k \ge \pi_k(\xi_{k-1})_s$ for every tracked statistic s, with equality on a replacement day;

$$g_k = s_k \ge \pi_k(\xi_{k-1})_g \tag{214}$$

on an intervention day;

$$g_k = g_{k-1} = \pi_k(\xi_{k-1})_g \tag{215}$$

on any other day. So $\xi_k \ge \pi_k(\xi_{k-1})$ for $k \ne H_1 + 1$. On day $H_1 + 1$ the rows read y, so $\xi_{H_1+1} \ge \pi_{H_1+1}(y)$ by (18), and $\pi_{H_1+1}(y) \ge \pi_{H_1+1}(\xi_{H_1})$ by (17) and Lemma 1.2. So $\xi_k \ge \pi_k(\xi_{k-1})$ on every day. Let

$$\theta_{k-1} \le \zeta_{k-1} \le \xi_{k-1} \tag{216}$$

(true for $k = 1$ by (M2), since $\zeta_0 = \xi_0$). Then, by (M2) and Lemma 1.2,

$$\theta_k \le \pi_k(\theta_{k-1}) \le \pi_k(\zeta_{k-1}) = \zeta_k \le \pi_k(\xi_{k-1}) \le \xi_k. \tag{217}$$

Therefore

$$\theta_k \le \zeta_k \le \xi_k \tag{218}$$

for every k, by induction. □

Derivation of the reliability on the days of the horizon. On every day $k = 1, \dots, T$ the tracked values satisfy the reliability row in the form chosen for the component. Every form lies inside $\mathcal{C}_{i\ell}$ (Proposition 2.3), so $\xi_k \in \mathcal{C}_{i\ell}$. By Proposition 1.3, $\theta_k \le \xi_k$, so (M3) gives $\Pr(D_{i\ell k} > \tau) \le \varepsilon$ on the days of the horizon. □

Proof of Theorem 1.4. Fix a component. Let ξ_k be its full tracked state, and θ_k and ζ_k the true statistics and the reference trajectory of the endless schedule. By (M2) and Lemma 1.2, the induction of Proposition 1.3 gives $\theta_k \le \zeta_k$ on every day $k \ge 1$.

- Days $k \le T$: $\zeta_k \le \xi_k$ (Proposition 1.3) and $\xi_k \in \mathcal{C}_{i\ell}$ (Section 1.3.1).
- By Proposition 1.3 and (17),
$$\zeta_{H_1} \le \xi_{H_1} \le y. \tag{219}$$
- By (18), Proposition 1.3 and Lemma 1.2,
$$\xi_{H_1+h} \ge \pi_{H_1+h}(\xi_{H_1+h-1}) \ge \cdots \ge \Pi_h(y) \tag{220}$$
for $h = 1, \dots, H_2$. With (19),
$$\Pi(y) \le \xi_T \le y. \tag{221}$$
- Let $\Pi^n(y) \le y$ (true for $n = 0$). Then
$$\Pi^{n+1}(y) = \Pi(\Pi^n(y)) \le \Pi(y) \le y \tag{222}$$
by Lemma 1.2. So $\Pi^n(y) \le y$ for every $n \ge 0$.
- Day $k = H_1 + nH_2 + h$ with $n \ge 0$ and $1 \le h \le H_2$:
$$\zeta_k = \Pi_h\big(\Pi^n(\zeta_{H_1})\big) \le \Pi_h\big(\Pi^n(y)\big) \le \Pi_h(y) \le \xi_{H_1+h} \in \mathcal{C}_{i\ell}. \tag{223}$$

Therefore, on every day $k \ge 1$,
$$\theta_k \le \zeta_k \le s \tag{224}$$
for a tracked value $s \in \mathcal{C}_{i\ell}$. Since $\zeta_k \ge 0$ and $\mathcal{C}_{i\ell}$ is a down-set, $\zeta_k \in \mathcal{C}_{i\ell}$, and (M3) gives $\Pr(D_{i\ell k} > \tau) \le \varepsilon$. □

Derivation that the loop constraint needs the anchors. The third step of the proof of Theorem 1.4 needs $\Pi(y) \le y$ in every coordinate. Under ARD_1 a larger anchor gives a weaker repair, so an anchor that grows over one repetition weakens the repairs of the next one. Example:

- **Data.** $\rho = 0.5$; the repetitions start at mean 0.4 and anchor 0.1 (the start y); the operating phase is a mission (+0.1), a repair, and a mission (+0.05).
- **First repetition.** The phase ends at mean 0.35 and anchor 0.3, because the repair gives $0.5 \cdot 0.5 + 0.5 \cdot 0.1 = 0.3$. The loop row on the mean holds ($0.35 \le 0.4$), and the loop row on the anchor fails ($0.3 > 0.1$).
- **Later repetitions.** Each repetition raises the mean after the repair by 0.075, so the mean grows without bound: the third repetition already peaks at 0.525, above the checked peak 0.5.

A loop row on the mean alone would accept this unsafe schedule. □

Proof of Proposition 1.5. (a) The values $\xi_k = \zeta_k$ satisfy every row of day k with equality at binary points. They lie in $\mathcal{C}_{i\ell}$ because the schedule is safe. Take $y = \bar{\zeta}$: the start rows hold with equality, the rows of day $H_1 + 1$ read $\zeta_{H_1+1} = \pi_{H_1+1}(y)$, and the loop rows read
$$\zeta_T = \Pi(\bar{\zeta}) \le \bar{\zeta} = y. \tag{225}$$

(b) Let $y_n = \Pi^n(\bar{\zeta})$, the reference state at the end of the n-th repetition. By Lemma 1.2 and $\Pi(\bar{\zeta}) \ge \bar{\zeta}$,
$$y_{n+1} = \Pi(y_n) \ge \Pi(y_{n-1}) = y_n. \tag{226}$$
The sequence is non-decreasing and lies in the natural ranges (anchors included), so it converges to some y^* in the natural ranges with $y^* \ge \bar{\zeta}$. Each day map is affine or constant, so Π and its prefixes Π_h are continuous:
$$\Pi(y^*) = \lim_n y_{n+1} = y^* \tag{227}$$
and
$$\Pi_h(y^*) = \lim_n \Pi_h(y_n). \tag{228}$$

Each $\Pi_h(y_n)$ is a reference value of a later day, so it lies in the closed set $\mathcal{C}_{i\ell}$; hence $\Pi_h(y^*) \in \mathcal{C}_{i\ell}$ for $h = 1, \dots, H_2$, and

$$y^* = \Pi_{H_2}(y^*) \in \mathcal{C}_{i\ell}. \tag{229}$$

Take $\xi_k = \zeta_k$ for $k \le H_1$, $y = y^*$ and $\xi_{H_1+h} = \Pi_h(y^*)$. The days $k \le H_1$ satisfy their rows with equality. The start rows hold, since $y^* \ge \bar{\zeta} = \xi_{H_1}$. The rows of day $H_1 + 1$ and of the later days hold with equality, and the loop rows read

$$\xi_T = \Pi(y^*) = y^* = y. \tag{230}$$

□

Derivation of the acceptance of every safe schedule for one tracked statistic. A scalar state y is always comparable with its image: $\Pi(y) \le y$ or $\Pi(y) \ge y$. So a component in the exact form with one tracked statistic under ARD_∞ accepts every safe schedule, with no condition on day H_1. □

Proof of Proposition 1.8. Suppose that the component is not replaced in the operating phase. Let μ_k be its tracked mean, g_k the anchor of the mean, and $\delta\mu_k$ the mean increment of day k. The rows of day $H_1 + 1$ read the start y (Section 1.3). So let $\bar{\mu}_{H_1} = y^\mu$ and $\bar{g}_{H_1} = y^g$ be the state before day $H_1 + 1$, and $\bar{\mu}_k = \mu_k$, $\bar{g}_k = g_k$ for $k > H_1$. On a day $k > H_1$ of the operating phase, a repair day gives

$$\bar{g}_k = \bar{\mu}_k \;\ge\; (1-\rho)\,\bar{\mu}_{k-1} + \rho\,\bar{g}_{k-1} \;=\; \bar{g}_{k-1} + (1-\rho)\,(\bar{\mu}_{k-1} - \bar{g}_{k-1}), \tag{231}$$

and any other day gives $\bar{\mu}_k \ge \bar{\mu}_{k-1} + \delta\mu_k$ and $\bar{g}_k = \bar{g}_{k-1}$. The loop rows read $\mu_T \le y^\mu$ and $g_T \le y^g$. Let $\Delta = \sum_{k=H_1+1}^{T} \delta\mu_k > 0$.

- **No repair.** $\mu_T \ge y^\mu + \Delta > y^\mu$, which violates the loop row.
- **A repair, and $y^g > y^\mu$.** Let r be the first repair day. Since $\bar{\mu}_{r-1} \ge y^\mu$ and $\rho > 0$,

  $$g_r \;\ge\; (1-\rho)\,\bar{\mu}_{r-1} + \rho\,y^g \;>\; y^\mu. \tag{232}$$

  From day r on, $\bar{\mu}_k \ge \bar{g}_k$ (equality on a repair day; μ does not decrease and g stays constant on other days). So every later repair day has a non-negative term $\bar{\mu}_{k-1} - \bar{g}_{k-1}$, and g does not decrease. On the last repair day s, $\mu_s = g_s$, and μ does not decrease afterwards. Therefore $\mu_T \ge \mu_s = g_s \ge g_r > y^\mu$, which violates the loop row.
- **A repair, and $y^g \le y^\mu$.** Every repair day has $\bar{\mu}_{k-1} - \bar{g}_{k-1} \ge 0$: on the first one because $\bar{\mu}_{r-1} \ge y^\mu \ge y^g = \bar{g}_{r-1}$, and on the later ones as above. So

  $$g_T \ge y^g + (1-\rho) \sum (\bar{\mu}_{k-1} - \bar{g}_{k-1}), \tag{233}$$

  the sum over the repair days of the operating phase, with non-negative terms. The loop row $g_T \le y^g$ and $\rho < 1$ force every term to 0. Then every repair day gives $\mu_k = g_k \ge \bar{g}_{k-1} = \bar{\mu}_{k-1}$, every other day gives $\mu_k \ge \bar{\mu}_{k-1} + \delta\mu_k$, and

  $$\mu_T \;\ge\; y^\mu + \Delta \;>\; y^\mu, \tag{234}$$

  which violates the loop row.

□

Proof of Proposition 1.9. The first inequality is (35) with $x^m_{i\ell k} = 1$. Under ARD_∞,

$$g^\mu = \zeta^g = 0, \tag{235}$$

and $\mu_{k-1} \ge \zeta^\mu_{k-1}$ by Proposition 1.3. Under ARD_1, let $t < k$ be the last intervention day before k, or $t = 0$ if there is none. Then $g^\mu_{k-1} = g^\mu_t$ and $\zeta^g_{k-1} = \zeta^g_t$.

- If $t = 0$: $g_0^\mu = \zeta_0^g$ (the starting anchor), so

$$g_{k-1}^\mu - \zeta_{k-1}^g = 0 \leq \mu_{k-1} - \zeta_{k-1}^\mu. \tag{236}$$

- If $t \geq 1$: $g_t^\mu = \mu_t$ and $\zeta_t^g = \zeta_t^\mu$. On the days $j = t+1, \ldots, k-1$ there is no intervention:

$$\mu_j \geq \mu_{j-1} + \delta\mu_j \tag{237}$$

and

$$\zeta_j^\mu = \zeta_{j-1}^\mu + \delta\mu_j. \tag{238}$$

So $\mu_j - \zeta_j^\mu$ does not decrease, and

$$\mu_{k-1} - \zeta_{k-1}^\mu \geq \mu_t - \zeta_t^\mu = g_{k-1}^\mu - \zeta_{k-1}^g. \tag{239}$$

Rearranging gives

$$\mu_{k-1} - g_{k-1}^\mu \geq \zeta_{k-1}^\mu - \zeta_{k-1}^g. \tag{240}$$

The mean recursion is linear (Section 2.3.4); with an exact starting anchor, ζ^μ and ζ^g are the true means, so

$$\rho\,(\zeta_{k-1}^\mu - \zeta_{k-1}^g) = \mathbb{E}[D^- - D^+]. \tag{241}$$

□

A.2 Stochastic degradation and fatigue models

Proof of Proposition 2.1. By the coupling characterisation of $\leq_{\text{st}}$ (Definition 2.7), for each t there exist $\hat{X}_t \overset{d}{=} \Delta D_t$ and $\hat{Y}_t \overset{d}{=} \widetilde{\Delta D}_t$ on a common probability space with $\hat{X}_t \leq \hat{Y}_t$ almost surely. Realise these couplings independently across t. Then

$$\sum_t \hat{X}_t \leq \sum_t \hat{Y}_t \tag{242}$$

almost surely. Independence gives

$$\sum_t \hat{X}_t \overset{d}{=} \sum_t \Delta D_t \quad \text{and} \quad \sum_t \hat{Y}_t \overset{d}{=} \sum_t \widetilde{\Delta D}_t. \tag{243}$$

Hence

$$\sum_t \Delta D_t \leq_{\text{st}} \sum_t \widetilde{\Delta D}_t. \tag{244}$$

Any non-decreasing g preserves the almost-sure inequality, which gives the general statement. □

Derivation of the Markov, Cantelli and Chernoff bounds. With $D \geq 0$ and $\tau > 0$, Markov's inequality (64) follows from

$$\tau\,\mathbf{1}\{D > \tau\} \leq D. \tag{245}$$

Applying Markov to $(D - \mathbb{E}[D] + c)^2$ and optimising over $c > 0$ gives the one-sided Chebyshev (Cantelli) bound (65), for $\tau > \mathbb{E}[D]$. The exponential (Cramér–Chernoff) bound (66) follows from Markov's inequality applied to e^{sD}, and it is valid for every fixed $s > 0$. □

Derivation of the properties of the Bernstein constraint (73). By (62) the mean and the variance are *affine* in the counts. So the square root is concave, and the feasible set is reverse-convex (its complement is convex); it cannot be written as a second-order cone.

Every tangent line to the concave square root lies *above* it. So the replacement of the square root by a fixed tangent, or by the pointwise minimum of several tangents, gives constraints in the counts that imply (73): one linear constraint for a fixed tangent, and linear constraints with an SOS2 set for the minimum.

For an increment whose range has width $b_{\ell j}$, Bernstein's bound of its cumulant generating function reads, for $0 < s < 3/b_{\ell j}$,

$$\psi_{\ell j}(s) \;\le\; s\,\mu_{\ell j} + \frac{s^2 v_{\ell j}/2}{1 - s\,b_{\ell j}/3}. \tag{246}$$

Used in place of $\psi_{\ell j}$ in (70), it keeps a support per mission, and the bound still needs the minimisation over s. With one constant $b_\ell \ge \max_j b_{\ell j}$ in every term, the choice $s = a/(\mathrm{Var}[D] + b_\ell\, a/3)$, with $a = \tau - \mathbb{E}[D]$, gives the closed form (73), which has no s. □

Derivation of the properties of the width of a range. Let the increment X lie in an interval $[l, l+b]$ almost surely. Bernstein's inequality needs an upper bound of the centred term, and the width is one:

$$X - \mathbb{E}X \le l + b - l = b, \tag{247}$$

since $\mathbb{E}X \ge l$.

For Popoviciu's inequality, let X take values in an interval of width w with midpoint m_c; then

$$\mathrm{Var}\, X \le \mathbb{E}\,(X - m_c)^2 \le (w/2)^2. \tag{248}$$

□

Proof of Proposition 2.2. Item 1: the mean and R are additive by definition, the variance and the cumulant generating function by independence. Item 2: m and V scale by $1-\rho$ and $(1-\rho)^2$; every weight is multiplied by $1-\rho$, so R scales by $(1-\rho)^2$. A cumulant generating function is convex with value 0 at 0, so

$$K\big((1-\rho)D\big) = K_D\big((1-\rho)s\big) \le (1-\rho)\,K_D(s) + \rho\,K_D(0) = (1-\rho)\,K(D), \tag{249}$$

with K_D the cumulant generating function of D as a function of its argument. Item 3: by independence,

$$\theta(D^-) = \theta(G) + \theta(\Delta); \tag{250}$$

for R the weights of the terms of G are unchanged and those of Δ are multiplied by $1-\rho$. Hence

$$\begin{aligned} \theta(D^+) &= \theta(G) + a_\theta\,\theta(\Delta) \\ &= \theta(G) + a_\theta\big(\theta(D^-) - \theta(G)\big) \\ &= a_\theta\,\theta(D^-) + (1 - a_\theta)\,\theta(G), \end{aligned} \tag{251}$$

and, with item 2 applied to Δ,

$$\begin{aligned} K(D^+) &= K(G) + K_\Delta\big((1-\rho)s\big) \\ &\le K(G) + (1-\rho)\big(K(D^-) - K(G)\big) \\ &= (1-\rho)\,K(D^-) + \rho\,K(G). \end{aligned} \tag{252}$$

□

Derivation of the rows of Table 8. Write $d_k = \tau - \mu_k$.

- **Markov.** By A0 and Markov's inequality,

$$\Pr(D_k > \tau) \le m/\tau \le \mu_k/\tau, \tag{253}$$

so the row $\mu_k \le \varepsilon\,\tau$ suffices. It uses the mean only and does not need A1 or A2.

- **Cantelli.** Cantelli's inequality gives

$$\Pr(D - m \ge a) \le V/(V + a^2) \quad \text{for } a > 0; \tag{254}$$

the row is equivalent to

$$\mu_k + \kappa_C \sqrt{v_k} \le \tau \quad \text{with} \quad \kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}. \tag{255}$$

If $d_k > 0$,

$$\Pr(D_k > \tau) \le \Pr(D_k - m \ge \tau - m) \le \frac{V}{V + (\tau - m)^2} \le \frac{v_k}{v_k + d_k^2} \le \varepsilon, \tag{256}$$

because $v/(v + d^2)$ increases in v and decreases in d, and $\tau - m \ge d_k$. If $d_k = 0$, the row gives $v_k = 0$, so $V = 0$ and

$$D_k = m \le \tau. \tag{257}$$

- **Hoeffding.** $D_k - m$ is a sum of independent centred terms $w_t(X_t - \mathbb{E}X_t)$ with ranges of widths $w_t\sqrt{r_t}$, and the sum of their squares is R^{t}. Hoeffding's inequality gives

$$\Pr(D_k - m \ge a) \le \exp(-2a^2/R^{\mathrm{t}}), \tag{258}$$

and $\exp(-2d^2/R) \le \varepsilon$ exactly when $d^2 \ge (\kappa/2)\,R$. If $R^{\mathrm{t}} = 0$, every term is constant and

$$D_k = m \le \tau; \tag{259}$$

this covers $d_k = 0$, where the row gives $R_k = 0$. Otherwise $d_k > 0$, and

$$\Pr(D_k > \tau) \le \exp\bigl(-2(\tau - m)^2/R^{\mathrm{t}}\bigr) \le \exp(-2d_k^2/R_k) \le \varepsilon, \tag{260}$$

because $\exp(-2d^2/R)$ increases in R and decreases in d.

- **Bernstein.** The support constant b_ℓ of (73) is a constant, not a decision variable; the replacement support enters it only when replacements are allowed. Every weighted term satisfies

$$w_t(X_t - \mathbb{E}X_t) \le w_t\, b_t \le b_\ell, \tag{261}$$

with b_t the width of the range of X_t, whatever the schedule, because $w_t \le 1$. Bernstein's inequality for independent terms bounded above by b_ℓ gives

$$\Pr(D - m \ge a) \le \exp\bigl(-a^2/(2(V + b_\ell\, a/3))\bigr) \quad \text{for } a > 0; \tag{262}$$

the right side is at most ε exactly when $a^2 - 2c\,a - 2\kappa V \ge 0$. With $v_k \ge 0$, the row and its side row give $d_k \ge 2c$. The function

$$a^2/(2(V + b_\ell\, a/3)) \tag{263}$$

increases in $a > 0$ and decreases in V, and

$$\tau - m \ge d_k \ge 2c; \tag{264}$$

so $\Pr(D_k > \tau) \le \varepsilon$ when $b_\ell > 0$. When $b_\ell = 0$, every term is constant and

$$D_k = m \le \tau. \tag{265}$$

- **Chernoff.** For $s > 0$, Markov's inequality applied to e^{sD} gives

$$\Pr(D_k \geq \tau) \leq \exp(K^{\mathrm{t}} - s\tau). \tag{266}$$

Since $K^{\mathrm{t}} \leq K_k$, the row gives

$$\Pr(D_k > \tau) \leq \exp(K_k - s\tau) \leq \varepsilon. \tag{267}$$

The row bounds K by $s\tau - \kappa$, its natural range, and $K \geq 0$, because

$$K(X) \geq s\,\mathbb{E}X \geq 0 \tag{268}$$

by Jensen's inequality and A0. So a known zero damage satisfies the row only if $s\tau \geq \kappa$, and a replacement day satisfies it only if the replacement value of K is at most $s\tau - \kappa$.

□

Derivation of the initial and replacement values of the statistic K. By Jensen's inequality,

$$\ln \mathbb{E}[e^{sD}] \geq s\,\mathbb{E}[D], \tag{269}$$

with equality exactly when D is almost surely constant. So a known initial damage μ_0 gives the initial value $K_0 = s\,\mu_0$, and a known replacement damage μ^{new} gives the reset value $s\,\mu^{\mathrm{new}}$. A random initial or replacement damage needs its own cumulant generating function at s, which is larger.

For the same reason, the cumulant generating function of every increment at s is at least s times its mean. Also the initial value and the replacement value of K are at least s times the corresponding mean. □

Derivation that $U^{(1)} \leq U^{(0)}$ does not chain (Section 2.1.5). The bound U is a many-to-one function of the parameters of the state, and its level sets are not invariant under the dynamics. Bernstein's bound, for example, depends on $(\mathbb{E}[D], \mathrm{Var}[D])$ through a single scalar. So $U^{(1)} = U^{(0)}$ can hold with

$$\big(\mathbb{E}[D_T], \mathrm{Var}[D_T]\big) \;\neq\; \big(\mathbb{E}[D_{H_1}], \mathrm{Var}[D_{H_1}]\big), \tag{270}$$

for example with a larger mean balanced by a smaller variance. One more repetition of the dynamics can map these two parameter points to different bound values, the larger one at the end of the second repetition. The Cantelli instance of Section 2.1.5 with $\sum_j n_{ij} v_{\ell j} = 4$ gives $U^{(1)} = U^{(0)} = 0.2000$ and $U^{(2)} = 0.2155$. So the relation (92) is possible. □

Derivation that the moment loop (94) chains. Under the dynamics, the mean and the variance change by one of three operations.

- **Increment.** The moments grow as

$$\mathbb{E} \mathrel{+}= \sum_j n_{ij}\mu_{\ell j} \qquad \text{and} \qquad \mathrm{Var} \mathrel{+}= \sum_j n_{ij} v_{\ell j}. \tag{271}$$

- **Imperfect repair.** A repair maps the moments as follows under ARD_∞ and under ARD_1:

$$\mathrm{ARD}_\infty\colon \quad \mathbb{E} \mapsto (1-\rho)\,\mathbb{E}, \qquad \mathrm{Var} \mapsto (1-\rho)^2\,\mathrm{Var}, \tag{272}$$

$$\mathrm{ARD}_1\colon \quad \mathbb{E} \mapsto (1-\rho)\,\mathbb{E} + \rho\, g^{\mu}, \qquad \mathrm{Var} \mapsto (1-\rho)^2\,\mathrm{Var} + \big(1-(1-\rho)^2\big)\, g^{v}. \tag{273}$$

- **Replacement.** A reset to the constants of D^{new}.

Each operation is non-decreasing in every moment and anchor separately, so the moments evolve as monotone recursions.

Componentwise domination therefore chains. Write Π for the map of one repetition of the operating phase. Then (94), with the rows on the anchors under ARD_1, states that Π does not increase either moment, nor either anchor, at the start of the repetition y. Hence the moments at the end of each repetition are at most those at the end of the previous one (Theorem 1.4).

The Bernstein bound (73), like the Cantelli bound (65), is increasing in both $\mathbb{E}[D]$ and $\mathrm{Var}[D]$. So the certified risk does not increase across repetitions, as in (95), and the schedule is repeatable. □

Derivation of the loop (96) on the cumulant generating function. Without imperfect repair, K_k is its value at the last replacement (or on day 0) plus the count-weighted sum

$$\sum_j n_{ij}^{(k)}\, \psi_{\ell j}(s). \tag{274}$$

Here $n_{ij}^{(k)}$ is the number of days on which mission j is served after the last replacement and up to day k. Then $M_D(s)$ propagates multiplicatively,

$$M_{D+\Delta D}(s) \;=\; M_D(s)\, M_{\Delta D}(s). \tag{275}$$

So the value of K on a day depends only on its value on the day before, and (96) carries over by induction.

An imperfect (ARD) repair does not keep this scalar exact. With $D^+ = (1-\rho)D^-$,

$$M_{D^+}(s) \;=\; \mathbb{E}\big[e^{s(1-\rho)D^-}\big] \;=\; M_{D^-}\big((1-\rho)s\big)\,: \tag{276}$$

the transform of the repaired state at s is the old transform at the *shifted* argument $(1-\rho)s$. The value $M_{D^-}(s)$ does not determine it: the repair moves the state along the transform curve. However, $M_{D^-}(s)$ determines an upper bound: the cumulant generating function is convex with value 0 at 0, so (Proposition 2.2)

$$K_{D^+}(s) \;\le\; (1-\rho)\, K_{D^-}(s) \qquad \text{under } \mathrm{ARD}_\infty, \text{ and} \tag{277}$$

$$K_{D^+}(s) \;\le\; (1-\rho)\, K_{D^-}(s) + \rho\, K_G(s) \qquad \text{under } \mathrm{ARD}_1. \tag{278}$$

So the tracked statistic follows the maps (97), with the anchor g^K. These maps are non-decreasing in (K, g^K), and the tracked K stays an upper bound on the true cumulant generating function, which is all that the Chernoff certificate needs. □

Derivation of the validity of the support $b_\ell(\zeta)$ and of its rows (Section 2.1.6).

- **Validity of $b_\ell(\zeta)$.** Between resets, the state is a sum, with weights in $[0,1]$, of one reset draw and of increments of *served* missions only. So $b_\ell(\zeta)$ bounds every summand, and Bernstein's inequality applies without change, on every day and in every repetition of the schedule.
- **Equivalence of the rows.** The certified value $\mu + c + \sqrt{c^2 + 2\kappa v}$ has the derivative

  $$1 + c/\sqrt{c^2 + 2\kappa v} \;\in\; [1, 2] \tag{279}$$

  in c. So it increases with the support constant, and the Bernstein row written for a larger constant implies every row written for a smaller one. The active row with the largest constant dominates the others, and that largest constant is exactly $\kappa b_\ell(\zeta)/3$. So the conditional rows at c_j and the unconditional row at c_0 are exactly equivalent to the single row with the schedule-dependent maximum.

□

Derivation of the remaining gap (112) *of the union of tangents.* The gap of tangent r is

$$f(\mu) - T_r(\mu) \;=\; (\mu - \mu_r)^2, \tag{280}$$

so the remaining gap of the union is (112) where the union admits $v \geq 0$. This gap is quadratic in the distance to the nearest tangent point, with worst case $(\Delta/2)^2$ for a spacing Δ. Where the union admits no $v \geq 0$, the gap is $f(\mu) \leq (\Delta/2)^2$. So doubling the number of tangents divides it by four. □

Proof of Proposition 2.3. f is a polynomial of degree 2 in μ with leading coefficient c_2, so its Taylor expansion at μ_r is exact:

$$f(\mu) = f(\mu_r) + f'(\mu_r)(\mu - \mu_r) + c_2(\mu - \mu_r)^2. \tag{281}$$

Since $c_2 > 0$, the tangent lies below f. Each linear form caps χ by one such tangent and keeps the side row. Therefore it implies (113). □

Derivation of the conditions on the tangent point of a Bernstein row (Section 2.1.7). A tangent is linear in μ, so its largest value on $[0, \tau - c]$ is at an end point. At $\tau - c$ it is at most

$$f(\tau - c) \;=\; -c^2/(2\kappa) \;<\; 0. \tag{282}$$

At 0 it equals

$$f(0) - \mu_r^2/(2\kappa) \;=\; \big(\tau(\tau - 2c) - \mu_r^2\big)/(2\kappa), \tag{283}$$

by Proposition 2.3 with $c_2 = 1/(2\kappa)$. So the tangent row admits a point with $v \geq 0$ exactly when $\mu_r^2 \leq \tau(\tau - 2c)$, and at equality only $\mu = 0$ and $v = 0$. Since $\sqrt{\tau(\tau - 2c)} < \tau - c$, this covers the case $\mu_r \geq \tau - c$, in which the tangent has a non-negative slope.

The `pwl` form accepts the points below the largest of its tangents, a convex function. At $\mu = 0$ the largest is the tangent at the midpoint $\tau/(2P)$ of the first segment, so the same argument applies with $\mu_r = \tau/(2P)$. Otherwise no segment admits a point with $v > 0$: the first segment fails as above, and on every later segment

$$\mu \;\geq\; \tau/P \;>\; \tau - 2c, \tag{284}$$

where $f(\mu) < 0$. □

Derivation of the inverse moments (119).

- **Log-normal.** The first line is the normal moment generating function at the argument $-r$.
- **Weibull.** The second line follows from the substitution $u = (x/\lambda)^k$, which turns $\mathbb{E}[N_f^{-r}]$ into
$$\lambda^{-r} \int_0^\infty u^{-r/k} e^{-u}\, \mathrm{d}u; \tag{285}$$
this integral diverges *at the origin* when $r \geq k$.
- **Gumbel-minimum.** The third line writes $\ln N_f = \mu + \beta G$ with $\Pr(G \leq g) = 1 - e^{-e^g}$. The transform
$$\mathbb{E}[e^{tG}] \;=\; \int_0^\infty u^t e^{-u}\, \mathrm{d}u \;=\; \Gamma(1 + t) \tag{286}$$
(substitute $u = e^g$) is finite if and only if $t > -1$.

□

Derivation that the loop condition (129) *chains.* Unlike the scalar bound of Section 2.1.5, which loses information, the condition (129) *chains*, for two reasons. First, M is a sufficient statistic: the failure probability is the fixed increasing function F_A of the scalar state. Second, the map Π of the operating phase is a composition of $m \mapsto m + c$ ($c \geq 0$), $m \mapsto (1-\rho)m$ and $m \mapsto 0$, and each of these maps is non-decreasing. So, from

$$M_T \;=\; \Pi(M_{H_1}) \;\leq\; M_{H_1}, \tag{287}$$

induction gives

$$\Pi^{n+1}(M_{H_1}) \;\leq\; \Pi^n(M_{H_1}) \qquad \text{for all } n. \tag{288}$$

The same monotonicity, applied to the partial maps, bounds every trajectory within a repetition by the trajectory of the operating phase on the horizon. So imposing (125) on all $k = 1, \ldots, T$ extends the per-day reliability to every repetition, replacements (which draw a new A) included. □

A.3 Stochastic orderings and maintenance strategies

Derivation of the Gamma rows of Tables 12–14. The log-likelihood ratio of the law $\mathrm{Ga}(\alpha_2, \beta_2)$ over the law $\mathrm{Ga}(\alpha_1, \beta_1)$ has derivative

$$(\alpha_2 - \alpha_1)/x - (\beta_2 - \beta_1), \tag{289}$$

which is non-negative for all $x > 0$ if and only if $\alpha_2 \geq \alpha_1$ and $\beta_2 \leq \beta_1$. Conversely, if $\alpha_1 > \alpha_2$, then

$$F_X(t) \sim (\beta_1 t)^{\alpha_1}/(\alpha_1\,\Gamma(\alpha_1)) \ll (\beta_2 t)^{\alpha_2}/(\alpha_2\,\Gamma(\alpha_2)) \sim F_Y(t) \tag{290}$$

as $t \to 0^+$, which violates $\leq_{\mathrm{st}}$. If $\beta_1 < \beta_2$, then $\bar{F}_X(t)/\bar{F}_Y(t) \to \infty$ as $t \to \infty$, which again violates $\leq_{\mathrm{st}}$. Hence all three orders coincide for the Gamma family [32, Sec. 1.C]. □

Derivation of the Gaussian rows of Tables 12–14. The CDF of $\mathcal{N}(\mu, \sigma^2)$ is $\Phi\big((t-\mu)/\sigma\big)$. The inequality

$$\Phi\big((t-\mu_1)/\sigma_1\big) \geq \Phi\big((t-\mu_2)/\sigma_2\big) \tag{291}$$

for all t requires

$$(t-\mu_1)/\sigma_1 \geq (t-\mu_2)/\sigma_2 \tag{292}$$

for all t. This condition is linear in t, and it can hold for all t only if $\sigma_1 = \sigma_2$. It then reduces to $\mu_1 \leq \mu_2$. The likelihood ratio is then

$$\exp\big((\mu_2 - \mu_1)\,x/\sigma^2 + \mathrm{const}\big), \tag{293}$$

non-decreasing in x. Thus all three orders coincide: $\sigma_1 = \sigma_2$ and $\mu_1 \leq \mu_2$. □

Derivation of the inverse Gaussian row of Table 12. The log-likelihood ratio of $\mathrm{IG}(\mu_2, \lambda_2)$ over $\mathrm{IG}(\mu_1, \lambda_1)$ has derivative

$$\frac{\lambda_1}{2\mu_1^2} - \frac{\lambda_2}{2\mu_2^2} + \frac{\lambda_2 - \lambda_1}{2x^2}. \tag{294}$$

This is non-negative for all $x > 0$ if and only if $\lambda_2 \geq \lambda_1$ (from $x \to 0^+$) and $\lambda_1/\mu_1^2 \geq \lambda_2/\mu_2^2$ (from $x \to \infty$). □

Derivation of the compound Poisson row of Table 14. Sufficiency follows from the superposition coupling

$$\mathrm{CP}(\nu_2, F_Y) \overset{d}{=} \mathrm{CP}(\nu_1, F_Y) + \mathrm{CP}(\nu_2 - \nu_1, F_Y), \tag{295}$$

with independent non-negative summands. Necessity follows from the atom at zero, $\Pr(X = 0) = e^{-\nu \Pr(Y>0)}$. □

Proof of Proposition 2.12. Let $c_t = w_t a_t$ and $A = \sum_t c_t$. If $A = 0$, then $D = 0$. Otherwise, the map $x \mapsto (x-d)_+$ is convex and non-decreasing, and $(A-d)_+ \le \mathbb{E}(AZ-d)_+$ by Jensen's inequality, since $\mathbb{E}Z = 1$. Hence, for every $d \ge 0$,

$$\mathbb{E}(D-d)_+ = \mathbb{E}\Big(\sum_t \tfrac{c_t}{A}\,(AZ_t) - d\Big)_+ \le \sum_t \tfrac{c_t}{A}\,\mathbb{E}(AZ_t - d)_+ \le \mathbb{E}(AZ-d)_+ \le \mathbb{E}(\mu Z - d)_+. \tag{296}$$

□

Proof of Theorem 2.13. If $\mu = 0$, then $\mathbb{E}D \le 0$ and $D = 0$. Let $\mu > 0$ and $d = \mu q$; then

$$d < \mu\bar{q} \le \tau. \tag{297}$$

For every x,

$$\mathbf{1}\{x > \tau\} \le (x-d)_+/(\tau - d), \tag{298}$$

and

$$\mathbb{E}(Z-q)_+ = \varepsilon\,(\bar{q} - q). \tag{299}$$

Therefore

$$\Pr(D > \tau) \le \frac{\mathbb{E}(D-d)_+}{\tau - d} \le \frac{\mu\,\mathbb{E}(Z-q)_+}{\tau - \mu q} = \frac{\varepsilon\,\mu\,(\bar{q}-q)}{\tau - \mu q} \le \varepsilon, \tag{300}$$

where the last step is $\mu\bar{q} \le \tau$. □

Derivation of the tail mean of the inverse Gaussian law. Substituting into the density f_Z of Z gives $z\,f_Z(z) = z^{-2} f_Z(1/z)$, and the change of variable $u = 1/z$ gives

$$\int_q^\infty z\,f_Z(z)\,\mathrm{d}z = \int_0^{1/q} f_Z(u)\,\mathrm{d}u = F_Z(1/q). \tag{301}$$

□

Derivation of the sharpness of the CVaR cap. Take $\mu > \bar{\mu}$ and a level $\varepsilon' > \varepsilon$ close to ε, with quantile q'; then still $\mu\,\mathbb{E}[Z \mid Z > q'] > \tau$. The two-point law

$$\mu\,\mathbb{E}\big[Z \mid \mathbf{1}\{Z > q'\}\big] \tag{302}$$

lies below μZ in the order of Definition 2.11, by Jensen's inequality, and exceeds τ with probability at least ε'. □

Derivation of the search interval of the inverse Gaussian quantile. The right end of the interval in $u = \ln q$ is $\ln b$, with

$$b = 1 + \sqrt{\eta\,(2/\varepsilon - 1)} > 1. \tag{303}$$

Z has the mean 1 and the variance η, so Cantelli's inequality (65) at the level b gives

$$\Pr(Z > b) \le \frac{\eta}{\eta + (b-1)^2} = \frac{\eta}{\eta + \eta\,(2/\varepsilon - 1)} = \frac{\varepsilon}{2} < \varepsilon. \tag{304}$$

Therefore $\ln \Pr(Z > e^u) - \ln \varepsilon < 0$ at $u = \ln b$, and the quantile satisfies $q < b$. □

Proof of Lemma 2.15. Since the last replacement (or day 0),

$$D_k = \delta + \sum_t w_t X_t. \tag{305}$$

The X_t are the independent Gamma terms added since then, $\delta \geq 0$ collects the known constants, and each weight $w_t \in [0,1]$ is a product of factors $1-\rho$. A term of weight 0, possible when $\rho = 1$, is dropped. A mission adds a term with weight 1. ARD_∞ multiplies every weight, constants included, by $1-\rho$. ARD_1 multiplies by $1-\rho$ the weights of the terms and constants added since the last intervention, and keeps the others. A replacement starts a new sum with the replacement draw. When the anchor before the first intervention is below the initial mean, the initial damage is read as two independent terms of its rate, with means equal to the anchor and to the rest. Their sum has the law of the initial damage. A known initial damage is read as two constants in the same way. Write $X_t = Y_t/\beta_t$ with $Y_t \sim \mathrm{Ga}(a_t, 1)$ and $\beta_t \geq \beta^*$. Then

$$\beta^* D_k = \beta^* \delta + \sum_t (w_t \beta^*/\beta_t)\, Y_t, \tag{306}$$

and every scale $w_t\beta^*/\beta_t$ lies in $(0,1]$. The Y_t are independent, so the law of $\beta^* D_k$ is in $\mathcal{G}(\beta^* \mathbb{E}[D_k])$. □

Proof of Lemma 2.16. S is infinitely divisible, with drift δ and Lévy density

$$\Lambda(y) = \sum_i (m_i/c_i)\, e^{-y/c_i}/y \tag{307}$$

on $y > 0$. A change of the parameters at the rates $(\dot\delta, \dot\Lambda)$ changes the Laplace transform

$$\mathbb{E}e^{-\vartheta S} = \exp\bigl(-\vartheta\delta - \int_0^\infty (1 - e^{-\vartheta y})\, \Lambda(y)\, \mathrm{d}y\bigr) \tag{308}$$

at the rate

$$\int_0^\infty \mathbb{E}\bigl[e^{-\vartheta(S+y)} - e^{-\vartheta S}\bigr]\, \dot\Lambda(y)\, \mathrm{d}y - \vartheta\dot\delta\, \mathbb{E}e^{-\vartheta S}. \tag{309}$$

Since $\int_0^\infty e^{-\vartheta t} \Pr(S > t)\, \mathrm{d}t = (1 - \mathbb{E}e^{-\vartheta S})/\vartheta$, this rate times $-1/\vartheta$ is the transform of the rate

$$\int_0^\infty \Pr(t - y < S \leq t)\, \dot\Lambda(y)\, \mathrm{d}y + \dot\delta\, f_S(t) \tag{310}$$

of $\Pr(S > t)$. Case (a) is $\dot\Lambda(y) = c^{-1}e^{-y/c}/y$, or $\dot\delta = 1$ when $c = 0$. Case (b) is

$$\dot\Lambda(y) = m_i\, \partial_c\bigl(c^{-1}e^{-y/c}/y\bigr) \tag{311}$$

at $c = c_i$. □

Proof of Theorem 2.17. Let $\mathcal{G}_N(m)$ be the elements with at most N atoms, the constant included, and let $N \geq 3$.

1. *A maximiser exists.* $\mathcal{G}_N(m)$ is the image of the compact set of atoms (m_i, c_i) with $m_i \geq 0$, $\sum_i m_i = m$, $c_i \in [0,1]$. The law of S depends continuously on the atoms, and only the constant element $S = m$ has an atom of probability, at $m \neq t$. So $\Pr(S > t)$ is continuous and attains its maximum at some S^*. Since

$$\Pr(m > t) = 0 < Q(m,t), \tag{312}$$

S^* has at least one term. Every element has finitely many atoms, so

$$W(m,t) = \sup_N \max_{\mathcal{G}_N(m)} \Pr(S > t). \tag{313}$$

2. *First-order conditions.* Write $\omega^\circ = \omega_{S^*}$. Lemma 2.16 gives three conditions.
   (i) Moving mean between two atoms of positive mean must not raise the tail, so ω° takes one value ω^* at every atom of positive mean.
   (ii) At an atom with $c \in (0,1)$, $\partial\omega^\circ/\partial c = 0$.
   (iii) If S^* has fewer than N atoms of positive mean, a new atom can be opened at any $c \in [0,1]$, so $\omega^\circ(c) \le \omega^*$ on $[0,1]$.

3. *A single-peak function.* Put $\gamma = 1/c$ and
$$h(u) = \Pr(t-u < S^* \le t)/u. \tag{314}$$
Then
$$\omega^\circ(\gamma) = \gamma \int_0^\infty h(u)\, e^{-\gamma u}\, \mathrm{d}u, \tag{315}$$
and
$$\omega^\circ(\infty) = h(0^+) = f_{S^*}(t) \tag{316}$$
is the value at the constant atom. S^* is a constant plus a sum of independent Gamma variables, so its density is unimodal [29, Theorem 4, p. 11], proved in [34, proof of Theorem 4, pp. 194–195]; the constant shifts the mode only. Then $p(v) = f_{S^*}(t-v)$ for $v < t$ and $p(v) = 0$ for $v \ge t$ is unimodal on $[0,\infty)$, and so is its running average
$$h(u) = u^{-1} \int_0^u p : \tag{317}$$
before the mode of p, $p \ge h$; after it,
$$u\,(p(u) - h(u)) = u\,p(u) - \int_0^u p \tag{318}$$
is non-increasing. Since $h' = (p-h)/u$, h rises and then falls.

4. *Zero count.* For every constant ω', $h - \omega'$ changes sign at most twice. The Laplace transform of a function with at most n sign changes has at most n zeros in $\gamma > 0$, counted with multiplicity. Indeed, at a sign change u_1, multiply the transform by $e^{\gamma u_1}$. Its derivative in γ is the transform of a function with one sign change fewer, and Rolle's theorem and induction conclude. So
$$\omega^\circ - \omega' = \gamma \int_0^\infty (h - \omega')\, e^{-\gamma u}\, \mathrm{d}u \tag{319}$$
has at most two zeros in $\gamma > 0$. Also $\omega^\circ(\gamma) \to 0$ as $\gamma \to 0^+$. Hence ω° cannot rise above a level $\omega' > 0$, fall below it and rise above it again: with $\omega^\circ(0^+) = 0$ that would give three zeros. So ω° is quasi-concave. It is analytic in $\gamma > 0$ and not constant, so its maximum over $\gamma \in [1,\infty]$ is attained at one point only: two maximisers would make ω° constant between them.

5. *One term.* By (i) and (ii), an atom with $c \in (0,1)$ is a zero of $\omega^\circ - \omega^*$ of multiplicity at least 2, and an atom at $c = 1$ a zero of multiplicity at least 1. The constant atom is at $\gamma = \infty$. By step 4, S^* has at most two atoms of positive mean. As $N \ge 3$, (iii) applies, so ω^* is the maximum of ω° over $[1,\infty]$, attained at every atom of positive mean. By step 4 there is only one such atom, and by step 1 it is not the constant. So $S^* = c^*\,\mathrm{Ga}(m/c^*, 1)$ with $c^* \in (0,1]$.

Therefore
$$\max_{\mathcal{G}_N(m)} \Pr(S > t) = \max_{c \in (0,1]} Q(m/c, t/c) \tag{320}$$
for every $N \ge 3$, and the claim follows from step 1. □

Proof of Lemma 2.18. Let $m < m' < t$ and let the law of S be in $\mathcal{G}(m)$. Then the law of $S+(m'-m)$ is in $\mathcal{G}(m')$, and $\Pr(S+m'-m > t) \geq \Pr(S > t)$. Therefore $W(m,t) \leq W(m',t)$. □

Derivation of the properties of the function w.

- w is non-decreasing in A, because $Q(a,x)$ increases in a. It is continuous on $[0, A']$ for every $A' < t$. By Chebyshev's inequality $Q(A/c, t/c) \leq cA/(t-A)^2$, so $w(A) \leq A/(t-A)^2 \to 0 = w(0)$ as $A \to 0$. On $[\delta, A']$, with $\delta > 0$, the maximum can therefore be taken over a compact interval $[c_0, 1]$, on which $Q(A/c, t/c)$ is continuous in (A, c).
- $w(A) < 1/2$ for every $A < t$: with $r = A/t < 1$ and $s = t/c$,

$$Q(rs, s) \leq Q(rs, rs) < 1/2, \tag{321}$$

  because the median of a Gamma law lies below its mean.
- $w(A) \to 1/2$ as $A \to t$, because $Q(s,s) \to 1/2$ as $s \to \infty$.

□

Proof of Proposition 2.19. Let $m = \beta^* \mathbb{E}[D_k]$ and $A = A_1(\beta^*\tau, \varepsilon)$, so

$$m \leq \beta^* \mu_k \leq A < t. \tag{322}$$

If $m = 0$, then $D_k = 0$. Otherwise

$$\Pr(D_k > \tau) = \Pr(\beta^* D_k > t) \leq W(m,t) \leq W(A,t) = w(A) \leq \varepsilon, \tag{323}$$

by Lemma 2.15, Lemma 2.18, Theorem 2.17 and the choice of A: in both cases of the definition, $A < t$ and $w(A) \leq \varepsilon$. □

Derivation of the computation of the Gamma cap. Three facts justify the computation.

1. A bisection on the first argument of Q finds the exact one-rate cap $\hat{A}$, the root of $Q(\hat{A}, t) = \varepsilon$, because Q increases in its first argument. The inverse routine `gammaincinv` solves for the second argument and does not apply.
2. The function $c \mapsto Q(A/c, t/c)$ tends to 0 as $c \to 0$ and has a single peak on $(0, 1]$. With $r = A/t$ and $s = t/c$, the function is $s \mapsto Q(rs, s)$ on $s \geq t$. A single peak of this function on $s \in [10^{-12}, 10^8]$ was checked numerically for r from 10^{-5} to 0.999, which covers the bisection for $\varepsilon \geq 10^{-5}$. A second check, on 200 values of r from 10^{-10} to 10^{-5}, found no second peak either. So a logarithmic grid of values of c, refined by a golden-section search between the neighbours of the best grid point, finds the maximum.
3. The bisection on A starts from the lower end εt, which satisfies $w(\varepsilon t) \leq \varepsilon$ by Markov's inequality $Q(a,x) \leq a/x$. It returns the lower end of its last interval, so $w(A_1) \leq \varepsilon$ holds at the returned value.

□

Derivation of the sharp cap below the threshold t_1. With $r = A/t$ and $s = t/c$, $Q(A/c, t/c) = Q(rs, s)$, and $c \in (0, 1]$ means $s \geq t$. Let $r^*(\varepsilon)$ be the value of r at which the peak value of $s \mapsto Q(rs, s)$ equals ε. This peak lies at $s = t_1(\varepsilon)$, so $r^*(\varepsilon) = \hat{A}(t_1, \varepsilon)/t_1$. For $t \leq t_1(\varepsilon)$ the peak lies at $s \geq t$, so the maximum does not depend on t, and

$$A_1(t, \varepsilon) = r^*(\varepsilon)\, t. \tag{324}$$

□

Derivation of the monotonicity of the Gamma mean cap in β^*. A larger t shrinks the range $s \geq t$, so $A_1(t,\varepsilon)/t$ is non-decreasing in t, and the mean cap

$$A_1(\beta^*\tau,\varepsilon)/\beta^* = \tau\, A_1(t,\varepsilon)/t \tag{325}$$

is non-decreasing in β^*. □

Derivation of facts (M1)–(M3) for the Gamma model. (M1) The repair factor $1-\rho$ lies in $[0,1]$, and the mean increments are non-negative.

(M2) The true mean equals the day map applied to the true statistics of the day before, and $\theta_0 \leq \xi_0$ holds as for the Gaussian model below.

(M3) If a value $\mu \leq A_1/\beta^*$ is at least the true mean of day k, then $\Pr(D_k > \tau) \leq \varepsilon$, because the proof of Proposition 2.19 uses only these two facts.

□

Proof of Proposition 2.20. A mission adds an independent Gaussian term, and ARD_∞ scales the damage by $1-\rho$; both keep the law Gaussian and give the first two cases. Under ARD_1, $D^- = G + \Delta$, with G the damage just after the last intervention and Δ the increments added since; G and Δ are independent Gaussian variables by N1. Then $D^+ = G + (1-\rho)\Delta$ is Gaussian, with mean $m_G + (1-\rho)(m^- - m_G)$ and variance

$$V_G + (1-\rho)^2\, V_\Delta = V_G + (1-\rho)^2\,(V^- - V_G) = (1-\rho)^2\, V^- + \big(1-(1-\rho)^2\big)\, V_G. \tag{326}$$

A replacement starts from the replacement draw. Therefore the law is Gaussian on every day, with the moments of the maps. □

Derivation of the validity of the Gaussian row. Let $(m_k, V_k) \leq (\mu_k, v_k)$ be the true moments (Proposition 2.20 and Proposition 1.3). If $V_k = 0$, then

$$D_k = m_k \leq \tau. \tag{327}$$

Otherwise $\mu_k < \tau$, because $\mu_k = \tau$ forces $v_k = 0$, and

$$\Pr(D_k > \tau) = 1 - \Phi\big((\tau - m_k)/\sqrt{V_k}\big) \leq 1 - \Phi\big((\tau-\mu_k)/\sqrt{v_k}\big) \leq \varepsilon, \tag{328}$$

because $(\tau-\mu)/\sqrt{v}$ decreases in μ and in v. □

Derivation of facts (M1)–(M3) for the Gaussian model. (M1) The repair factors $1-\rho$ and $(1-\rho)^2$ lie in $[0,1]$, and the mean and variance increments are non-negative.

(M2) $\theta_0 \leq \xi_0$: the initial values are the moments of the initial damage, and each starting anchor is at least the moment of the part of the damage present just after the last intervention before day 1 (Section 2.3.4). On every later day, the true moments equal the maps applied to the true statistics of the day before (Proposition 2.20).

(M3) If values (μ, v) that satisfy the row are at least the true moments of day k, then $\Pr(D_k > \tau) \leq \varepsilon$ (the derivation of the validity of the Gaussian row, Appendix A.3, page 88).

□

Derivation for a repair of the Wiener damage path. A repair of the damage path, as in the Gaussian model, would give the same mean and the variance

$$(1-\rho)^2 V \quad \text{under } \mathrm{ARD}_\infty, \qquad (1-\rho)^2 V + \big(1-(1-\rho)^2\big)\, V_G \quad \text{under } \mathrm{ARD}_1, \tag{329}$$

with $V_G \leq V$ the variance just after the last intervention. Both are at most V, and the reliability row increases in v, so the rows of the Wiener model are safe for that reading as well. □

Derivation of facts (M1)–(M3) for the inverse Gaussian model. (M1) The repair factor $1-\rho$ lies in $[0,1]$, and the mean increments are non-negative.

(M2) $\theta_0 \le \xi_0$ holds as for the Gaussian model, and the true mean equals the day map applied to the true statistics of the day before.

(M3) If a value $\mu \le \bar{\mu}$ is at least the true mean of day k, then $\Pr(D_k > \tau) \le \varepsilon$, by Proposition 2.12 with this μ and Theorem 2.13.

Each day of each repetition has its own set of terms. The argument holds for each of them, because it needs neither independence nor a fixed set of terms. □

Derivation of facts (M1)–(M3) for the rainflow model. (M1) The repair factors $1-\rho$ (for the mean and the cumulant generating function) and $(1-\rho)^2$ (for the variance and the Hoeffding budget) lie in $[0,1]$, and the increments of the four statistics are non-negative. The cumulant generating function of an increment is at least s times its mean, which is non-negative, by Jensen's inequality and A0.

(M2) $\theta_0 \le \xi_0$ holds as for the Gaussian model. On every later day the true statistics are at most the day maps applied to the true statistics of the day before, with equality for the mean, the variance and the Hoeffding budget, and an inequality for the cumulant generating function (Proposition 2.2).

(M3) If values that satisfy the row of the bound are at least the true statistics of day k, then $\Pr(D_k > \tau) \le \varepsilon$ (the derivations of Section 2.1.4).

□

Derivation of the repair maps of the mean and the variance.

- **Mean.** $a = 1-\rho$, because expectation is linear: Eq. (175) gives

$$\mathbb{E}[D^+] = (1-\rho)\,\mathbb{E}[D^-] + \rho\,\mathbb{E}[D^{\text{last}}]. \tag{330}$$

- **Variance and Hoeffding budget.** $a = (1-\rho)^2$. Under ARD_1 the reason is that $D^+ = G + (1-\rho)\Delta$, where G is the damage just after the last intervention and Δ the damage added since, independent of G. So

$$\operatorname{Var}[D^+] = \operatorname{Var}[G] + (1-\rho)^2 \operatorname{Var}[\Delta] = (1-\rho)^2 \operatorname{Var}[D^-] + \big(1-(1-\rho)^2\big) \operatorname{Var}[G], \tag{331}$$

because $\operatorname{Var}[D^-] = \operatorname{Var}[G] + \operatorname{Var}[\Delta]$. Under ARD_∞, $\operatorname{Var}[(1-\rho)D^-] = (1-\rho)^2 \operatorname{Var}[D^-]$. The Hoeffding budget $R = \sum_t w_t^2 r_t$ follows the same map, because a repair multiplies by $1-\rho$ the weight w_t of each term that it repairs (Proposition 2.2).

□

A.4 Electric vehicles and implementation

Derivation of the validity of the big-M constants of Section 5.1. An intervention day is a depot day, so $\delta s_k = 0$ on it.

- The carry row with an intervention reads $s_k \ge s_{k-1} - M_s$, whose right-hand side is at most 0.
- The repair row without a repair has a right-hand side

$$a\,s_{k-1} + (1-a)\,g_{k-1} - M_s \le 0, \tag{332}$$

because the repair map is a convex combination of values in $[0, M_s]$.
- The replacement rows without a replacement allow

$$s_k \in [s^{\text{new}} - M_s,\ s^{\text{new}} + M_s] \supseteq [0, M_s]. \tag{333}$$

- The relaxed anchor rows allow every difference in $[-M_s, M_s]$.

□

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